

Space and Time Decompositions for LQG Problems in Decision-Hazard Information Structure

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Abstract

In this paper, we implement spatial decomposition, temporal block decomposition, and the combination of these two decompositions for a Linear Quadratic Gaussian (LQG) problem that can be decomposed into subproblems linked by linear coupling constraints, with an optimization span comprising two time scales. This allows us to solve only quadratic problems under linear constraints during the decompositions, which we address by establishing the Riccati equation adapted to the different cases considered, and thus to be able to efficiently solve the different subproblems appearing in the decompositions. We can also compute the solution of the global problem using Riccati and thus measure the quality of the solutions obtained by the decompositions.

1 Problem statement and Riccati equations

To motivate this paper, we consider the following energy-type problem. Over a time span $\mathbb{T} = \{0, \dots, \mathcal{T}-1\}$, we have a dam whose volume at the beginning of the time interval $[t, t+1[$ is denoted by x_t , supplied by a random inflow a_{t+1} during $[t, t+1[$, and allowing the production of a quantity u_t of (hydraulic) energy in a nearly cost-free manner. We also have another source of energy production (fossil) producing the quantity v_{t+1} at a known price p_t . Both energy sources must meet a random demand D_{t+1} . The initial state x_0 is assumed to be known, and the deviation of the final state $x_{\mathcal{T}}$ from a desired volume $\hat{x}$ is penalized by a quadratic term with coefficient α . The state and control variables are subject to bound constraints.

The production-demand balance constraint

$$u_t + v_{t+1} = D_{t+1} ,$$

allows us to eliminate the variable v_{t+1} . The production cost associated with fossil energy in the problem's criterion becomes $p_t(D_{t+1} - u_t)$. Furthermore, the random variable D_{t+1} , which then appears only in the criterion, is eliminated by integration and generates a constant term that has no influence on the optimization.

In order to use duality techniques in decomposition, we introduce into the criterion of this problem quadratic terms with coefficient ϵ (small) on the controls u_t ,¹ which allows us to have a strongly convex (rather than linear) criterion.

To simplify the problem further, the following assumptions are made:

- the boundary constraints on the state x_t and the control u_t are ignored,
- the input a_{t+1} follows a normal distribution with mean c_t and variance Σ_{t+1} .

The resulting optimization problem takes the form

$$\min_{x,u} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} \epsilon u_t^2 - p_t u_t \right) + \frac{1}{2} \alpha (x_{\mathcal{T}} - \hat{x})^2 \right],$$

subject to the dynamic constraints

$$x_0 \text{ given}, \quad x_{t+1} = x_t - u_t + c_t + w_{t+1}, \quad t = 0, \dots, \mathcal{T}-1,$$

where w_{t+1} is a Gaussian noise with zero mean and variance Σ_{t+1} , with $a_{t+1} = c_t + w_{t+1}$. This problem belongs to the class of *linear quadratic Gaussian* (LQG) problems, which will be studied in this note. Note the importance of the linear term $-p_t u_t$ in the problem's criterion, which allows modeling the cost of the fossil energy source, and the importance of the constant term c_t in the dynamics, which allows modeling inputs into the dam with non-zero mean.

1.1 Problem statement

We now formulate the general optimization problem that interests us in this paper. Over a finite discrete optimization span $\mathbb{T} = \{0, \dots, \mathcal{T}-1\}$, we consider I linear quadratic Gaussian subproblems indexed by $i \in \mathbb{I} = \{1, \dots, I\}$, and minimize their sum:

$$\min_{x,u} \sum_{i \in \mathbb{I}} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} \begin{pmatrix} x_{i,t}^\top & u_{i,t}^\top \end{pmatrix} \begin{pmatrix} M_{i,t} & S_{i,t} \\ S_{i,t}^\top & N_{i,t} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ u_{i,t} \end{pmatrix} + m_{i,t}^\top x_{i,t} + n_{i,t}^\top u_{i,t} + d_{i,t} \right) \right. \\ \left. + \frac{1}{2} x_{i,\mathcal{T}}^\top M_{i,\mathcal{T}} x_{i,\mathcal{T}} + m_{i,\mathcal{T}}^\top x_{i,\mathcal{T}} + d_{i,\mathcal{T}} \right], \quad (1a)$$

subject to the dynamic constraints of all subproblems

$$x_{i,0} \text{ donné}, \quad x_{i,t+1} = A_{i,t} x_{i,t} + B_{i,t} u_t + C_{i,t} w_{t+1} + c_{i,t}, \quad t \in \mathbb{T}, \quad i \in \mathbb{I}, \quad (1b)$$

where w_{t+1} is a centered Gaussian vector noise with covariance Σ_{t+1} , independent t by t , and under the constraints coupling the I subproblems together

$$\theta_t - \sum_{i \in \mathbb{I}} \Theta_{i,t} u_{i,t} = 0, \quad t \in \mathbb{T}. \quad (1c)$$

¹we can also introduce quadratic terms on the state x_t

Denoting $\lambda = (\lambda_t)_{t \in \mathbb{T}}$ the multipliers associated with the coupling constraints (1c), we want to tackle Problem (1) by the Uzawa algorithm and thus decompose the problem in the spatial dimension $\mathbb{I}$, the solution of each subproblem at $\lambda = \lambda^{(k)}$ fixed being obtained by solving a Riccati equation. We also want to be able to handle the case where the optimization span $\mathbb{T}$ consists of D consecutive temporal blocks and perform a decomposition in the temporal dimension according to these blocks.

1.2 Recall of the classical Riccati equation

We first consider the classical LQG problem

$$\min_{x,u} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} + \frac{1}{2} x_{\mathcal{T}}^\top M_{\mathcal{T}} x_{\mathcal{T}} \right], \quad (2a)$$

subject to the dynamic constraints

$$x_0 \text{ given}, \quad x_{t+1} = A_t x_t + B_t u_t + C_t w_{t+1}, \quad t \in \mathbb{T}, \quad (2b)$$

the w_{t+1} being centered Gaussian noises with covariance Σ_{t+1} , *independent*. We assume that the matrices $\begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix}$ and $M_{\mathcal{T}}$ are symmetric positive semi-definite, and that the matrices N_t are positive definite (see [8] for more details on LQ and LQG problems).

Proposition 1. *Under the assumptions of symmetry and positive semi-definiteness of the matrices M_t , symmetry and positivity of the matrices N_t , normality and independence of the noises w_{t+1} , the optimal solution of Problem (2) is obtained by solving the associated Bellman equation. The Bellman functions take the form*

$$V_t(x_t) = \frac{1}{2} x_t^\top P_t x_t + r_t, \quad t \in \mathbb{T} \cup \{\mathcal{T}\}, \quad (3)$$

where P_t is a symmetric positive semi-definite matrix and r_t is a scalar. With the notations given by (6), the matrices P_t and the scalars r_t are calculated by the recursions (8). The optimal control of Problem (2) is given by (7).

Proof. We show by induction that the Bellman functions associated with the resolution of Problem (2) take the form (3). The proposition holds true at $t = \mathcal{T}$, with $P_{\mathcal{T}} = M_{\mathcal{T}}$ and $r_{\mathcal{T}} = 0$. Assuming it is true at $t + 1$, the optimization problem giving the value $V_t(x_t)$ of the Bellman function at time step t is

$$\begin{aligned} \min_{u_t} \mathbb{E} \left[\frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} \right. \\ \left. + \frac{1}{2} [A_t x_t + B_t u_t + C_t w_{t+1}]^\top P_{t+1} [A_t x_t + B_t u_t + C_t w_{t+1}] + r_{t+1} \right], \quad (4) \end{aligned}$$

and its optimality condition is

$$(N_t + B_t^\top P_{t+1} B_t) u_t^* + (S_t^\top + B_t^\top P_{t+1} A_t) x_t = 0 . \quad (5)$$

Introducing the three matrices:

$$E_{t+1} = S_t^\top + B_t^\top P_{t+1} A_t , \quad (6a)$$

$$F_{t+1} = N_t + B_t^\top P_{t+1} B_t , \quad (6b)$$

$$G_{t+1} = M_t + A_t^\top P_{t+1} A_t , \quad (6c)$$

the expression of the optimal control u_t^* as a function of the state x_t is

$$u_t^* = -F_{t+1}^{-1} E_{t+1} x_t , \quad (7)$$

and substituting this expression into (4) provides the value of the Bellman function at t :²

$$\begin{aligned} V_t(x_t) &= \frac{1}{2} x_t^\top G_{t+1} x_t + \frac{1}{2} u_t^{*\top} F_{t+1} u_t^* + x_t^\top E_{t+1}^\top u_t^* + \frac{1}{2} \mathbb{E}[w_{t+1}^\top C_t^\top P_{t+1} C_t w_{t+1}] + r_{t+1} , \\ &= \frac{1}{2} x_t^\top (G_{t+1} - E_{t+1}^\top F_{t+1}^{-1} E_{t+1}) x_t + r_{t+1} + \frac{1}{2} \text{trace}(\Sigma_{t+1} C_t^\top P_{t+1} C_t) , \end{aligned}$$

where we used the fact that $\mathbb{E}[w_{t+1}^\top C_t^\top P_{t+1} C_t w_{t+1}] = \text{trace}(\Sigma_{t+1} C_t^\top P_{t+1} C_t)$.

We deduce the Riccati equations defining the Bellman functions:

$$P_{\mathcal{T}} = M_{\mathcal{T}} , \quad (8a)$$

$$P_t = G_{t+1} - E_{t+1}^\top F_{t+1}^{-1} E_{t+1} , \quad t \in \mathbb{T} , \quad (8b)$$

and

$$r_{\mathcal{T}} = 0 , \quad (8c)$$

$$r_t = r_{t+1} + \frac{1}{2} \text{trace}(\Sigma_{t+1} C_t^\top P_{t+1} C_t) , \quad t \in \mathbb{T} . \quad (8d)$$

From equations (8), we deduce that the matrix P_t is symmetric. Furthermore, the function V_t defined by (4) is the marginal function of a function that is jointly convex in (x_t, u_t) , and is therefore itself convex in x_t , which implies that the matrix P_t is positive semidefinite $\square$

Obtaining these formulas uses the fact that there is no linear term in the criterion, no constant term in the dynamics, and that the noise has zero mean. Therefore, these formulas are not suitable for Problem (1). Moreover, in the context of the decompositions that we develop in this paper, dualizing the coupling constraint or dualizing the dynamic constraint linking two temporal blocks introduces linear terms in the control or the state in the criterion of the subproblems. A more general Riccati equation is needed, which we establish now.

²Recall that, by assumption, $\mathbb{E}[w_{t+1}] = 0$ and $\mathbb{E}[w_{t+1} w_{t+1}^\top] = \Sigma_{t+1}$.

1.3 General Riccati equation

We then consider the LQG problem with linear and constant terms

$$\min_{x,u} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} + m_t^\top x_t + n_t^\top u_t + d_t \right) \right. \\ \left. + \frac{1}{2} x_{\mathcal{T}}^\top M_{\mathcal{T}} x_{\mathcal{T}} + m_{\mathcal{T}}^\top x_{\mathcal{T}} + d_{\mathcal{T}} \right], \quad (9a)$$

subject to the dynamic constraints

$$x_0 \text{ given}, \quad x_{t+1} = A_t x_t + B_t u_t + C_t w_{t+1} + c_t, \quad t \in \mathbb{T}, \quad (9b)$$

that is, a problem of the same type as (1), but without the coupling constraints (1c). The solution of this problem is given by the following proposition.

Proposition 2. *Under the assumptions of symmetry and positive semi-definiteness of the matrices M_t , symmetry and positivity of the matrices N_t , normality and independence of the noises w_{t+1} , the optimal solution of Problem (9) is obtained by solving the associated Bellman equation, the Bellman functions being of the form*

$$V_t(x_t) = \frac{1}{2} x_t^\top P_t x_t + p_t^\top x_t + r_t, \quad t \in \mathbb{T} \cup \{\mathcal{T}\}, \quad (10)$$

where P_t is a symmetric positive semi-definite matrix, p_t a vector and r_t a scalar. With the notations defined by Equations (13), the matrices P_t , the vectors p_t and the scalars r_t are computed by the recursion equations (15). The optimal control is given by Equation (14).

Proof. We show by induction that the Bellman functions associated with Problem (9) are of the form (10). The proposition is true at $t = \mathcal{T}$, with $P_{\mathcal{T}} = M_{\mathcal{T}}$, $p_{\mathcal{T}} = m_{\mathcal{T}}$ and $r_{\mathcal{T}} = d_{\mathcal{T}}$. Assuming it true at $t + 1$, the calculation of the value $V_t(x_t)$ by solving

$$\min_{u_t} \mathbb{E} \left[\frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} \right. \\ \left. + \frac{1}{2} [A_t x_t + B_t u_t + C_t w_{t+1} + c_t]^\top P_{t+1} [A_t x_t + B_t u_t + C_t w_{t+1} + c_t] \right. \\ \left. + m_t^\top x_t + n_t^\top u_t + d_t + p_{t+1}^\top [A_t x_t + B_t u_t + C_t w_{t+1} + c_t] + r_{t+1} \right], \quad (11)$$

has as optimality condition:

$$(N_t + B_t^\top P_{t+1} B_t) u_t^* + (S_t^\top + B_t^\top P_{t+1} A_t) x_t + (n_t + B_t^\top p_{t+1} + B_t^\top P_{t+1} c_t) = 0. \quad (12)$$

Introducing the three matrices:

$$E_{t+1} = S_t^\top + B_t^\top P_{t+1} A_t, \quad (13a)$$

$$F_{t+1} = N_t + B_t^\top P_{t+1} B_t, \quad (13b)$$

$$G_{t+1} = M_t + A_t^\top P_{t+1} A_t, \quad (13c)$$

and the three vectors

$$f_{t+1} = n_t + B_t^\top p_{t+1} + B_t^\top P_{t+1} c_t, \quad (13d)$$

$$g_{t+1} = m_t + A_t^\top p_{t+1} + A_t^\top P_{t+1} c_t, \quad (13e)$$

$$h_{t+1} = \frac{1}{2} \text{trace}(\Sigma_{t+1} C_t^\top P_{t+1} C_t) + \frac{1}{2} c_t^\top P_{t+1} c_t + p_{t+1}^\top c_t + d_t + r_{t+1}, \quad (13f)$$

the optimal control is given by:

$$u_t^* = -F_{t+1}^{-1} E_{t+1} x_t - F_{t+1}^{-1} f_{t+1}, \quad (14)$$

and substituting this value into (11) provides the value of the Bellman function at t :

$$\begin{aligned} V_t(x_t) &= \frac{1}{2} x_t^\top G_{t+1} x_t + \frac{1}{2} u_t^{*\top} F_{t+1} u_t^* + x_t^\top E_{t+1}^\top u_t^* + g_{t+1}^\top x_t + f_{t+1}^\top u_t^* + h_{t+1}, \\ &= \frac{1}{2} x_t^\top (G_{t+1} - E_{t+1}^\top F_{t+1}^{-1} E_{t+1}) x_t + (g_{t+1} - E_{t+1}^\top F_{t+1}^{-1} f_{t+1})^\top x_t + h_{t+1} - \frac{1}{2} f_{t+1}^\top F_{t+1}^{-1} f_{t+1}. \end{aligned}$$

We deduce the Riccati equations:

$$P_{\mathcal{T}} = M_{\mathcal{T}}, \quad (15a)$$

$$P_t = G_{t+1} - E_{t+1}^\top F_{t+1}^{-1} E_{t+1}, \quad t \in \mathbb{T}, \quad (15b)$$

$$p_{\mathcal{T}} = m_{\mathcal{T}}, \quad (15c)$$

$$p_t = g_{t+1} - E_{t+1}^\top F_{t+1}^{-1} f_{t+1}, \quad t \in \mathbb{T}, \quad (15d)$$

$$r_{\mathcal{T}} = d_{\mathcal{T}}, \quad (15e)$$

$$r_t = h_{t+1} - \frac{1}{2} f_{t+1}^\top F_{t+1}^{-1} f_{t+1}, \quad t \in \mathbb{T}, \quad (15f)$$

and this recursion can be represented in a functional manner

$$(P_t, p_t, r_t) = \text{RiccatiG}(P_{t+1}, p_{t+1}, r_{t+1}), \quad t \in \mathbb{T}.$$

From Equations (15), we deduce that the matrix P_t is symmetric. Moreover, the function V_t defined by (11) is the marginal function of a jointly convex function in (x_t, u_t) , and is therefore itself convex in x_t , which implies that the matrix P_t is positive semidefinite. $\square$

We thus have a general Riccati equation that takes into account linear terms in x_t and u_t in the cost, and a constant term in the dynamics. This equation will be used in Sections 3, 4 and 6 within the framework of the spatial decomposition methods and that of mixed spatial and temporal decomposition by block.

1.4 General constrained Riccati equation

Finally, we consider the LQG problem with linear and constant terms similar to (1),

$$\min_{x,u} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} + m_t^\top x_t + n_t^\top u_t + d_t \right) \right. \\ \left. + \frac{1}{2} x_{\mathcal{T}}^\top M_{\mathcal{T}} x_{\mathcal{T}} + m_{\mathcal{T}}^\top x_{\mathcal{T}} + d_{\mathcal{T}} \right], \quad (16a)$$

under the dynamic constraints

$$x_0 \text{ given}, \quad x_{t+1} = A_t x_t + B_t u_t + C_t w_{t+1} + c_t, \quad t \in \mathbb{T}, \quad (16b)$$

and under the linear constraints on the state and the control as in §1.2.

$$\theta_t - \Phi_t x_t - \Theta_t u_t = 0, \quad t \in \mathbb{T}. \quad (16c)$$

Proposition 3. *Under the assumptions of symmetry and positive semi-definiteness of the matrices M_t , symmetry and positivity of the matrices N_t , normality and independence of the noises w_{t+1} , assuming that the matrices Θ_t are of full rank and that the matrices Φ_t are zero, the optimal solution of Problem (16) is obtained by solving the associated Bellman equation, with the Bellman functions of the form*

$$V_t(x_t) = \frac{1}{2} x_t^\top P_t x_t + p_t^\top x_t + r_t, \quad t \in \mathbb{T} \cup \{\mathcal{T}\}, \quad (17)$$

where P_t is a symmetric positive semi-definite matrix, p_t is a vector, and r_t is a scalar. With the notations defined by (19) and (22), the matrices P_t , the vectors p_t , and the scalars r_t are calculated by the recursions (24). The optimal control is given by (23).

Proof. We show by induction that the associated Bellman functions are of the form (17). The proposition is true at $t = \mathcal{T}$, with $P_{\mathcal{T}} = M_{\mathcal{T}}$, $p_{\mathcal{T}} = m_{\mathcal{T}}$, and $r_{\mathcal{T}} = d_{\mathcal{T}}$. Assuming it is true at $t + 1$, the calculation of the value $V_t(x_t)$ by solving

$$\min_{u_t} \mathbb{E} \left[\frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} \right. \\ \left. + \frac{1}{2} [A_t x_t + B_t u_t + C_t w_{t+1} + c_t]^\top P_{t+1} [A_t x_t + B_t u_t + C_t w_{t+1} + c_t] \right. \\ \left. + m_t^\top x_t + n_t^\top u_t + d_t + p_{t+1}^\top [A_t x_t + B_t u_t + C_t w_{t+1} + c_t] + r_{t+1} \right], \quad (18a)$$

subject to the constraint

$$\theta_t - \Phi_t x_t - \Theta_t u_t = 0, \quad (18b)$$

has the following optimality conditions:

$$\begin{aligned} (N_t + B_t^\top P_{t+1} B_t) u^* + (S_t^\top + B_t^\top P_{t+1} A_t) x + (n_t + B_t^\top p_{t+1} + B_t^\top P_{t+1} c_t) - \Theta_t^\top \lambda_t^* &= 0, \\ \Theta_t u_t^* + \Phi_t x_t - \theta_t &= 0, \end{aligned}$$

where λ_t^* is the multiplier associated with the constraint (18b). Introducing the matrices

$$E_{t+1} = S_t^\top + B_t^\top P_{t+1} A_t, \quad (19a)$$

$$F_{t+1} = N_t + B_t^\top P_{t+1} B_t, \quad (19b)$$

$$G_{t+1} = M_t + A_t^\top P_{t+1} A_t, \quad (19c)$$

and the vectors

$$f_{t+1} = n_t + B_t^\top p_{t+1} + B_t^\top P_{t+1} c_t, \quad (19d)$$

$$g_{t+1} = m_t + A_t^\top p_{t+1} + A_t^\top P_{t+1} c_t, \quad (19e)$$

$$h_{t+1} = \frac{1}{2} \text{trace}(\Sigma_{t+1} C_t^\top P_{t+1} C_t) + \frac{1}{2} c_t^\top P_{t+1} c_t + p_{t+1}^\top c_t + d_t + r_{t+1}, \quad (19f)$$

the first optimality condition can be written as:

$$u_t^* = -F_{t+1}^{-1} E_{t+1} x_t - F_{t+1}^{-1} f_{t+1} + F_{t+1}^{-1} \Theta_t^\top \lambda_t^*.$$

Substituting this value into the second optimality condition, we get

$$(\Theta_t F_{t+1}^{-1} \Theta_t^\top) \lambda_t^* = (\Theta_t F_{t+1}^{-1} E_{t+1} - \Phi_t) x_t + \Theta_t F_{t+1}^{-1} f_{t+1} + \theta_t.$$

Since the matrix Θ_t is full rank, with the notations

$$L_{t+1} = (\Theta_t F_{t+1}^{-1} \Theta_t^\top)^{-1} (\Theta_t F_{t+1}^{-1} E_{t+1} - \Phi_t), \quad (20a)$$

$$\ell_{t+1} = (\Theta_t F_{t+1}^{-1} \Theta_t^\top)^{-1} (\Theta_t F_{t+1}^{-1} f_{t+1} + \theta_t), \quad (20b)$$

the expression for the optimal multiplier λ_t^* is

$$\lambda_t^* = L_{t+1} x_t + \ell_{t+1}, \quad (21)$$

which induces the expression for the optimal control u_t^* :

$$u_t^* = F_{t+1}^{-1} (\Theta_t^\top L_{t+1} - E_{t+1}) x_t + F_{t+1}^{-1} (\Theta_t^\top \ell_{t+1} - f_{t+1}).$$

Since the optimal control u_t^* is affine in x_t , we are in the same situation as in §1.3, and the Bellman function V_t is therefore quadratic, characterized by a matrix P_t , a vector p_t and a scalar r_t . With the notations

$$K_{t+1} = \Theta_t^\top L_{t+1} - E_{t+1}, \quad (22a)$$

$$k_{t+1} = \Theta_t^\top \ell_{t+1} - f_{t+1}, \quad (22b)$$

the optimal control is written as:

$$u_t^* = F_{t+1}^{-1} K_{t+1} x_t + F_{t+1}^{-1} k_{t+1} , \quad (23)$$

and substituting this value into (18a) yields the value of the Bellman function at t :

$$V_t(x_t) = \frac{1}{2} x_t^\top G_{t+1} x_t + \frac{1}{2} u_t^{*\top} F_{t+1} u_t^* + x_t^\top E_{t+1}^\top u_t^* + g_{t+1}^\top x_t + f_{t+1}^\top u_t^* + h_{t+1} ,$$

and thus

$$\begin{aligned} V_t(x_t) = & \frac{1}{2} x_t^\top (G_{t+1} + 2E_{t+1}^\top F_{t+1}^{-1} K_{t+1} + K_{t+1}^\top F_{t+1}^{-1} K_{t+1}) x_t \\ & + (g_{t+1} + K_{t+1}^\top F_{t+1}^{-1} f_{t+1} + K_{t+1}^\top F_{t+1}^{-1} k_{t+1} + E_{t+1}^\top F_{t+1}^{-1} k_{t+1})^\top x_t \\ & + h_{t+1} + \frac{1}{2} k_{t+1}^\top F_{t+1}^{-1} k_{t+1} + f_{t+1}^\top F_{t+1}^{-1} k_{t+1} . \end{aligned}$$

We deduce the Riccati equations:

$$P_{\mathcal{T}} = M_{\mathcal{T}} , \quad (24a)$$

$$P_t = G_{t+1} + 2E_{t+1}^\top F_{t+1}^{-1} K_{t+1} + K_{t+1}^\top F_{t+1}^{-1} K_{t+1} , \quad t \in \mathbb{T} , \quad (24b)$$

$$p_{\mathcal{T}} = m_{\mathcal{T}} , \quad (24c)$$

$$p_t = g_{t+1} + K_{t+1}^\top F_{t+1}^{-1} f_{t+1} + K_{t+1}^\top F_{t+1}^{-1} k_{t+1} + E_{t+1}^\top F_{t+1}^{-1} k_{t+1} , \quad t \in \mathbb{T} , \quad (24d)$$

$$r_{\mathcal{T}} = d_{\mathcal{T}} , \quad (24e)$$

$$r_t = h_{t+1} + \frac{1}{2} k_{t+1}^\top F_{t+1}^{-1} k_{t+1} + f_{t+1}^\top F_{t+1}^{-1} k_{t+1} , \quad t \in \mathbb{T} . \quad (24f)$$

This recursion can be represented in a functional manner

$$(P_t, p_t, r_t) = \text{RiccatiCG}(P_{t+1}, p_{t+1}, r_{t+1}) , \quad t \in \mathbb{T} .$$

The only term that poses a problem for the symmetry of the matrix P_t in (24b) is $E_{t+1}^\top F_{t+1}^{-1} K_{t+1}$. By the definition of K_{t+1} in (22), the problematic term is $E_{t+1}^\top F_{t+1}^{-1} \Theta_t^\top L_{t+1}$. Following the expression of L_{t+1} in (20), this last term cannot be symmetric in the presence of an arbitrary matrix Φ_t , hence the necessity of the assumption $\Phi_t = 0$. The remaining term is then $E_{t+1}^\top F_{t+1}^{-1} \Theta_t^\top (\Theta_t F_{t+1}^{-1} \Theta_t^\top)^{-1} \Theta_t F_{t+1}^{-1} E_{t+1}$ which is symmetric, hence the symmetry of P_t . Moreover, the Bellman function V_t defined by (18) is the marginal function of a jointly convex function in (x, u) , hence itself convex in x , which implies that the matrix P_t is positive semi-definite $\square$

We now have a Riccati equation that takes into account linear terms in x_t and u_t in the cost, with constant terms in the cost and dynamics, as well as an additional linear constraint on u_t . This equation will be used in Section 2 to solve the global problem and in Section 5 to compute the intraday functions in the temporal block decomposition. Moreover, the expression of the optimal control (23) will be used during simulations to evaluate the quality of the results of the different decompositions.

2 Solving the global problem

We return to Problem (1), namely,

$$\min_{x,u} \sum_{i \in \mathbb{I}} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} \begin{pmatrix} x_{i,t}^\top & u_{i,t}^\top \end{pmatrix} \begin{pmatrix} M_{i,t} & S_{i,t} \\ S_{i,t}^\top & N_{i,t} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ u_{i,t} \end{pmatrix} + m_{i,t}^\top x_{i,t} + n_{i,t}^\top u_{i,t} + d_{i,t} \right) \right. \\ \left. + \frac{1}{2} x_{i,\mathcal{T}}^\top M_{i,\mathcal{T}} x_{i,\mathcal{T}} + m_{i,\mathcal{T}}^\top x_{i,\mathcal{T}} + d_{i,\mathcal{T}} \right],$$

subject to the constraints

$$x_{i,0} \text{ given}, \quad x_{i,t+1} = A_{i,t}x_{i,t} + B_{i,t}u_{i,t} + C_{i,t}w_{t+1} + c_{i,t}, \quad t \in \mathbb{T}, \quad i \in \mathbb{I}, \\ \theta_t - \sum_{i \in \mathbb{I}} \Theta_{i,t} u_{i,t} = 0, \quad t \in \mathbb{T}.$$

This problem can be written in the form (16) with $x_t^\top = [x_{1,t}^\top \cdots x_{I,t}^\top]$, $u_t^\top = [u_{1,t}^\top \cdots u_{I,t}^\top]$, and

$$M_t = \begin{bmatrix} M_{1,t} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & M_{I,t} \end{bmatrix}, \quad S_t = \begin{bmatrix} S_{1,t} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & S_{I,t} \end{bmatrix}, \quad N_t = \begin{bmatrix} N_{1,t} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & N_{I,t} \end{bmatrix}, \\ A_t = \begin{bmatrix} A_{1,t} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_{I,t} \end{bmatrix}, \quad B_t = \begin{bmatrix} B_{1,t} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & B_{I,t} \end{bmatrix}, \quad C_t = \begin{bmatrix} C_{1,t} \\ \vdots \\ C_{I,t} \end{bmatrix}, \\ m_t = \begin{bmatrix} m_{1,t} \\ \vdots \\ m_{I,t} \end{bmatrix}, \quad n_t = \begin{bmatrix} n_{1,t} \\ \vdots \\ n_{I,t} \end{bmatrix}, \quad c_t = \begin{bmatrix} c_{1,t} \\ \vdots \\ c_{I,t} \end{bmatrix}, \\ d_t = \sum_{i=1}^I d_{i,t}, \quad \Phi_t = [0 \cdots 0], \quad \Theta_t = [\Theta_{1,t} \cdots \Theta_{I,t}].$$

There is therefore no particular difficulty in computing the solution of Problem (1) by solving the general constrained Riccati equation presented in §1.4, under the assumption that the matrices $\Theta_{i,t}$ are surjective (Θ has full rank) and with the remark that the coupling constraint does not contain linear terms in the states $x_{i,t}$ ($\Phi_t = 0$).

3 Spatial decomposition with deterministic multipliers

Consider Problem (1) which we treat by dualizing the coupling constraint (1c). The associated dual problem is

$$\begin{aligned} \max_{\lambda} \min_{x,u} \sum_{i \in \mathbb{I}} \left(\mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} \begin{pmatrix} x_{i,t}^\top & u_{i,t}^\top \end{pmatrix} \begin{pmatrix} M_{i,t} & S_{i,t} \\ S_{i,t}^\top & N_{i,t} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ u_{i,t} \end{pmatrix} \right. \right. \\ \left. \left. + m_{i,t}^\top x_{i,t} + n_{i,t}^\top u_{i,t} + d_{i,t} - \lambda_t^\top \Theta_{i,t} u_{i,t} \right) + \sum_{t \in \mathbb{T}} \lambda_t^\top \theta_t \right. \\ \left. \left. + \frac{1}{2} x_{i,\mathcal{T}}^\top M_{i,\mathcal{T}} x_{i,\mathcal{T}} + m_{i,\mathcal{T}}^\top x_{i,\mathcal{T}} + d_{i,\mathcal{T}} \right] \right), \quad (25a) \end{aligned}$$

subject to the dynamics constraints

$$x_{i,0} \text{ given}, \quad x_{i,t+1} = A_{i,t}x_{i,t} + B_{i,t}u_{i,t} + C_{i,t}w_{t+1} + c_{i,t}, \quad t \in \mathbb{T}, \quad i \in \mathbb{I}. \quad (25b)$$

and it is solved by a Uzawa-type algorithm to update the multiplier λ . As in the most usual application of spatial decomposition, we begin by limiting ourselves to *deterministic* multipliers λ_t . We then use the methodology developed in [5].

3.1 Formulation of subproblems

At iteration k of the decomposition algorithm with deterministic prices, knowing the value $\lambda^{(k)}$ of the multiplier, we solve for each subproblem $i \in \mathbb{I}$ the Linear Quadratic Gaussian subproblem

$$\begin{aligned} \min_{x_i, u_i} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} \begin{pmatrix} x_{i,t}^\top & u_{i,t}^\top \end{pmatrix} \begin{pmatrix} M_{i,t} & S_{i,t} \\ S_{i,t}^\top & N_{i,t} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ u_{i,t} \end{pmatrix} + m_{i,t}^\top x_{i,t} + n_{i,t}^\top u_{i,t} + d_{i,t} - \lambda_t^{(k)\top} \Theta_{i,t} u_{i,t} \right) \right. \\ \left. + \frac{1}{2} x_{i,\mathcal{T}}^\top M_{i,\mathcal{T}} x_{i,\mathcal{T}} + m_{i,\mathcal{T}}^\top x_{i,\mathcal{T}} + d_{i,\mathcal{T}} \right], \quad (26a) \end{aligned}$$

subject to the dynamic constraints

$$x_{i,0} \text{ given}, \quad x_{i,t+1} = A_{i,t}x_{i,t} + B_{i,t}u_{i,t} + C_{i,t}w_{t+1} + c_{i,t}, \quad t \in \mathbb{T}. \quad (26b)$$

We can easily obtain the solution of Subproblem (26) by solving the general Riccati equation developed in §1.3. It should be noted that the presence of the *duality linear terms* in u in the criterion means that the classic Riccati equation is never applicable to this subproblem.

3.2 Uzawa algorithm

The Uzawa algorithm applies without particular difficulty to the resolution of the dual problem (25). At iteration k of the algorithm, knowing the values $\lambda_t^{(k)}$ of the multipliers,

1. we have to solve the subproblems (26) for all $i \in \mathbb{I}$, the optimal controls of subproblem i being denoted $u_{i,t}^{(k)}$ for $t \in \mathbb{T}$,
2. we then update the multipliers by a gradient-type formula:

$$\lambda_t^{(k+1)} = \lambda_t^{(k)} + \rho \mathbb{E} \left[\theta_t - \sum_{i \in \mathbb{I}} \Theta_{i,t} u_{i,t}^{(k)} \right], \quad t \in \mathbb{T}. \quad (27)$$

It should be noted that the calculation of the expectation in the multiplier update step is done subproblem by subproblem, the term $\mathbb{E}[\Theta_{i,t} u_{i,t}^{(k)}]$ being evaluated during the resolution of subproblem i (see §3.3).

This algorithm converges for well-chosen values of the step ρ , and provides:

- a lower bound of the optimal cost of Problem (1) by summing the optimal costs of each of the subproblems and the duality terms $\lambda_t^{(k)\top} \theta_t$,
- Bellman functions $V_{i,t}$ for each subproblem i at each time step t , each function $V_{i,t}$ being characterized, as seen in §1.3, by a matrix $P_{i,t}$, a vector $p_{i,t}$ and a constant $r_{i,t}$:

$$V_{i,t}(x_{i,t}) = \frac{1}{2} x_{i,t}^\top P_{i,t} x_{i,t} + p_{i,t}^\top x_{i,t} + r_{i,t}.$$

Note that the Bellman functions $V_{i,t}$ depend *implicitly* on the multipliers $(\lambda_0, \dots, \lambda_{\mathcal{T}-1})$.

3.3 Gradient calculations

Updating the multipliers λ_t by the gradient formula (27) requires each subproblem i to compute the terms $\mathbb{E}[\Theta_{i,t} u_{i,t}^{(k)}]$. The standard way to perform this expectation calculation is the Monte Carlo method, which consists of simulating the optimal control $u_{i,t}^{(k)}$ along a set of noise scenarios and averaging the terms $\Theta_{i,t} u_{i,t}^{(k)}$ obtained over the scenarios at each time step t to obtain their empirical mean. This method is generally costly in CPU time because computing gradients requires good precision for implementing ascent direction methods, and thus it is necessary to work with a large set of scenarios.

Another way is to apply the methodology developed in [6] which allows, for an optimal control problem depending on a parameter, to compute in the Bellman recursion both the value functions and the partial gradients of these value functions with respect to the parameter. We recall the principle of such a computation in the next paragraph.

3.3.1 Bellman mechanism for computing gradients

We recall the methodology developed in [6], and we use here standard notations for multi-stage stochastic optimization, that is,

- a discrete span $\mathbb{T} = \{0, \dots, \mathcal{T}-1\}$,
- a parameter $\lambda = (\lambda_0, \dots, \lambda_{\mathcal{T}})$ indexed by time,
- instantaneous cost functions L_t , $t \in \mathbb{T}$, depending on λ_t ,
- a final cost K depending on $\lambda_{\mathcal{T}}$,
- dynamics functions f_t , $t \in \mathbb{T}$, *not depending* on the parameter.

The optimal value of the problem is

$$\Phi(\lambda) = \min_{x,u} \mathbb{E} \left[\sum_{t \in \mathbb{T}} L_t(x_t, \lambda_t, u_t, \mathbf{w}_{t+1}) + K(x_{\mathcal{T}}, \lambda_{\mathcal{T}}) \right], \quad (28a)$$

subject to the dynamic constraint

$$x_0 \text{ given}, \quad x_{t+1} = f_t(x_t, u_t, \mathbf{w}_{t+1}), \quad t \in \mathbb{T}. \quad (28b)$$

The Bellman functions V_t associated to Problem (28) are parameterized by λ and are thus denoted by $V_t[\lambda]$. The recursion allowing their computation is

$$V_{\mathcal{T}}[\lambda](x_{\mathcal{T}}) = K(x_{\mathcal{T}}, \lambda_{\mathcal{T}}), \quad (29a)$$

$$V_t[\lambda](x_t) = \min_{u_t} \mathbb{E} \left[L_t(x_t, \lambda_t, u_t, \mathbf{w}_{t+1}) + V_{t+1}[\lambda](f_t(x_t, u_t, \mathbf{w}_{t+1})) \right]. \quad (29b)$$

Noting u_t^* the optimal controls obtained by solving the dynamic programming equations (29), and under technical assumptions detailed in [6], the partial gradients of the Bellman functions with respect to the parameter λ are obtained by the recursion

$$\nabla_{\lambda} V_{\mathcal{T}}[\lambda](x_{\mathcal{T}}) = \nabla_{\lambda} K(x_{\mathcal{T}}, \lambda_{\mathcal{T}}), \quad (30a)$$

$$\nabla_{\lambda} V_t[\lambda](x_t) = \mathbb{E} \left[\nabla_{\lambda} L_t(x_t, \lambda_t, u_t^*, \mathbf{w}_{t+1}) + \nabla_{\lambda} V_{t+1}[\lambda](f_t(x_t, u_t^*, \mathbf{w}_{t+1})) \right]. \quad (30b)$$

It should be noted that many components of these gradients are zero. Indeed, the only non-zero component of $\nabla_{\lambda} V_{\mathcal{T}}[\lambda]$ in (30a) is the one corresponding to the component $\lambda_{\mathcal{T}}$ of the parameter λ . In (30b), the only non-zero component of $\nabla_{\lambda} L_t$ is the one corresponding to λ_t , and the non-zero components of $\nabla_{\lambda} V_{t+1}[\lambda]$ correspond to the components of the sub-vector $(\lambda_{t+1}, \dots, \lambda_{\mathcal{T}})$ since the function $V_{t+1}[\lambda]$ does not depend on the multipliers $\lambda_0, \dots, \lambda_t$. More precisely,

- the partial gradients at time step $\mathcal{T}$ are

$$\nabla_{\lambda_{\tau}} V_{\mathcal{T}}[\lambda](x_{\mathcal{T}}) = 0, \quad \tau \in \{0, \dots, \mathcal{T}-1\}, \quad (31a)$$

$$\nabla_{\lambda_{\mathcal{T}}} V_{\mathcal{T}}[\lambda](x_{\mathcal{T}}) = \nabla_{\lambda_{\mathcal{T}}} K(x_{\mathcal{T}}, \lambda_{\mathcal{T}}), \quad (31b)$$

- the gradients at time step $t \in \{0, \dots, \mathcal{T}-1\}$ are

$$\nabla_{\lambda_\tau} V_t[\lambda](x_t) = 0, \quad \tau \in \{0, \dots, t-1\}, \quad (31c)$$

$$\nabla_{\lambda_t} V_t[\lambda](x_t) = \mathbb{E} \left[\nabla_{\lambda_t} L_t(x_t, \lambda_t, u_t^*, \mathbf{w}_{t+1}) \right], \quad (31d)$$

$$\nabla_{\lambda_\tau} V_t[\lambda](x_t) = \mathbb{E} \left[\nabla_{\lambda_\tau} V_{t+1}[\lambda](f_t(x_t, u_t^*, \mathbf{w}_{t+1})) \right], \quad \tau \in \{t+1, \dots, \mathcal{T}-1\}, \quad (31e)$$

Under the assumption of independent noise, the optimal value $\Phi(\lambda)$ of Problem (28) is given by the Bellman function $V_0[\lambda]$ evaluated at the initial state x_0 , and the gradients of Φ are

$$\nabla_{\lambda_0} \Phi(\lambda) = \mathbb{E} \left[\nabla_{\lambda_0} L_0(x_0, \lambda_0, u_0^*, \mathbf{w}_1) \right], \quad (31f)$$

$$\nabla_{\lambda_t} \Phi(\lambda) = \mathbb{E} \left[\nabla_{\lambda_t} V_1[\lambda](f_0(x_0, u_0^*, \mathbf{w}_1)) \right], \quad t \in \{1, \dots, \mathcal{T}-1\}. \quad (31g)$$

3.3.2 Application to spatial decomposition

Returning to Subproblem (26), we denote λ the sequence of multipliers $(\lambda_0, \dots, \lambda_{\mathcal{T}-1})$, and we consider λ as the parameter of the problem. We are in the particular situation where

- the final cost does not depend on the parameter λ (no multiplier $\lambda_{\mathcal{T}}$),
- the instantaneous cost depends on the parameter λ only through the term $-\lambda_t^\top \Theta_{i,t} u_{i,t}$,
- the dynamics are linear, of the form $A_{i,t} x_{i,t} + B_{i,t} u_{i,t} + C_{i,t} w_{t+1} + c_{i,t}$,
- the noises $w_{i,t+1}$ have zero mean.

Moreover, the optimal control is given by (14) and is written as

$$u_{i,t}^* = \Gamma_{i,t} x_{i,t} + \gamma_{i,t}, \quad (32)$$

with $\Gamma_{i,t} = -F_{i,t+1}^{-1} E_{i,t+1}$ and $\gamma_{i,t} = -F_{i,t+1}^{-1} f_{i,t+1}$ ($F_{i,t+1}$, $E_{i,t+1}$ and $f_{i,t+1}$ being defined in §1.3). The partial gradients of the Bellman functions are given by the next proposition.

Proposition 4. *The partial gradients with respect to λ of the Bellman functions $V_{i,t}[\lambda]$ of Subproblem (26) are, for $t = 0, \dots, \mathcal{T}$, of the form*

$$\nabla_{\lambda_\tau} V_{i,t}[\lambda](x_{i,t}) = 0, \quad \tau = 0, \dots, t-1, \quad (33a)$$

$$\nabla_{\lambda_\tau} V_{i,t}[\lambda](x_{i,t}) = \alpha_{i,\tau,t} x_{i,t} + \beta_{i,\tau,t}, \quad \tau = t, \dots, \mathcal{T}-1. \quad (33b)$$

where the matrices $\alpha_{i,\tau,t}$ and the vectors $\beta_{i,\tau,t}$ are obtained by the recursions (36).

Proof. The proof is by induction. At the last time step $\mathcal{T}$, the final cost does not depend on the parameter, so that

$$\nabla_{\lambda_\tau} V_{i,\mathcal{T}}[\lambda](x_{i,\mathcal{T}}) = 0, \quad \tau = 0, \dots, \mathcal{T}-1, \quad (34)$$

and thus of the form (33) with $\alpha_{i,\tau,\mathcal{T}} = 0$ and $\beta_{i,\tau,\mathcal{T}} = 0$. Suppose that at time step $t+1$, the partial gradients $\nabla_{\lambda_\tau} V_{i,t+1}[\lambda]$ are of the form (33).

- By construction, the Bellman function $V_{i,t}[\lambda]$ does not depend on the multipliers λ_τ with $\tau < t$. We thus have

$$\nabla_{\lambda_\tau} V_{i,t}[\lambda](x_{i,t}) = 0, \quad \tau = 0, \dots, t-1. \quad (35a)$$

- The Bellman function $V_{i,t}[\lambda]$ depends on λ_t only through the instantaneous cost $L_{i,t}$ and its term $\lambda_t^\top \Theta_{i,t} u_{i,t}$, we deduce from (31d) and (32) that

$$\begin{aligned} \nabla_{\lambda_t} V_{i,t}[\lambda](x_{i,t}) &= -\Theta_{i,t} u_{i,t}^*, \\ &= -\Theta_{i,t} \Gamma_{i,t} x_{i,t} - \Theta_{i,t} \gamma_{i,t}. \end{aligned} \quad (35b)$$

- For $\tau = t+1, \dots, \mathcal{T}-1$, the Bellman function $V_{i,t}[\lambda]$ depends on λ_τ only through the function $V_{i,t+1}[\lambda]$. From (31e), we deduce that

$$\begin{aligned} \nabla_{\lambda_\tau} V_{i,t}[\lambda](x_{i,t}) &= \mathbb{E} \left[\nabla_{\lambda_\tau} V_{i,t+1}[\lambda] (A_{i,t} x_{i,t} + B_{i,t} u_{i,t}^* + C_{i,t} w_{t+1} + c_{i,t}) \right], \\ &= \alpha_{i,\tau,t+1} (A_{i,t} + B_{i,t} \Gamma_{i,t}) x_{i,t} \\ &\quad + (\alpha_{i,\tau,t+1} (B_{i,t} \gamma_{i,t} + c_{i,t}) + \beta_{i,\tau,t+1}). \end{aligned} \quad (35c)$$

We deduce the recurrence relations

$$\alpha_{i,\tau,t} = 0, \quad \tau = 0, \dots, t-1, \quad (36a)$$

$$\alpha_{i,t,t} = -\Theta_{i,t} \Gamma_{i,t}, \quad (36b)$$

$$\alpha_{i,\tau,t} = \alpha_{i,\tau,t+1} (A_{i,t} + B_{i,t} \Gamma_{i,t}), \quad \tau = t+1, \dots, \mathcal{T}-1, \quad (36c)$$

$$\beta_{i,\tau,t} = 0, \quad \tau = 0, \dots, t-1, \quad (36d)$$

$$\beta_{i,t,t} = -\Theta_{i,t} \gamma_{i,t}, \quad (36e)$$

$$\beta_{i,\tau,t} = \alpha_{i,\tau,t+1} (B_{i,t} \gamma_{i,t} + c_{i,t}) + \beta_{i,\tau,t+1}, \quad \tau = t+1, \dots, \mathcal{T}-1, \quad (36f)$$

which completes the proof. $\square$

Let $\Phi_i(\lambda^{(k)})$ be the optimal value of Subproblem (26). After computing the matrices $\alpha_{i,\tau,t}$ and the vectors $\beta_{i,\tau,t}$ using the relations (36), we obtain

$$\nabla_{\lambda_t} \Phi_i(\lambda^{(k)}) = \alpha_{i,t,0} x_{i,0} + \beta_{i,t,0}, \quad t = 0, \dots, \mathcal{T}-1. \quad (37)$$

We are then able to update the multipliers λ_t using relation (27) since we have

$$\nabla_{\lambda_t} \Phi_i(\lambda^{(k)}) = -\mathbb{E}[\Theta_{i,t} u_{i,t}^{(k)}].$$

3.4 Simulation of an admissible control

At the end of the Uzawa algorithm, we have at disposal a value λ^* of the multiplier as well as Bellman functions $V_{i,t}[\lambda^*]$, at each time step t and for each subproblem i . By summing the $V_{i,t}[\lambda^*]$, we can generate surrogate functions $\tilde{V}_t$ for the true Bellman functions V_t of Problem (1), namely,

$$\tilde{V}_t(x_{1,t}, \dots, x_{I,t}) = \sum_{i \in \mathbb{I}} \frac{1}{2} x_{i,t}^\top P_{i,t} x_{i,t} + p_{i,t}^\top x_{i,t} + r_{i,t} = \frac{1}{2} [x_{1,t}^\top \dots x_{I,t}^\top] \tilde{P}_t \begin{bmatrix} x_{1,t} \\ \vdots \\ x_{I,t} \end{bmatrix} + \tilde{p}_t^\top \begin{bmatrix} x_{1,t} \\ \vdots \\ x_{I,t} \end{bmatrix} + \tilde{r}_t ,$$

with

$$\tilde{P}_t = \begin{bmatrix} P_{1,t} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & P_{I,t} \end{bmatrix} , \quad \tilde{p}_t = \begin{bmatrix} p_{1,t} \\ \vdots \\ p_{I,t} \end{bmatrix} , \quad \tilde{r}_t = \sum_{i=1}^I r_{i,t} .$$

We then replace the true Bellman functions V_t given by the triplet (P_t, p_t, r_t) by the surrogate Bellman functions $\tilde{V}_t$ given by the triplet $(\tilde{P}_t, \tilde{p}_t, \tilde{r}_t)$.

Using the results obtained in §1.4, one can then calculate admissible controls, that is, controls that satisfy all the constraints of the global problem (1). Indeed, at time step t , starting from state x_t , the admissible control to be applied is calculated by solving an optimization problem of type (18), namely

$$\begin{aligned} \min_{u_t} \mathbb{E} & \left(\frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} + m_t^\top x_t + n_t^\top u_t + d_t \right. \\ & + \frac{1}{2} [A_t x_t + B_t u_t + C_t w_{t+1} + c_t]^\top \tilde{P}_{t+1} [A_t x_t + B_t u_t + C_t w_{t+1} + c_t] \\ & \left. + \tilde{p}_{t+1}^\top [A_t x_t + B_t u_t + C_t w_{t+1} + c_t] + \tilde{r}_{t+1} \right) , \end{aligned}$$

subject to the constraint

$$\theta_t - \Theta_t u_t = 0 , \quad t \in \mathbb{T} .$$

As seen in §1.4, the admissible control to be applied at time step t is of the form (see Equation (23))

$$u_t^* = F_{t+1}^{-1} K_{t+1} x_t + F_{t+1}^{-1} k_{t+1} , \quad (38)$$

where the matrices K_{t+1} and the vectors k_{t+1} are given by Equation (22).

Simulating this admissible control along a set of noise trajectories obtained by Monte Carlo sampling provides a statistical upper bound on the optimal cost of Problem (1).

Remark 1. *Plugging the controls (38) in Problem (1) would allow the upper bound of the optimal cost to be accurately assessed. This has not been done yet but see Appendix A.*

4 Spatial decomposition with AR(1) multipliers

Consider again Problem (1) treated by dualizing the coupling constraint (1c), that is,

$$\begin{aligned} \max_{\lambda} \min_{x,u} \sum_{i \in \mathbb{I}} \left(\mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} \begin{pmatrix} x_{i,t}^\top & u_{i,t}^\top \end{pmatrix} \begin{pmatrix} M_{i,t} & S_{i,t} \\ S_{i,t}^\top & N_{i,t} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ u_{i,t} \end{pmatrix} \right. \right. \\ \left. \left. + m_{i,t}^\top x_{i,t} + n_{i,t}^\top u_{i,t} + d_{i,t} - \lambda_t^\top \Theta_{i,t} u_{i,t} \right) + \sum_{t \in \mathbb{T}} \lambda_t^\top \theta_t \right. \\ \left. \left. + \frac{1}{2} x_{i,\mathcal{T}}^\top M_{i,\mathcal{T}} x_{i,\mathcal{T}} + m_{i,\mathcal{T}}^\top x_{i,\mathcal{T}} + d_{i,\mathcal{T}} \right] \right), \quad (39a) \end{aligned}$$

subject to the dynamics constraints

$$x_{i,0} \text{ given}, \quad x_{i,t+1} = A_{i,t} x_{i,t} + B_{i,t} u_{i,t} + C_{i,t} w_{t+1} + c_{i,t}, \quad t \in \mathbb{T}, \quad i \in \mathbb{I}. \quad (39b)$$

In Section 3, we have detailed how to solve such a problem by an Uzawa-type algorithm, in the case that we restrict ourselves to *deterministic* multipliers λ .

It can be observed from §1.4 that, using Equation (21), it is possible to define an *extended* dynamics for the pair (x, λ) , namely, with x_0 given, $\lambda_0 = L_1 x_0 + \ell_1$, and for $t \in \mathbb{T}$,

$$\begin{pmatrix} x_{t+1} \\ \lambda_{t+1} \end{pmatrix} = \begin{pmatrix} A_t & 0 \\ L_{t+2} A_t & 0 \end{pmatrix} \begin{pmatrix} x_t \\ \lambda_t \end{pmatrix} + \begin{pmatrix} B_t \\ L_{t+2} B_t \end{pmatrix} u_t + \begin{pmatrix} C_t \\ L_{t+2} C_t \end{pmatrix} w_{t+1} + \begin{pmatrix} c_t \\ L_{t+2} c_t + \ell_{t+2} \end{pmatrix}. \quad (40)$$

This extended dynamics allows to solve Problem (1) without taking into account the coupling constraint, but it is not useful in a spatial decomposition approach because the multiplier λ_{t+1} depends on the whole vector x_t , and thus couples all the subproblems arising in decomposition. A way to allow decomposition is to try to obtain an *autonomous* dynamics for λ , for example an autoregressive AR(1) model:

$$\lambda_0 \text{ given}, \quad \lambda_{t+1} = \Xi_t \lambda_t + \Omega_t w_{t+1} + \xi_t. \quad (41)$$

Such an approach has been explored in [9], where it has been shown that an exact AR(1) dynamics exists for the multiplier for a very specific LQG problem, thus allowing for spatial decomposition with an extended state incorporating the multiplier in each subproblem. Based on that, another idea is to approximate the multiplier using an AR(1) dynamics as in (41) (rather than a deterministic multiplier as done in Section 3), and to identify the best possible parameters λ_0 and $(\Xi_t, \xi_t, \Omega_t)_{t \in \mathbb{T}}$ by maximization. It is of course expected that using AR(1) multipliers rather than deterministic multipliers should improve the lower bound obtained by decomposition, at the price of an higher complexity when solving subproblems. Using AR(1) multipliers has been investigated in [3] for a power management problem and in [2] for a problem involving a cascade of dams.

Throughout Section 4, we make the following assumption.

Assumption 1. *For all $t \in \mathbb{T}$, the matrices Ξ_t are diagonal, and we denote by Ξ_t^v the vector formed by the diagonal elements of Ξ_t , that is, $\Xi_t = \text{diag}(\Xi_t^v)$.*

4.1 Formulation of subproblems

At iteration k of the decomposition algorithm, the values $\lambda_0^{(k)}$ and $(\Xi_t^{(k)}, \Omega_t^{(k)}, \xi_t^{(k)})_{t \in \mathbb{T}}$ of the parameters in the AR(1) model are given. For each subproblem $i \in \mathbb{I}$, we have to solve the linear quadratic Gaussian subproblem

$$\min_{x_i, u_i} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} (x_{i,t}^\top, \lambda_t^\top, u_{i,t}^\top) \begin{pmatrix} M_{i,t} & 0 & S_{i,t} \\ 0 & 0 & -\Theta_{i,t} \\ S_{i,t}^\top & -\Theta_{i,t}^\top & N_{i,t} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ \lambda_t \\ u_{i,t} \end{pmatrix} + m_{i,t}^\top x_{i,t} + n_{i,t}^\top u_{i,t} + d_{i,t} \right) \right. \\ \left. + \frac{1}{2} x_{i,\mathcal{T}}^\top M_{i,\mathcal{T}} x_{i,\mathcal{T}} + m_{i,\mathcal{T}}^\top x_{i,\mathcal{T}} + d_{i,\mathcal{T}} \right], \quad (42a)$$

subject to the dynamic constraints, with $x_{i,0}$ given and $\lambda_0 = \lambda_0^{(k)}$,

$$\begin{pmatrix} x_{i,t+1} \\ \lambda_{t+1} \end{pmatrix} = \begin{pmatrix} A_{i,t} & 0 \\ 0 & \Xi_t^{(k)} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ \lambda_t \end{pmatrix} + \begin{pmatrix} B_{i,t} \\ 0 \end{pmatrix} u_{i,t} + \begin{pmatrix} C_{i,t} \\ \Omega_t^{(k)} \end{pmatrix} w_{t+1} + \begin{pmatrix} c_{i,t} \\ \xi_t^{(k)} \end{pmatrix}, \quad t \in \mathbb{T}. \quad (42b)$$

Note that we have added a fictitious multiplier $\lambda_{\mathcal{T}} = 0$, that is, $\Xi_{\mathcal{T}-1}^{(k)} = 0$, $\xi_{\mathcal{T}-1}^{(k)} = 0$ and $\Omega_{\mathcal{T}-1}^{(k)} = 0$ for all k . Subproblem (42) can be solved by using the general Riccati equation developed in §1.3.

We have also to deal with the sum

$$\mathbb{E} \left[\sum_{t \in \mathbb{T}} \lambda_t^\top \theta_t \right], \quad (43a)$$

subject to the dynamics

$$\lambda_0 = \lambda_0^{(k)}, \quad \lambda_{t+1} = \Xi_t^{(k)} \lambda_t + \Omega_t^{(k)} w_{t+1} + \xi_t^{(k)}, \quad t \in \mathbb{T}. \quad (43b)$$

The computation of this sum as well as the computation of its gradients with respect to the parameters $\lambda_0^{(k)}$ and $(\Xi_t^{(k)}, \Omega_t^{(k)}, \xi_t^{(k)})_{t \in \mathbb{T}}$ is detailed in §4.3.2.

4.2 Uzawa algorithm

The Uzawa algorithm can be applied to the resolution of the global problem. At iteration k of the algorithm, knowing the values $\lambda_0^{(k)}$ and $(\Xi_t^{(k)}, \Omega_t^{(k)}, \xi_t^{(k)})_{t \in \mathbb{T}}$ we have to

1. solve Subproblem (42) for each $i \in \mathbb{I}$, the optimal controls of subproblem i being denoted $u_{i,t}^{(k)}$, and solve Subproblem (43),
2. update the parameters λ_0 and $(\Xi_t, \Omega_t, \xi_t)_{t \in \mathbb{T}}$ by gradient-type formulas.

This algorithm, if converging, provides:

- a lower bound of the optimal cost of Problem (1) by summing the optimal costs of all subproblems and all the terms $\lambda_t^\top \theta_t$,

- Bellman functions $\mathfrak{V}_{i,t}$ for each subproblem i at each time step t , each function $\mathfrak{V}_{i,t}$ being characterized, as seen in §1.3, by a matrix $\mathfrak{P}_{i,t}$, a vector $\mathfrak{p}_{i,t}$ and a constant $\mathfrak{r}_{i,t}$:

$$\mathfrak{V}_{i,t}(x_{i,t}, \lambda_t) = \frac{1}{2} \begin{pmatrix} x_{i,t}^\top & \lambda_t^\top \end{pmatrix} \begin{pmatrix} \mathfrak{P}_{i,t}^x & \mathfrak{P}_{i,t}^c \\ \mathfrak{P}_{i,t}^{c^\top} & \mathfrak{P}_{i,t}^\lambda \end{pmatrix} \begin{pmatrix} x_{i,t} \\ \lambda_t \end{pmatrix} + \begin{pmatrix} \mathfrak{p}_{i,t}^x & \mathfrak{p}_{i,t}^\lambda \end{pmatrix} \begin{pmatrix} x_{i,t} \\ \lambda_t \end{pmatrix} + \mathfrak{r}_{i,t}, \quad (44a)$$

so that its partial gradient of $\mathfrak{V}_{i,t}$ with respect to λ is

$$\nabla_{\lambda_t} \mathfrak{V}_{i,t}(x_{i,t}, \lambda_t) = \mathfrak{P}_{i,t}^{c^\top} x_{i,t} + \mathfrak{P}_{i,t}^\lambda \lambda_t + \mathfrak{p}_{i,t}^\lambda. \quad (44b)$$

We recall that, at time $\mathcal{T}$, the Bellman function $\mathfrak{V}_{i,\mathcal{T}}$ is of the form (44a), with

$$\mathfrak{P}_{i,\mathcal{T}} = \begin{pmatrix} M_{i,\mathcal{T}} & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathfrak{p}_{i,\mathcal{T}} = \begin{pmatrix} m_{i,\mathcal{T}} \\ 0 \end{pmatrix}, \quad \mathfrak{r}_{i,\mathcal{T}} = d_{i,\mathcal{T}},$$

and does not depend on λ .

From §1.3 and Equation (14), we know that the optimal control is of the form

$$u_{i,t} = \begin{pmatrix} \Gamma_{i,t}^x & \Gamma_{i,t}^\lambda \end{pmatrix} \begin{pmatrix} x_{i,t} \\ \lambda_t \end{pmatrix} + \gamma_{i,t}. \quad (45)$$

Remark 2. *The Bellman functions $\mathfrak{V}_{i,t}$ depends on the parameters λ_0 and $(\Xi_t, \Omega_t, \xi_t)_{t \in \mathbb{T}}$. According to §3.3.1, these functions should be noted $\mathfrak{V}_{i,t}[\lambda_0, \Xi, \Omega, \xi]$. To avoid such notational heaviness, we omit writing the dependency in terms of parameters and simply write them $\mathfrak{V}_{i,t}$. In calculating gradients with respect to a given type of parameter, for example ξ , we will also use the reduced notation $\mathfrak{V}_{i,t}[\xi]$ to emphasize the dependency of $\mathfrak{V}_{i,t}$ on ξ .*

4.3 Gradient calculations

We detail the computation of the partial gradients of the optimal cost of Subproblem (42) and of Subproblem (43) with respect to the parameters λ_0 and $(\Xi_t, \Omega_t, \xi_t)_{t \in \mathbb{T}}$.

We use again the methodology propose in [6], but without having theoretical guarantees because the parameters with respect to which the partial gradients are calculated come into play in the dynamics and not just in the cost functions (see §3.3.1).

4.3.1 Partial gradients of the optimal cost in Subproblem (42)

We recall the Bellman equation allowing to compute the optimal cost of Subproblem i :

$$\begin{aligned} \mathfrak{V}_{i,t}(x_{i,t}, \lambda_t) = \min_{u_{i,t}} \mathbb{E} \left[\frac{1}{2} \begin{pmatrix} x_{i,t} \\ \lambda_t \\ u_{i,t} \end{pmatrix}^\top \begin{pmatrix} M_{i,t} & 0 & S_{i,t} \\ 0 & 0 & -\Theta_{i,t} \\ S_{i,t}^\top & -\Theta_{i,t}^\top & N_{i,t} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ \lambda_t \\ u_{i,t} \end{pmatrix} + \begin{pmatrix} m_{i,t} \\ 0 \\ n_{i,t} \end{pmatrix}^\top \begin{pmatrix} x_{i,t} \\ \lambda_t \\ u_{i,t} \end{pmatrix} + d_{i,t} \right. \\ \left. + \mathfrak{V}_{i,t+1}(A_{i,t}x_{i,t} + B_{i,t}u_{i,t} + C_{i,t}w_{t+1} + c_{i,t}, \Xi_t\lambda_t + \Omega_t w_{t+1} + \xi_t) \right]. \quad (46) \end{aligned}$$

Partial gradient with respect to the initial condition λ_0 . From the Bellman equation (46), we observe that $\mathfrak{V}_{i,t}$ does not depend on the parameter λ_0 for $t \geq 1$. The partial gradient of $\mathfrak{V}_{i,0}$ with respect to λ_0 is given directly by (44b) for $t = 0$, namely,

$$\nabla_{\lambda_0} \mathfrak{V}_{i,0}(x_{i,0}, \lambda_0) = \mathfrak{P}_{i,0}^{c^\top} x_{i,0} + \mathfrak{P}_{i,0}^\lambda \lambda_0 + \mathfrak{p}_{i,0}^\lambda. \quad (47)$$

Partial gradient with respect to the parameters ξ_t . In order to highlight the dependence of the Bellman function $\mathfrak{V}_{i,t}$ on ξ , we denote it by $\mathfrak{V}_{i,t}[\xi]$ in this paragraph (see Remark 2).

Proposition 5. *The partial gradients with respect to ξ of the Bellman functions $\mathfrak{V}_{i,t}[\xi]$ of Subproblem (42) are, for $t = 0, \dots, \mathcal{T}$, of the form*

$$\nabla_{\xi_\tau} \mathfrak{V}_{i,t}[\xi](x_{i,t}, \lambda_t) = 0, \quad \tau = 0, \dots, t-1, \quad (48a)$$

$$\nabla_{\xi_\tau} \mathfrak{V}_{i,t}[\xi](x_{i,t}, \lambda_t) = \mathfrak{Q}_{i,\tau,t}^x x_{i,t} + \mathfrak{Q}_{i,\tau,t}^\lambda \lambda_t + \mathfrak{q}_{i,\tau,t}, \quad \tau = t, \dots, \mathcal{T}-1, \quad (48b)$$

where the matrices $\mathfrak{Q}_{i,\tau,t}^x$, $\mathfrak{Q}_{i,\tau,t}^\lambda$ and the vectors $\mathfrak{q}_{i,\tau,t}$ are obtained by the recursions (49).

Proof. The proof is by induction. The proposition holds true at $t = \mathcal{T}$ since the final cost and hence the final Bellman function $\mathfrak{V}_{i,\mathcal{T}}[\xi]$ does not depend on any of the parameters ξ_τ , $\tau \in \mathbb{T}$. Assume that it holds true at $t+1$.

- For $\tau = 0, \dots, t-1$, from the form (46) of the Bellman functions, we have that $\mathfrak{V}_{i,t}[\xi]$ does not depend on the parameter ξ_τ .
- For $\tau = t$, by differentiating (46) with respect to ξ_t , we obtain

$$\nabla_{\xi_t} \mathfrak{V}_{i,t}[\xi](x_{i,t}, \lambda_t) = \mathbb{E}[\nabla_{\xi_t} \lambda_{t+1} \nabla_{\lambda_{t+1}} \mathfrak{V}_{i,t+1}[\xi](x_{i,t+1}, \lambda_{t+1})],$$

since $\mathfrak{V}_{i,t+1}[\xi]$ does not depend on ξ_t

$$= \mathbb{E}[\mathfrak{P}_{i,t+1}^{c^\top} x_{i,t+1} + \mathfrak{P}_{i,t+1}^\lambda \lambda_{t+1} + \mathfrak{p}_{i,t+1}^\lambda],$$

so that using the fact that the noise has a zero mean, hence the terms involving it in the expectation above are zero, replacing $x_{i,t+1}$ and λ_{t+1} using the dynamics (42b) and replacing the optimal control $u_{i,t}$ using (45) leads to

$$\begin{aligned} \nabla_{\xi_t} \mathfrak{V}_{i,t}[\xi](x_{i,t}, \lambda_t) &= \mathfrak{P}_{i,t+1}^x (A_{i,t} x_{i,t} + B_{i,t} u_{i,t} + c_{i,t}) + \mathfrak{P}_{i,t+1}^\lambda (\Xi_t \lambda_t + \xi_t) + \mathfrak{p}_{i,t+1}, \\ &= \mathfrak{P}_{i,t+1}^{c^\top} (A_{i,t} + B_{i,t} \Gamma_{i,t}^x) x_{i,t} + (\mathfrak{P}_{i,t+1}^{c^\top} B_{i,t} \Gamma_{i,t}^\lambda + \mathfrak{P}_{i,t+1}^\lambda \Xi_t) \lambda_t \\ &\quad + \mathfrak{P}_{i,t+1}^{c^\top} B_{i,t} \gamma_{i,t} + \mathfrak{P}_{i,t+1}^{c^\top} c_{i,t} + \mathfrak{P}_{i,t+1}^\lambda \xi_t + \mathfrak{p}_{i,t+1}. \end{aligned}$$

- For $\tau = t+1, \dots, \mathcal{T}-1$, from (46) and thanks to the induction assumption, we have

$$\begin{aligned} \nabla_{\xi_\tau} \mathfrak{V}_{i,t}[\xi](x_{i,t}, \lambda_t) &= \mathbb{E}[\nabla_{\xi_\tau} \mathfrak{V}_{i,t+1}[\xi](x_{i,t+1}, \lambda_{t+1})], \\ &= \mathfrak{Q}_{i,\tau,t+1}^x (A_{i,t} x_{i,t} + B_{i,t} u_{i,t} + c_{i,t}) + \mathfrak{Q}_{i,\tau,t+1}^\lambda (\Xi_t \lambda_t + \xi_t) + \mathfrak{q}_{i,\tau,t+1}, \\ &= \mathfrak{Q}_{i,\tau,t+1}^x (A_{i,t} + B_{i,t} \Gamma_{i,t}^x) x_{i,t} + (\mathfrak{Q}_{i,\tau,t+1}^x B_{i,t} \Gamma_{i,t}^\lambda + \mathfrak{Q}_{i,\tau,t+1}^\lambda \Xi_t) \lambda_t \\ &\quad + \mathfrak{Q}_{i,\tau,t+1}^x B_{i,t} \gamma_{i,t} + \mathfrak{Q}_{i,\tau,t+1}^x c_{i,t} + \mathfrak{Q}_{i,\tau,t+1}^\lambda \xi_t + \mathfrak{q}_{i,\tau,t+1}. \end{aligned}$$

The proposition thus holds true at t , with, for $\tau = 0, \dots, t-1$,

$$\mathfrak{Q}_{i,\tau,t}^x = 0, \quad \mathfrak{Q}_{i,\tau,t}^\lambda = 0, \quad \mathfrak{q}_{i,\tau,t} = 0, \quad (49a)$$

for $\tau = t$,

$$\mathfrak{Q}_{i,t,t}^x = \mathfrak{P}_{i,t+1}^{c^\top} (A_{i,t} + B_{i,t} \Gamma_{i,t}^x), \quad (49b)$$

$$\mathfrak{Q}_{i,t,t}^\lambda = (\mathfrak{P}_{i,t+1}^{c^\top} B_{i,t} \Gamma_{i,t}^\lambda + \mathfrak{P}_{i,t+1}^\lambda \Xi_t), \quad (49c)$$

$$\mathfrak{q}_{i,t,t} = \mathfrak{P}_{i,t+1}^{c^\top} B_{i,t} \gamma_{i,t} + \mathfrak{P}_{i,t+1}^{c^\top} c_{i,t} + \mathfrak{P}_{i,t+1}^\lambda \xi_t + \mathfrak{p}_{i,t+1}^\lambda, \quad (49d)$$

and, for $\tau = t+1, \dots, T-1$,

$$\mathfrak{Q}_{i,\tau,t}^x = \mathfrak{Q}_{i,\tau,t+1}^x (A_{i,t} + B_{i,t} \Gamma_{i,t}^x), \quad (49e)$$

$$\mathfrak{Q}_{i,\tau,t}^\lambda = (\mathfrak{Q}_{i,\tau,t+1}^x B_{i,t} \Gamma_{i,t}^\lambda + \mathfrak{Q}_{i,\tau,t+1}^\lambda \Xi_t), \quad (49f)$$

$$\mathfrak{q}_{i,\tau,t} = \mathfrak{Q}_{i,\tau,t+1}^x B_{i,t} \gamma_{i,t} + \mathfrak{Q}_{i,\tau,t+1}^x c_{i,t} + \mathfrak{Q}_{i,\tau,t+1}^\lambda \xi_t + \mathfrak{q}_{i,\tau,t+1}. \quad (49g)$$

The proof is complete. $\square$

Partial gradient with respect to the parameters Ξ_t . By Assumption 1, the matrices Ξ_t are diagonal. In the sequel, the gradients with respect to Ξ_t should be understood as the gradients with respect to the vector Ξ_t^v (diagonal elements of Ξ_t). In order to highlight the dependence of the Bellman function $\mathfrak{V}_{i,t}$ on Ξ , we denote it by $\mathfrak{V}_{i,t}[\Xi]$.

Proposition 6. *The partial gradients with respect to Ξ of the Bellman functions $\mathfrak{V}_{i,t}[\Xi]$ of Subproblem (42) are, for $t = 0, \dots, \mathcal{T}$, of the form*

$$\nabla_{\Xi_t^v} \mathfrak{V}_{i,t}[\Xi](x_{i,t}, \lambda_t) = 0, \quad \tau = 0, \dots, t-1, \quad (50a)$$

$$\begin{aligned} \nabla_{\Xi_t^v} \mathfrak{V}_{i,t}[\Xi](x_{i,t}, \lambda_t) = & \text{diag}(\lambda_t) (\mathfrak{R}_{i,\tau,t}^x x_{i,t} + \mathfrak{R}_{i,\tau,t}^\lambda \lambda_t + \mathfrak{r}_{i,\tau,t}) \\ & + (\mathfrak{S}_{i,\tau,t}^x x_{i,t} + \mathfrak{S}_{i,\tau,t}^\lambda \lambda_t + \mathfrak{s}_{i,\tau,t}), \quad \tau = t, \dots, \mathcal{T}-1, \end{aligned} \quad (50b)$$

where the $\mathfrak{R}_{i,\tau,t}^x$, $\mathfrak{R}_{i,\tau,t}^\lambda$, $\mathfrak{r}_{i,\tau,t}$, $\mathfrak{S}_{i,\tau,t}^x$, $\mathfrak{S}_{i,\tau,t}^\lambda$ and $\mathfrak{s}_{i,\tau,t}$ are obtained by the recursions (52).

Proof. The proof is by induction. The proposition holds true at $t = \mathcal{T}$ since $\mathfrak{V}_{i,\mathcal{T}}[\Xi]$ does not depend on any of the parameters Ξ_τ , $\tau \in \mathbb{T}$. Assume that it holds true at $t+1$.

- For $\tau = 0, \dots, t-1$, from the form (46) of the Bellman functions, we have that $\mathfrak{V}_{i,t}[\Xi]$ does not depend on the parameter Ξ_τ^v , so that the proposition holds true at t for $\tau < t$.
- For $\tau = t$, by differentiating (46) with respect to Ξ_t^v and since $\mathfrak{V}_{i,t+1}[\Xi]$ does not depend on Ξ_t , we obtain

$$\begin{aligned} \nabla_{\Xi_t^v} \mathfrak{V}_{i,t}[\Xi](x_{i,t}, \lambda_t) &= \mathbb{E} [\nabla_{\Xi_t^v} \lambda_{t+1} \nabla_{\lambda_{t+1}} \mathfrak{V}_{i,t+1}[\Xi](x_{i,t+1}, \lambda_{t+1})], \\ &= \text{diag}(\lambda_t) \mathbb{E} [\mathfrak{P}_{i,t+1}^{c^\top} x_{i,t+1} + \mathfrak{P}_{i,t+1}^\lambda \lambda_{t+1} + \mathfrak{p}_{i,t+1}^\lambda], \end{aligned}$$

since $\nabla_{\Xi_t^v} \lambda_{t+1} = \text{diag}(\lambda_t)$. The computation of the expectation has been performed when studying the partial gradient with respect to the parameter ξ_t for the case $\tau = t$, and so

$$\nabla_{\Xi_t^v} \mathfrak{V}_{i,t}[\Xi](x_{i,t}, \lambda_t) = \text{diag}(\lambda_t)(\mathfrak{Q}_{i,t,t}^x x_{i,t} + \mathfrak{Q}_{i,t,t}^\lambda \lambda_t + \mathfrak{q}_{i,t,t}) ,$$

where $\mathfrak{Q}_{i,t,t}^x$, $\mathfrak{Q}_{i,t,t}^\lambda$ and $\mathfrak{q}_{i,t,t}$ are given by (49). The proposition holds true at t for $\tau = t$.

- For $\tau = t+1, \dots, T-1$, from (46) and thanks to the induction assumption, we have

$$\begin{aligned} \nabla_{\Xi_\tau^v} \mathfrak{V}_{i,t}[\Xi](x_{i,t}, \lambda_t) &= \mathbb{E}[\nabla_{\Xi_\tau^v} \mathfrak{V}_{i,t+1}[\Xi](x_{i,t+1}, \lambda_{t+1})] , \\ &= \mathbb{E}[\text{diag}(\lambda_{t+1})(\mathfrak{R}_{i,\tau,t+1}^x x_{i,t+1} + \mathfrak{R}_{i,\tau,t+1}^\lambda \lambda_{t+1} + \mathfrak{r}_{i,\tau,t+1}) \\ &\quad + (\mathfrak{S}_{i,\tau,t+1}^x x_{i,t+1} + \mathfrak{S}_{i,\tau,t+1}^\lambda \lambda_{t+1} + \mathfrak{s}_{i,\tau,t+1})] . \end{aligned}$$

From the dynamics (42b) of x_i and λ and from the expression (45) of the control, we have that the second part of the right-hand side in the equation above is

$$\begin{aligned} &\mathbb{E}[\mathfrak{S}_{i,\tau,t+1}^x x_{i,t+1} + \mathfrak{S}_{i,\tau,t+1}^\lambda \lambda_{t+1} + \mathfrak{s}_{i,\tau,t+1}] \\ &= \mathfrak{S}_{i,\tau,t+1}^x (A_{i,t} + B_{i,t} \Gamma_{i,t}^x) x_{i,t} + (\mathfrak{S}_{i,\tau,t+1}^x B_{i,t} \Gamma_{i,t}^\lambda + \mathfrak{S}_{i,\tau,t+1}^\lambda \Xi_t) \lambda_t \\ &\quad + (\mathfrak{S}_{i,\tau,t+1}^x B_{i,t} \gamma_{i,t} + \mathfrak{S}_{i,\tau,t+1}^x c_{i,t} + \mathfrak{S}_{i,\tau,t+1}^\lambda \xi_t + \mathfrak{s}_{i,\tau,t+1}) , \\ &= \mathfrak{B}_{i,\tau,t+1}^x x_{i,t} + \mathfrak{B}_{i,\tau,t+1}^\lambda \lambda_t + \mathfrak{b}_{i,\tau,t+1} , \end{aligned}$$

with

$$\mathfrak{B}_{i,\tau,t+1}^x = \mathfrak{S}_{i,\tau,t+1}^x (A_{i,t} + B_{i,t} \Gamma_{i,t}^x) , \quad (51a)$$

$$\mathfrak{B}_{i,\tau,t+1}^\lambda = \mathfrak{S}_{i,\tau,t+1}^x B_{i,t} \Gamma_{i,t}^\lambda + \mathfrak{S}_{i,\tau,t+1}^\lambda \Xi_t , \quad (51b)$$

$$\mathfrak{b}_{i,\tau,t+1} = \mathfrak{S}_{i,\tau,t+1}^x B_{i,t} \gamma_{i,t} + \mathfrak{S}_{i,\tau,t+1}^x c_{i,t} + \mathfrak{S}_{i,\tau,t+1}^\lambda \xi_t + \mathfrak{s}_{i,\tau,t+1} . \quad (51c)$$

In a similar way, we obtain that

$$\mathbb{E}[\mathfrak{R}_{i,\tau,t+1}^x x_{i,t+1} + \mathfrak{R}_{i,\tau,t+1}^\lambda \lambda_{t+1} + \mathfrak{r}_{i,\tau,t+1}] = \mathfrak{A}_{i,\tau,t+1}^x x_{i,t} + \mathfrak{A}_{i,\tau,t+1}^\lambda \lambda_t + \mathfrak{a}_{i,\tau,t+1} ,$$

with

$$\mathfrak{A}_{i,\tau,t+1}^x = \mathfrak{R}_{i,\tau,t+1}^x (A_{i,t} + B_{i,t} \Gamma_{i,t}^x) , \quad (51d)$$

$$\mathfrak{A}_{i,\tau,t+1}^\lambda = \mathfrak{R}_{i,\tau,t+1}^x B_{i,t} \Gamma_{i,t}^\lambda + \mathfrak{R}_{i,\tau,t+1}^\lambda \Xi_t , \quad (51e)$$

$$\mathfrak{a}_{i,\tau,t+1} = \mathfrak{R}_{i,\tau,t+1}^x B_{i,t} \gamma_{i,t} + \mathfrak{R}_{i,\tau,t+1}^x c_{i,t} + \mathfrak{R}_{i,\tau,t+1}^\lambda \xi_t + \mathfrak{r}_{i,\tau,t+1} . \quad (51f)$$

Since Ξ_t is diagonal, we have

$$\text{diag}(\lambda_{t+1}) = \Xi_t \text{diag}(\lambda_t) + \text{diag}(\xi_t) + \text{diag}(\Omega_t w_{t+1}) .$$

We are then able to compute

$$\begin{aligned} &\mathbb{E}[\Xi_t \text{diag}(\lambda_t)(\mathfrak{R}_{i,\tau,t+1}^x x_{i,t+1} + \mathfrak{R}_{i,\tau,t+1}^\lambda \lambda_{t+1} + \mathfrak{r}_{i,\tau,t+1})] \\ &= \text{diag}(\lambda_t) \Xi_t (\mathfrak{A}_{i,\tau,t+1}^x x_{i,t} + \mathfrak{A}_{i,\tau,t+1}^\lambda \lambda_t + \mathfrak{a}_{i,\tau,t+1}) , \end{aligned}$$

$$\begin{aligned} \mathbb{E}[\text{diag}(\xi_t)(\mathfrak{R}_{i,\tau,t+1}^x x_{i,t+1} + \mathfrak{R}_{i,\tau,t+1}^\lambda \lambda_{t+1} + \mathfrak{r}_{i,\tau,t+1})] \\ = \text{diag}(\xi_t)(\mathfrak{A}_{i,\tau,t+1}^x x_{i,t} + \mathfrak{A}_{i,\tau,t+1}^\lambda \lambda_t + \mathfrak{a}_{i,\tau,t+1}) , \end{aligned}$$

and

$$\mathbb{E}[\text{diag}(\Omega_t w_{t+1})(\mathfrak{R}_{i,\tau,t+1}^x x_{i,t+1} + \mathfrak{R}_{i,\tau,t+1}^\lambda \lambda_{t+1} + \mathfrak{r}_{i,\tau,t+1})] = \mathfrak{c}_{i,\tau,t+1} .$$

Using the dynamics (42b) of x_i and λ , as well as the centered assumption of the noises, we obtain that

$$\mathfrak{c}_{i,\tau,t+1} = \mathbb{E}[\text{diag}(\Omega_t w_{t+1})(\mathfrak{R}_{i,\tau,t+1}^x C_{i,t} + \mathfrak{R}_{i,\tau,t+1}^\lambda \Omega_t) w_{t+1}] . \quad (51g)$$

Gathering these results, we have that

$$\begin{aligned} \nabla_{\Xi_\tau^v} \mathfrak{V}_{i,t}[\Xi](x_{i,t}, \lambda_t) &= \text{diag}(\lambda_t) (\Xi_t \mathfrak{A}_{i,\tau,t+1}^x x_{i,t} + \Xi_t \mathfrak{A}_{i,\tau,t+1}^\lambda \lambda_t + \Xi_t \mathfrak{a}_{i,\tau,t+1}) \\ &\quad + (\text{diag}(\xi_t) \mathfrak{A}_{i,\tau,t+1}^x + \mathfrak{B}_{i,\tau,t+1}^x) x_{i,t} \\ &\quad + (\text{diag}(\xi_t) \mathfrak{A}_{i,\tau,t+1}^\lambda + \mathfrak{B}_{i,\tau,t+1}^\lambda) \lambda_t \\ &\quad + (\text{diag}(\xi_t) \mathfrak{a}_{i,\tau,t+1} + \mathfrak{b}_{i,\tau,t+1} + \mathfrak{c}_{i,\tau,t+1}) , \end{aligned}$$

the variables $\mathfrak{A}_{i,\tau,t+1}^x$, $\mathfrak{A}_{i,\tau,t+1}^\lambda$, $\mathfrak{a}_{i,\tau,t+1}$, $\mathfrak{B}_{i,\tau,t+1}^x$, $\mathfrak{B}_{i,\tau,t+1}^\lambda$, $\mathfrak{b}_{i,\tau,t+1}$ and $\mathfrak{c}_{i,\tau,t+1}$ being given by (51). The proposition holds true at t for $\tau > t$.

The proposition thus holds true at t , with, for $\tau = 0, \dots, t-1$,

$$\mathfrak{R}_{i,\tau,t}^x = 0 , \quad \mathfrak{R}_{i,\tau,t}^\lambda = 0 , \quad \mathfrak{r}_{i,\tau,t} = 0 , \quad \mathfrak{S}_{i,\tau,t}^x = 0 , \quad \mathfrak{S}_{i,\tau,t}^\lambda = 0 , \quad \mathfrak{s}_{i,\tau,t} = 0 , \quad (52a)$$

for $\tau = t$,

$$\mathfrak{R}_{i,t,t}^x = \mathfrak{Q}_{i,t,t}^x , \quad \mathfrak{R}_{i,t,t}^\lambda = \mathfrak{Q}_{i,t,t}^\lambda , \quad \mathfrak{r}_{i,t,t} = \mathfrak{q}_{i,t,t} , \quad (52b)$$

$$\mathfrak{S}_{i,t,t}^x = 0 , \quad \mathfrak{S}_{i,t,t}^\lambda = 0 , \quad \mathfrak{s}_{i,t,t} = 0 , \quad (52c)$$

and, for $\tau = t+1, \dots, \mathcal{T}-1$,

$$\mathfrak{R}_{i,\tau,t}^x = \Xi_t \mathfrak{A}_{i,\tau,t+1}^x , \quad (52d)$$

$$\mathfrak{R}_{i,\tau,t}^\lambda = \Xi_t \mathfrak{A}_{i,\tau,t+1}^\lambda , \quad (52e)$$

$$\mathfrak{r}_{i,\tau,t} = \Xi_t \mathfrak{a}_{i,\tau,t+1} \quad (52f)$$

$$\mathfrak{S}_{i,\tau,t}^x = \text{diag}(\xi_t) \mathfrak{A}_{i,\tau,t+1}^x + \mathfrak{B}_{i,\tau,t+1}^x , \quad (52g)$$

$$\mathfrak{S}_{i,\tau,t}^\lambda = \text{diag}(\xi_t) \mathfrak{A}_{i,\tau,t+1}^\lambda + \mathfrak{B}_{i,\tau,t+1}^\lambda , \quad (52h)$$

$$\mathfrak{s}_{i,\tau,t} = \text{diag}(\xi_t) \mathfrak{a}_{i,\tau,t+1} + \mathfrak{b}_{i,\tau,t+1} + \mathfrak{c}_{i,\tau,t+1} . \quad (52i)$$

The proof is complete. $\square$

Remark 3. The vectors $\mathbf{c}_{i,\tau,t+1}$ are easy to compute when the covariance matrices Σ_{t+1} are diagonal, that is, when the components of the noise w_{t+1} are independent. With this assumption, all the terms involving a product $w_{k,t+1}w_{k',t+1}$ with $k \neq k'$ in (51g) are equal to 0. Denoting by σ_{t+1} the vector formed by the diagonal elements of Σ_{t+1} , we obtain that

$$\mathbf{c}_{i,\tau,t+1} = (\Omega_t \odot (\mathfrak{R}_{i,\tau,t+1}^x C_{i,t} + \mathfrak{R}_{i,\tau,t+1}^\lambda \Omega_t)) \sigma_{t+1},$$

where $\odot$ denotes the Hadamard product³ of the two matrices involved in (51g). Indeed, denoting by $\varpi_{j,k}$ (resp. $z_{j,k}$) the coefficients of the matrix Ω_t (resp. $\mathfrak{R}_{i,\tau,t+1}^x C_{i,t} + \mathfrak{R}_{i,\tau,t+1}^\lambda \Omega_t$), the j -th component of vector $\mathbf{c}_{i,\tau,t+1}$, that is,

$$\mathbb{E} \left[\left(\sum_k \varpi_{j,k} w_{k,t+1} \right) \left(\sum_{k'} z_{j,k'} w_{k',t+1} \right) \right],$$

reduces thanks to the assumption on Σ_{t+1} to

$$\mathbb{E} \left[\sum_k \varpi_{j,k} z_{j,k} w_{k,t+1}^2 \right],$$

hence the result since $\mathbb{E}[w_{k,t+1}^2] = \sigma_{k,t+1}$.

Remark 4. If we assume that the matrices Ξ_t do not depend on time t , that is, $\Xi_t = \Xi_0$ for all $t \in \mathbb{T}$, and with $\Xi_0^v = \text{diag}(\Xi_0)$, we obtain that, for $t = 0, \dots, T-1$,

$$\nabla_{\Xi_0^v} \mathfrak{V}_{i,t}[\Xi](x_{i,t}, \lambda_t) = \text{diag}(\lambda_t) (\hat{\mathfrak{R}}_{i,t}^x x_{i,t} + \hat{\mathfrak{R}}_{i,t}^\lambda \lambda_t + \hat{\mathbf{r}}_{i,t}) + (\hat{\mathfrak{S}}_{i,t}^x x_{i,t} + \hat{\mathfrak{S}}_{i,t}^\lambda \lambda_t + \hat{\mathbf{s}}_{i,t}), \quad (53)$$

the partial gradients of $\mathfrak{V}_{i,T}[\Xi]$ with respect to Ξ being zero by construction. Indeed, assuming that (53) holds true at $t+1$, and since $\mathfrak{V}_{i,t+1}[\Xi]$ now depends on Ξ_0 , we obtain

$$\begin{aligned} \nabla_{\Xi_0^v} \mathfrak{V}_{i,t}[\Xi](x_{i,t}, \lambda_t) &= \mathbb{E} [\nabla_{\Xi_0^v} \lambda_{t+1} \nabla_{\lambda_{t+1}} \mathfrak{V}_{i,t+1}[\Xi](x_{i,t+1}, \lambda_{t+1}) + \nabla_{\Xi_0^v} \mathfrak{V}_{i,t+1}[\Xi](x_{i,t+1}, \lambda_{t+1})], \\ &= \text{diag}(\lambda_t) (\mathfrak{Q}_{i,t,t}^x x_{i,t} + \mathfrak{Q}_{i,t,t}^\lambda \lambda_t + \mathbf{q}_{i,t,t}) + \mathbb{E} [\nabla_{\Xi_0^v} \mathfrak{V}_{i,t+1}[\Xi](x_{i,t+1}, \lambda_{t+1})], \end{aligned}$$

where $\mathfrak{Q}_{i,t,t}^x$, $\mathfrak{Q}_{i,t,t}^\lambda$ and $\mathbf{q}_{i,t,t}$ are given by (49),

$$\begin{aligned} &= \text{diag}(\lambda_t) (\mathfrak{Q}_{i,t,t}^x x_{i,t} + \mathfrak{Q}_{i,t,t}^\lambda \lambda_t + \mathbf{q}_{i,t,t}) \\ &\quad + \mathbb{E} [\text{diag}(\lambda_{t+1}) (\hat{\mathfrak{R}}_{i,t+1}^x x_{i,t+1} + \hat{\mathfrak{R}}_{i,t+1}^\lambda \lambda_{t+1} + \hat{\mathbf{r}}_{i,t+1}) \\ &\quad + (\hat{\mathfrak{S}}_{i,t+1}^x x_{i,t+1} + \hat{\mathfrak{S}}_{i,t+1}^\lambda \lambda_{t+1} + \hat{\mathbf{s}}_{i,t+1})], \end{aligned}$$

by the induction assumption. Using calculations similar to those performed in the non-stationary case, we obtained that

$$\mathbb{E} [\hat{\mathfrak{S}}_{i,t+1}^x x_{i,t+1} + \hat{\mathfrak{S}}_{i,t+1}^\lambda \lambda_{t+1} + \hat{\mathbf{s}}_{i,t+1}] = \hat{\mathfrak{B}}_{i,t+1}^x x_{i,t} + \hat{\mathfrak{B}}_{i,t+1}^\lambda \lambda_t + \hat{\mathbf{b}}_{i,t+1},$$

³The Hadamard product of two vectors x and y of $\mathbb{R}^n$ is the vector $x \odot y = [x_1 y_1 \dots x_n y_n]^\top$. Note that $x \odot y = \text{diag}(x) y$. The Hadamard product of two matrices $A = (a_{i,j})$ and $B = (b_{i,j})$ of the same dimension $n \times p$ is the matrix $A \odot B$ of the same dimension $n \times p$ obtained by taking the componentwise product of the coefficients of the two matrices, that is, $A \odot B = (a_{i,j} b_{i,j})$.

with

$$\begin{aligned}\hat{\mathfrak{B}}_{i,t+1}^x &= \hat{\mathfrak{S}}_{i,t+1}^x (A_{i,t} + B_{i,t} \Gamma_{i,t}^x) , \\ \hat{\mathfrak{B}}_{i,t+1}^\lambda &= \hat{\mathfrak{S}}_{i,t+1}^x B_{i,t} \Gamma_{i,t}^\lambda + \hat{\mathfrak{S}}_{i,t+1}^\lambda \Xi_t , \\ \hat{\mathfrak{b}}_{i,t+1} &= \hat{\mathfrak{S}}_{i,t+1}^x B_{i,t} \gamma_{i,t} + \hat{\mathfrak{S}}_{i,t+1}^x c_{i,t} + \hat{\mathfrak{S}}_{i,t+1}^\lambda \xi_t + \hat{\mathfrak{s}}_{i,t+1} ,\end{aligned}$$

$$\begin{aligned}\mathbb{E}[\text{diag}(\lambda_{t+1})(\hat{\mathfrak{R}}_{i,t+1}^x x_{i,t+1} + \hat{\mathfrak{R}}_{i,t+1}^\lambda \lambda_{t+1} + \hat{\mathfrak{r}}_{i,t+1})] \\ = \text{diag}(\lambda_t) \Xi_t (\hat{\mathfrak{Q}}_{i,t+1}^x x_{i,t} + \hat{\mathfrak{Q}}_{i,t+1}^\lambda \lambda_t + \hat{\mathfrak{a}}_{i,t+1}) \\ + \text{diag}(\xi_t) (\hat{\mathfrak{Q}}_{i,t+1}^x x_{i,t} + \hat{\mathfrak{Q}}_{i,t+1}^x \lambda_t + \hat{\mathfrak{a}}_{i,t+1}) + \hat{\mathfrak{c}}_{i,t+1} ,\end{aligned}$$

with

$$\begin{aligned}\hat{\mathfrak{Q}}_{i,t+1}^x &= \hat{\mathfrak{R}}_{i,t+1}^x (A_{i,t} + B_{i,t} \Gamma_{i,t}^x) , \\ \hat{\mathfrak{Q}}_{i,t+1}^\lambda &= \hat{\mathfrak{R}}_{i,t+1}^x B_{i,t} \Gamma_{i,t}^\lambda + \hat{\mathfrak{R}}_{i,t+1}^\lambda \Xi_t , \\ \hat{\mathfrak{a}}_{i,t+1} &= \hat{\mathfrak{R}}_{i,t+1}^x B_{i,t} \gamma_{i,t} + \hat{\mathfrak{R}}_{i,t+1}^x c_{i,t} + \hat{\mathfrak{R}}_{i,t+1}^\lambda \xi_t + \hat{\mathfrak{r}}_{i,t+1} ,\end{aligned}$$

and

$$\hat{\mathfrak{c}}_{i,t+1} = \mathbb{E}[\text{diag}(\Omega_t w_{t+1})(\hat{\mathfrak{R}}_{i,t+1}^x C_{i,t} + \hat{\mathfrak{R}}_{i,t+1}^\lambda \Omega_t) w_{t+1}] .$$

We thus obtain that (53) holds true at time t with

$$\begin{aligned}\hat{\mathfrak{R}}_{i,t}^x &= \Xi_t \hat{\mathfrak{Q}}_{i,t+1}^x + \mathfrak{Q}_{i,t,t}^x , \\ \hat{\mathfrak{R}}_{i,t}^\lambda &= \Xi_t \hat{\mathfrak{Q}}_{i,t+1}^\lambda + \mathfrak{Q}_{i,t,t}^\lambda , \\ \hat{\mathfrak{r}}_{i,t} &= \Xi_t \hat{\mathfrak{a}}_{i,t+1} + \mathfrak{q}_{i,t,t} , \\ \hat{\mathfrak{S}}_{i,t}^x &= \text{diag}(\xi_t) \hat{\mathfrak{Q}}_{i,t+1}^x + \hat{\mathfrak{B}}_{i,t+1}^x , \\ \hat{\mathfrak{S}}_{i,t}^\lambda &= \text{diag}(\xi_t) \hat{\mathfrak{Q}}_{i,t+1}^\lambda + \hat{\mathfrak{B}}_{i,t+1}^\lambda , \\ \hat{\mathfrak{s}}_{i,t} &= \text{diag}(\xi_t) \hat{\mathfrak{a}}_{i,t+1} + \hat{\mathfrak{b}}_{i,t+1} + \hat{\mathfrak{c}}_{i,t+1} .\end{aligned}$$

Partial gradient with respect to the parameters Ω_t . In order to highlight the dependence of the Bellman function $\mathfrak{V}_{i,t}$ on Ω , we denote it by $\mathfrak{V}_{i,t}[\Omega]$.

Proposition 7. *The partial gradients with respect to Ω of the Bellman functions $\mathfrak{V}_{i,t}[\Omega]$ of Subproblem (42) are, for $t = 0, \dots, \mathcal{T}$, of the form*

$$\nabla_{\Omega_\tau} \mathfrak{V}_{i,t}[\Omega](x_{i,t}, \lambda_t) = 0 , \quad \tau = 0, \dots, t-1 , \quad (54a)$$

$$\nabla_{\Omega_\tau} \mathfrak{V}_{i,t}[\Omega](x_{i,t}, \lambda_t) = \mathfrak{T}_{i,\tau,t} , \quad \tau = t, \dots, \mathcal{T}-1 , \quad (54b)$$

where the $\mathfrak{T}_{i,\tau,t}$ are obtained by the recursions (55).

Proof. The proof is by induction. The proposition holds true at $t = \mathcal{T}$ since $\mathfrak{V}_{i,\mathcal{T}}[\Omega]$ does not depend on any of the parameters Ω_τ , $\tau \in \mathbb{T}$. Assume that it holds true at $t+1$.

- For $\tau = 0, \dots, t-1$, from the form (46) of the Bellman functions, we have that $\mathfrak{V}_{i,t}[\Omega]$ does not depend on the parameter Ω_τ , so that the proposition holds true at t for $\tau < t$.
- For $\tau = t$, consider the j -th line $\Omega_{j,t}$ of Ω_t , and the mapping $\varphi : \Omega_{j,t} \mapsto \Omega_{j,t} w_{t+1}$. We have that $\nabla \varphi(\Omega_{j,t}) = w_{t+1}^\top$. Considering then the partial gradient of $\mathfrak{V}_{i,t+1}[\Omega]$ with respect to the j -th component of λ given by (44b), we obtain the partial gradient of $\mathfrak{V}_{i,t}[\Omega]$ with respect to $\Omega_{j,t}$ by differentiating (46):

$$\nabla_{\Omega_{j,t}} \mathfrak{V}_{i,t}[\Omega](x_{i,t}, \lambda_t) = \mathbb{E} \left[\left(\mathfrak{P}_{i,t+1}^{c^\top} x_{i,t+1} + \mathfrak{P}_{i,t+1}^\lambda \lambda_{t+1} + \mathfrak{p}_{i,t+1}^\lambda \right)_j w_{t+1}^\top \right].$$

The partial gradient of $\mathfrak{V}_{i,t+1}[\Omega]$ with respect to the whole matrix Ω_t is accordingly

$$\nabla_{\Omega_t} \mathfrak{V}_{i,t}[\Omega](x_{i,t}, \lambda_t) = \mathbb{E} \left[\left(\mathfrak{P}_{i,t+1}^{c^\top} x_{i,t+1} + \mathfrak{P}_{i,t+1}^\lambda \lambda_{t+1} + \mathfrak{p}_{i,t+1}^\lambda \right) w_{t+1}^\top \right].$$

Replacing $x_{i,t+1}$ and λ_{t+1} using the dynamics (42b) and since the noise w_{t+1} is centered, we obtain

$$\nabla_{\Omega_t} \mathfrak{V}_{i,t}[\Omega](x_{i,t}, \lambda_t) = (\mathfrak{P}_{i,t+1}^{c^\top} C_{i,t} + \mathfrak{P}_{i,t+1}^\lambda \Omega_t) \Sigma_{t+1},$$

so that the proposition holds true at t for $\tau = t$.

- For $\tau = t+1, \dots, T-1$, from (46) and thanks to the induction assumption, we have

$$\begin{aligned} \nabla_{\Omega_\tau} \mathfrak{V}_{i,t}[\Omega](x_{i,t}, \lambda_t) &= \mathbb{E} [\nabla_{\Omega_\tau} \mathfrak{V}_{i,t+1}[\Omega](x_{i,t+1}, \lambda_{t+1})], \\ &= \mathfrak{T}_{i,\tau,t+1}, \end{aligned}$$

so that the proposition holds true at t for $\tau > t$.

The proposition thus holds true at t , with

$$\mathfrak{T}_{i,\tau,t} = 0, \quad \tau = 0, \dots, t-1, \quad (55a)$$

$$\mathfrak{T}_{i,t,t} = (\mathfrak{P}_{i,t+1}^{c^\top} C_{i,t} + \mathfrak{P}_{i,t+1}^\lambda \Omega_t) \Sigma_{t+1}, \quad (55b)$$

$$\mathfrak{T}_{i,\tau,t} = \mathfrak{T}_{i,\tau,t+1}, \quad \tau = t+1, \dots, T-1. \quad (55c)$$

The proof is complete. $\square$

Remark 5. If we assume that the matrices Ω_t do not depend on time t , that is, $\Omega_t = \Omega_0$ for all $t \in \mathbb{T}$, we obtain by similar computations that, for $t = 0, \dots, T-1$,

$$\nabla_{\Omega_0} \mathfrak{V}_{i,t}[\Omega](x_{i,t}, \lambda_t) = \sum_{\tau=t}^{T-1} (\mathfrak{P}_{i,\tau+1}^{c^\top} C_{i,\tau} + \mathfrak{P}_{i,\tau+1}^\lambda \Omega) \Sigma_{\tau+1}, \quad (56)$$

the partial gradients of $\mathfrak{V}_{i,T}[\Omega]$ with respect to Ω being zero. Note that the partial gradient of $\mathfrak{V}_{i,T-1}[\Omega]$ with respect to Ω is also zero since $\mathfrak{P}_{i,T}^c$ and $\mathfrak{P}_{i,T}^\lambda$ are zero by construction.

4.3.2 The case of Subproblem (43)

Denoting by $\mathbb{T} = \{0, \dots, T-1\}$ the horizon, we consider the following expectation

$$\mathbb{E} \left[\sum_{t \in \mathbb{T}} \theta_t^\top \lambda_t \right], \quad (57a)$$

where λ is subject to the dynamics

$$\lambda_0 \text{ given, } \lambda_{t+1} = \Xi_t \lambda_t + \xi_t + \Omega_t w_{t+1}, \quad t \in \mathbb{T}. \quad (57b)$$

We aim at computing the value (57a) as well as its partial gradients with respect to the parameters λ_0 and $(\Xi_t, \Omega_t, \xi_t)_{t \in \mathbb{T}}$. We recall that, by Assumption 1, the matrices Ξ_t are also diagonal.

Value of the expectation. The value of (57a) is obtained by solving the Bellman equation

$$\mathfrak{V}_{\theta,T}(\lambda_T) = 0, \quad (58a)$$

$$\mathfrak{V}_{\theta,t}(\lambda_t) = \theta_t^\top \lambda_t + \mathbb{E}[V_{\theta,t+1}(\Xi_t \lambda_t + \xi_t + \Omega_t w_{t+1})], \quad t \in \mathbb{T}. \quad (58b)$$

It is easy to prove by induction that these Bellman functions are of the form

$$\mathfrak{V}_{\theta,t}(\lambda_t) = \alpha_{\theta,t}^\top \lambda_t + \beta_{\theta,t}, \quad t \in \mathbb{T}, \quad (59)$$

with

$$\begin{cases} \alpha_{\theta,T} = 0 \\ \beta_{\theta,T} = 0 \end{cases}, \quad \begin{cases} \alpha_{\theta,t} = \Xi_t^\top \alpha_{\theta,t+1} + \theta_t \\ \beta_{\theta,t} = \alpha_{\theta,t+1}^\top \xi_t + \beta_{\theta,t+1} \end{cases}, \quad t \in \mathbb{T}. \quad (60)$$

Partial gradient with respect to the initial condition λ_0 . The Bellman functions $\mathfrak{V}_{\theta,t}$ do not depend on the parameter λ_0 for $t \geq 1$. For $t = 0$, from the form (59) of the Bellman function $\mathfrak{V}_{\theta,0}$, we obtain $\mathfrak{V}_{\theta,0}(\lambda_0) = \alpha_{\theta,0}$, so that

$$\nabla_{\lambda_0} \mathfrak{V}_{\theta,0}(\lambda_0) = \alpha_{\theta,0}, \quad (61a)$$

$$\nabla_{\lambda_0} \mathfrak{V}_{\theta,t}(\lambda_t) = 0, \quad t \in \mathbb{T} \setminus \{0\}. \quad (61b)$$

Partial gradients with respect to the parameters ξ_t . In order to highlight the dependence of the Bellman function $\mathfrak{V}_{\theta,t}$ on ξ , we denote it by $\mathfrak{V}_{\theta,t}[\xi]$.

At $\mathcal{T}$, the Bellman function $\mathfrak{V}_{\theta,T}$ is equal to 0, so that its partial gradients w.r.t. ξ_τ , $\tau \in \mathbb{T}$ are all equal to 0. Let $t \in \mathbb{T}$ and assume that, at $t+1$, the partial gradients of $\mathfrak{V}_{\theta,t+1}[\xi]$ w.r.t. ξ_τ , $\tau \in \mathbb{T}$, are constant:

$$\nabla_{\xi_\tau} \mathfrak{V}_{\theta,t+1}[\xi](\lambda_{t+1}) = \gamma_{\theta,\tau,t+1}, \quad \tau \in \mathbb{T},$$

which is fulfilled at $t+1 = \mathcal{T}$. We proceed by induction.

- The Bellman function $\mathfrak{V}_{\theta,t}[\xi]$ does not depend on the parameter ξ_τ for $\tau \leq t-1$, so that the partial gradients of $\mathfrak{V}_{\theta,t}[\xi]$ w.r.t. ξ_τ are equal to 0.
- Thanks to (58) and (59), we obtain

$$\nabla_{\xi_t} \mathfrak{V}_{\theta,t}[\xi](\lambda_t) = \mathbb{E}[\nabla_{\xi_t} \lambda_{t+1} \nabla_{\lambda_{t+1}} \mathfrak{V}_{\theta,t+1}[\xi](\lambda_{t+1})] = \alpha_{\theta,t+1} .$$

- By (58) and the induction assumption, for $\tau \geq t+1$, we obtain

$$\nabla_{\xi_\tau} \mathfrak{V}_{\theta,t}[\xi](\lambda_t) = \mathbb{E}[\nabla_{\xi_\tau} \mathfrak{V}_{\theta,t+1}[\xi](\lambda_{t+1})] = \gamma_{\theta,\tau,t+1} .$$

We thus have that, for $t \in \mathbb{T} \cup \{\mathcal{T}\}$,

$$\nabla_{\xi_\tau} \mathfrak{V}_{\theta,t}[\xi](\lambda_t) = \gamma_{\theta,\tau,t} = \begin{cases} 0 & \text{for } \tau = 0, \dots, t-1 \\ \alpha_{\theta,t+1} & \text{for } \tau = t \\ \gamma_{\theta,\tau,t+1} & \text{for } \tau = t+1, \dots, \mathcal{T}-1 \end{cases} . \quad (62)$$

Partial gradients with respect to the parameters Ξ_t . Using Assumption 1, the gradients with respect to Ξ_t are obtained by computing the gradients with respect to the vector Ξ_t^v formed by the diagonal elements of Ξ_t . In order to highlight the dependence of the Bellman function $\mathfrak{V}_{\theta,t}$ on Ξ , we denote it by $\mathfrak{V}_{\theta,t}[\Xi]$.

At $\mathcal{T}$, the Bellman function $\mathfrak{V}_{\theta,\mathcal{T}}[\Xi]$ is equal to 0, so that its partial gradients with respect to Ξ_τ^v , $\tau \in \mathbb{T}$ are all equal to 0. Let $t \in \mathbb{T}$ and assume that, at $t+1$, the partial gradients of $\mathfrak{V}_{\theta,t+1}[\Xi]$ with respect to Ξ_τ^v , $\tau \in \mathbb{T}$, are of the form

$$\nabla_{\Xi_\tau^v} \mathfrak{V}_{\theta,t+1}[\Xi](\lambda_{t+1}) = \delta_{\theta,\tau,t+1} \lambda_{t+1} + \varepsilon_{\theta,\tau,t+1} , \quad \tau \in \mathbb{T} ,$$

with $\delta_{\theta,\tau,t+1}$ a diagonal matrix. This assumption is fulfilled at $t+1 = \mathcal{T}$. We proceed by induction.

- The Bellman function $\mathfrak{V}_{\theta,t}[\Xi]$ does not depend on the parameter Ξ_τ for $\tau \leq t-1$, so that the partial gradients of $\mathfrak{V}_{\theta,t}[\Xi]$ w.r.t. Ξ_τ^v are equal to 0.
- Thanks to (58) and (59), we obtain

$$\nabla_{\Xi_t^v} \mathfrak{V}_{\theta,t}[\Xi](\lambda_t) = \mathbb{E}[\nabla_{\Xi_t^v} \lambda_{t+1} \nabla_{\lambda_{t+1}} \mathfrak{V}_{\theta,t+1}[\Xi](\lambda_{t+1})] = \text{diag}(\lambda_t) \alpha_{\theta,t+1} = \text{diag}(\alpha_{\theta,t+1}) \lambda_t .$$

- By (58) and the induction assumption, for $\tau \geq t+1$, we obtain

$$\begin{aligned} \nabla_{\Xi_\tau^v} \mathfrak{V}_{\theta,t}[\Xi](\lambda_t) &= \mathbb{E}[\nabla_{\Xi_\tau^v} \mathfrak{V}_{\theta,t+1}[\Xi](\lambda_{t+1})] , \\ &= \delta_{\theta,\tau,t+1} (\Xi_t \lambda_t + \xi_t) + \varepsilon_{\theta,\tau,t+1} , \\ &= \delta_{\theta,\tau,t+1} \Xi_t \lambda_t + \delta_{\theta,\tau,t+1} \xi_t + \varepsilon_{\theta,\tau,t+1} . \end{aligned}$$

We thus have that, for $t \in \mathbb{T} \cup \{\mathcal{T}\}$,

$$\nabla_{\Xi_\tau^v} \mathfrak{V}_{\theta,t}[\Xi](\lambda_t) = \delta_{\theta,\tau,t} \lambda_t + \varepsilon_{\theta,\tau,t} , \quad (63a)$$

with $\delta_{\theta,\tau,t}$ a diagonal matrix and

$$\delta_{\theta,\tau,t} = \begin{cases} 0 & \text{for } \tau = 0, \dots, t-1 , \\ \text{diag}(\alpha_{\theta,t+1}) & \text{for } \tau = t , \\ \delta_{\theta,\tau,t+1} \Xi_t & \text{for } \tau = t+1, \dots, \mathcal{T}-1 , \end{cases} \quad (63b)$$

$$\varepsilon_{\theta,\tau,t} = \begin{cases} 0 & \text{for } \tau = 0, \dots, t-1 , \\ 0 & \text{for } \tau = t , \\ \delta_{\theta,\tau,t+1} \xi_t + \varepsilon_{\theta,\tau,t+1} & \text{for } \tau = t+1, \dots, \mathcal{T}-1 . \end{cases} \quad (63c)$$

Remark 6. If we assume that the matrices Ξ_t do not depend on time t , that is, $\Xi_t = \Xi_0$ for all $t \in \mathbb{T}$, and with $\Xi_0^v = \text{diag}(\Xi_0)$, we obtain that, for $t = 0, \dots, \mathcal{T}-1$,

$$\nabla_{\Xi_0^v} \mathfrak{V}_{\theta,t}[\Xi](\lambda_t) = \hat{\delta}_{\theta,t} \lambda_t + \hat{\epsilon}_{\theta,t} , \quad (64)$$

the partial gradients of $\mathfrak{V}_{\theta,T}$ with respect to Ξ_0^v being zero by construction. Indeed, assuming that (64) holds true at $t+1$ and from the form (59) of the Bellman function, we have

$$\begin{aligned} \nabla_{\Xi_0^v} \mathfrak{V}_{\theta,t}[\Xi](\lambda_t) &= \text{diag}(\alpha_{\theta,t+1}) \lambda_t + \mathbb{E}[\hat{\delta}_{\theta,t+1} \lambda_{t+1} + \hat{\epsilon}_{\theta,t+1}] , \\ &= \text{diag}(\alpha_{\theta,t+1}) \lambda_t + \hat{\delta}_{\theta,t+1} (\Xi \lambda_t + \xi_t) + \hat{\epsilon}_{\theta,t+1} , \end{aligned}$$

hence

$$\hat{\delta}_{\theta,t} = \text{diag}(\alpha_{\theta,t+1}) + \hat{\delta}_{\theta,t+1} \Xi , \quad \hat{\epsilon}_{\theta,t} = \hat{\delta}_{\theta,t+1} \xi_t + \hat{\epsilon}_{\theta,t+1} .$$

Partial gradients with respect to the parameters Ω_t . In order to highlight the dependence of the Bellman function $\mathfrak{V}_{\theta,t}$ on Ω , we denote it by $\mathfrak{V}_{\theta,t}[\Omega]$. From the form (59) of the Bellman functions and the centered assumption of the noises, it is easy to deduce that, for any $t \in \mathbb{T} \cup \{\mathcal{T}\}$, the Bellman function $\mathfrak{V}_{\theta,t}[\Omega]$ does not depend on the parameters Ω_τ , $\tau \in \mathbb{T}$, so that

$$\nabla_{\Omega_\tau} \mathfrak{V}_{\theta,t}[\Omega](\lambda_t) = 0 , \quad \tau \in \mathbb{T} , \quad t \in \mathbb{T} \cup \{\mathcal{T}\} . \quad (65)$$

The same holds true if we assume that the matrices Ω_t do not depend on time t .

4.3.3 Partial gradients of the whole problem

Gathering the results obtained in §4.3.1 and in §4.3.2, we obtain the partial gradients of the optimal cost $\mathfrak{V}_0(x_0, \lambda_0)$ with respect to the parameters λ_0 and $(\Xi_t, \Omega_t, \xi_t)_{t \in \mathbb{T}}$ by summing for each parameter the expression of the partial gradients obtained for Subproblem (42) for each $i \in \mathbb{I}$ and for Subproblem (43).

4.4 Simulation of an admissible control

At the end of the Uzawa algorithm, we have at disposal values λ_0^* and $(\Xi_t^*, \Omega_t^*, \xi_t^*)_{t \in \mathbb{T}}$ for the parameters of the multiplier process λ^* of the multiplier as well as Bellman functions $\mathfrak{V}_{i,t}$, at each time step t and for each subproblem i . By summing these Bellman functions,⁴ we generate surrogate functions $\tilde{\mathfrak{V}}_t$ for the true Bellman functions V_t of Problem (1), namely,

$$\begin{aligned} \tilde{\mathfrak{V}}_t(x_{1,t}, \dots, x_{I,t}, \lambda_t) &= \sum_{i \in \mathbb{I}} \left(\frac{1}{2} \begin{pmatrix} x_{i,t}^\top & \lambda_t^\top \end{pmatrix} \begin{pmatrix} \mathfrak{P}_{i,t}^x & \mathfrak{P}_{i,t}^c \\ \mathfrak{P}_{i,t}^{c^\top} & \mathfrak{P}_{i,t}^\lambda \end{pmatrix} \begin{pmatrix} x_{i,t} \\ \lambda_t \end{pmatrix} + \begin{pmatrix} \mathfrak{p}_{i,t}^x & \mathfrak{p}_{i,t}^{\lambda^\top} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ \lambda_t \end{pmatrix} + \mathfrak{r}_{i,t} \right) \\ &= \frac{1}{2} [x_{1,t}^\top \dots x_{I,t}^\top \lambda_t^\top] \tilde{\mathfrak{P}}_t \begin{bmatrix} x_{1,t} \\ \vdots \\ x_{I,t} \\ \lambda_t \end{bmatrix} + \tilde{\mathfrak{p}}_t^\top \begin{bmatrix} x_{1,t} \\ \vdots \\ x_{I,t} \\ \lambda_t \end{bmatrix} + \tilde{\mathfrak{r}}_t, \end{aligned}$$

with

$$\tilde{\mathfrak{P}}_t = \begin{bmatrix} \mathfrak{P}_{1,t}^x & \dots & 0 & \mathfrak{P}_{1,t}^c \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & \mathfrak{P}_{I,t}^x & \mathfrak{P}_{I,t}^c \\ \mathfrak{P}_{1,t}^{c^\top} & \dots & \mathfrak{P}_{I,t}^{c^\top} & \sum_{i \in \mathbb{I}} \mathfrak{P}_{i,t}^\lambda \end{bmatrix}, \quad \tilde{\mathfrak{p}}_t = \begin{bmatrix} \mathfrak{p}_{1,t}^x \\ \vdots \\ \mathfrak{p}_{I,t}^x \\ \sum_{i \in \mathbb{I}} \mathfrak{p}_{i,t}^\lambda \end{bmatrix}, \quad \tilde{\mathfrak{r}}_t = \sum_{i=1}^I \mathfrak{r}_{i,t}.$$

One can then calculate admissible controls, that is, controls that satisfy all the constraints of Problem (1). At time step t , starting from state (x_t, λ_t) , the admissible control to be applied is obtained by solving an optimization problem of type (18), namely

$$\begin{aligned} \min_{u_t} \mathbb{E} \left[\frac{1}{2} \begin{pmatrix} x_t^\top & \lambda_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & 0 & S_t \\ 0 & 0 & 0 \\ S_t^\top & 0 & N_t \end{pmatrix} \begin{pmatrix} x_t \\ \lambda_t \\ u_t \end{pmatrix} + \begin{pmatrix} m_t^\top & 0 & n_t^\top \end{pmatrix} \begin{pmatrix} x_t \\ \lambda_t \\ u_t \end{pmatrix} + d_t \right. \\ \left. + \tilde{\mathfrak{V}}_{t+1}(A_t x_t + B_t u_t + C_t w_{t+1} + c_t, \Xi_t^* \lambda_t + \Omega_t^* w_{t+1} + \xi_t^*) \right], \end{aligned}$$

subject to the constraint

$$\theta_t - \Theta_t u_t = 0, \quad t \in \mathbb{T},$$

The expression of this admissible control is given by Equation (45). Simulating this admissible control along a set of noise trajectories obtained by Monte Carlo sampling provides a statistical upper bound on the optimal cost of Problem (1). Note that it is also possible to compute an exact upper bound: see Remark 1 and Appendix A for further details.

⁴The functions $\mathfrak{V}_{\theta,t}$ are independent of both the control u_t and the state x_t , and thus can be omitted.

5 Temporal block decomposition

We are still interested in Problem (1), and we assume that the optimization span $\mathbb{T}$ consists of D temporal blocks of length T :⁵

$$\mathbb{T} = \left\{ \underbrace{0, \dots, T-1}_{\text{block 0}}, \underbrace{T, \dots, 2T-1}_{\text{block 1}}, \dots, \underbrace{(D-1)T, \dots, DT-1}_{\text{block } D-1} \right\},$$

to which we add a final instant $\mathcal{T} = DT$ to take into account the final state of the problem. We introduce the following notations:

- $\mathbb{D} = \{0, \dots, D-1\}$ is the set of blocks of the optimization span $\mathbb{T}$,
- $\mathbb{T}_d = \{dT, \dots, (d+1)T-1\}$ is the set of time steps of block $d \in \mathbb{D}$,
- $t_d^i = dT$ is the initial time step of block d , with the convention $t_D^i = \mathcal{T}$,
- $t_d^f - 1 = (d+1)T-1$ is the final time step of block d , and thus $t_d^f = t_{d+1}^i$.

We wish to solve Problem (1) by decomposing it by temporal blocks. In order to be able to compute local solutions by Riccati, we only implement decomposition by prices,⁶ the constraints to be dualized being the dynamic constraints (1b) at the interface between two consecutive temporal blocks. We denote ν_d the multiplier associated with the constraint coupling blocks d and $d+1$. For the calculation to be operational, we limit ourselves to *deterministic* multipliers. We then use the methodology developed in [7].

5.1 Recall of the temporal block decomposition mechanism

The temporal block decomposition is implemented as follows.

- Since $t_d^f = t_{d+1}^i$, the only equation coupling the temporal blocks d and $d+1$ is

$$x_{t_d^f} - x_{t_{d+1}^i} = 0, \tag{66}$$

and dualizing this constraint with a multiplier ν_d allows us to decouple the blocks.

- The value of the intraday function $\mathcal{L}_d$ of temporal block d evaluated at point $(x_{t_d^i}, \nu_d)$ is obtained by minimizing the sum of the instantaneous costs of block d and the duality term $\nu_d^\top x_{t_d^f}$ coming from (66), under the constraints specific to block d .
- Once the intraday functions $\mathcal{L}_d$ of each block are obtained, the Bellman recursion is done block by block and takes into account the remaining duality term $-\nu_d^\top x_{t_{d+1}^i}$.

⁵The temporal blocks are all of the same length here, but this assumption is not essential.

⁶Decomposition by resources would induce constraints incompatible with the solution by Riccati.

The Bellman function $\mathcal{V}_D$ of the final block is taken equal to the final cost function of the problem and, for $d = D-1, \dots, 0$, the Bellman function $\mathcal{V}_d$ is given by

$$\begin{aligned}\mathcal{V}_d(x_{t_d}^i) &= \max_{\nu_d} \min_{x_{t_{d+1}}^i} \left(\mathcal{L}_d(x_{t_d}^i, \nu_d) + \mathcal{V}_{d+1}(x_{t_{d+1}}^i) - \nu_d^\top x_{t_{d+1}}^i \right) \\ &= \max_{\nu_d} \left(\mathcal{L}_d(x_{t_d}^i, \nu_d) - \max_{x_{t_{d+1}}^i} (\nu_d^\top x_{t_{d+1}}^i - \mathcal{V}_{d+1}(x_{t_{d+1}}^i)) \right) \\ &= \max_{\nu_d} \left(\mathcal{L}_d(x_{t_d}^i, \nu_d) - (\mathcal{V}_{d+1})^*(\nu_d) \right),\end{aligned}$$

where $(\mathcal{V}_{d+1})^*$ is the Fenchel conjugate of $\mathcal{V}_{d+1}$.

5.2 Formulation of the problem with temporal blocks

To formulate Problem (1) by block, we rewrite the dynamics by block by adding the coupling constraints between the temporal blocks. For $d \in \mathbb{D}$, we write the dynamics of the block over the time span $\mathbb{T}_d$, with $\mathbb{T}_d = \{t_d^i, \dots, t_d^f - 1\}$. With the notations of Section 2, the dynamics of block d is

$$x_{t_d}^i \text{ given, } x_{t+1} = A_t x_t + B_t u_t + C_t w_{t+1} + c_t, \quad t \in \mathbb{T}_d,$$

the matching conditions of the problem dynamics between blocks d and $d+1$ being

$$x_{t_d}^f - x_{t_{d+1}}^i = 0, \quad d \in \mathbb{D}.$$

These latter constraints are dualized with multipliers ν_d , $d \in \mathbb{D}$, and the resulting dual problem is written, with $\nu = (\nu_0, \dots, \nu_{D-1})$,

$$\begin{aligned}\max_{\nu} \min_{x, u} \mathbb{E} \left[\sum_{d \in \mathbb{D}} \left(\sum_{t \in \mathbb{T}_d} \left(\frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} \right. \right. \\ \left. \left. + m_t^\top x_t + n_t^\top u_t + d_t \right) + \nu_d^\top (x_{t_d}^f - x_{t_{d+1}}^i) \right) \\ \left. + \frac{1}{2} x_{\mathcal{T}}^\top M_{\mathcal{T}} x_{\mathcal{T}} + m_{\mathcal{T}}^\top x_{\mathcal{T}} + d_{\mathcal{T}} \right], \quad (67a)\end{aligned}$$

subject to the dynamic constraints written per block

$$x_{t_d}^i \text{ given, } x_{t+1} = A_t x_t + B_t u_t + C_t w_{t+1} + c_t, \quad t \in \mathbb{T}_d, \quad d \in \mathbb{D}, \quad (67b)$$

and subject to the coupling constraints written per block

$$\theta_t - \Theta_t u_t = 0, \quad t \in \mathbb{T}_d, \quad d \in \mathbb{D}. \quad (67c)$$

5.3 Study of intraday functions

The intraday function $\mathcal{L}_d$ of block d is defined by

$$\mathcal{L}_d(x_{t_d^i}, \nu_d) = \min_{x, u} \mathbb{E} \left[\sum_{t \in \mathbb{T}_d} \left(\frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} + m_t^\top x_t + n_t^\top u_t + d_t \right) + \nu_d^\top x_{t_d^f} \right], \quad (68a)$$

subject to the dynamic constraints

$$x_{t_d^i} \text{ given}, \quad x_{t+1} = A_t x_t + B_t u_t + C_t w_{t+1} + c_t, \quad t \in \mathbb{T}_d. \quad (68b)$$

and the coupling constraint

$$\theta_t - \Theta_t u_t = 0, \quad t \in \mathbb{T}_d. \quad (68c)$$

Remark 7. *The main practical interest of temporal block decomposition comes from the notion of periodicity class: if all the parameters entering the definition of the intraday function of two blocks d and d' (matrices and vectors of instantaneous costs, dynamics, and coupling constraint) are identical, the intraday functions of blocks d and d' will be identical and it is therefore sufficient to compute it only once. Identical temporal blocks are then grouped into periodicity classes, and only one intraday function per class is computed, which can lead to considerable CPU time savings in the optimization process.*

5.3.1 Computation of the intraday function $\mathcal{L}_d$ by Riccati

In (68), the computation of the intraday function $\mathcal{L}_d$ is done while respecting the coupling constraint (68c), and this computation can therefore be carried out thanks to the general constrained Riccati equation developed in §1.4. The Bellman functions V_t , for $t \in \mathbb{T}_d$, associated with the computation of the intraday function $\mathcal{L}_d$ are thus quadratic, of the form

$$V_t(x_t) = \frac{1}{2} x_t^\top P_t x_t + p_t^\top x_t + r_t, \quad (69)$$

and we compute them by solving, for $t \in \mathbb{T}_d$, the Riccati recursion

$$P_{t_d^f} = 0, \quad P_t = G_{t+1} + 2E_{t+1}^\top F_{t+1}^{-1} K_{t+1} + K_{t+1}^\top F_{t+1}^{-1} K_{t+1}, \quad (70a)$$

$$p_{t_d^f} = \nu_d, \quad p_t = g_{t+1} + K_{t+1}^\top F_{t+1}^{-1} f_{t+1} + K_{t+1}^\top F_{t+1}^{-1} k_{t+1} + E_{t+1}^\top F_{t+1}^{-1} k_{t+1}, \quad (70b)$$

$$r_{t_d^f} = 0, \quad r_t = h_{t+1} + \frac{1}{2} k_{t+1}^\top F_{t+1}^{-1} k_{t+1} + f_{t+1}^\top F_{t+1}^{-1} k_{t+1}, \quad (70c)$$

with the notations from §1.4 which we recall, namely

$$E_{t+1} = S_t^\top + B_t^\top P_{t+1} A_t, \quad (71a)$$

$$F_{t+1} = N_t + B_t^\top P_{t+1} B_t, \quad (71b)$$

$$G_{t+1} = M_t + A_t^\top P_{t+1} A_t, \quad (71c)$$

$$L_{t+1} = (\Theta_t F_{t+1}^{-1} \Theta_t^\top)^{-1} \Theta_t F_{t+1}^{-1} E_{t+1}, \quad (71d)$$

$$K_{t+1} = \Theta_t^\top L_{t+1} - E_{t+1}, \quad (71e)$$

and

$$f_{t+1} = n_t + B_t^\top p_{t+1} + B_t^\top P_{t+1} c_t, \quad (72a)$$

$$g_{t+1} = m_t + A_t^\top p_{t+1} + A_t^\top P_{t+1} c_t, \quad (72b)$$

$$h_{t+1} = \frac{1}{2} \text{trace}(\Sigma_{t+1} C_t^\top P_{t+1} C_t) + \frac{1}{2} c_t^\top P_{t+1} c_t + p_{t+1}^\top c_t + d_t + r_{t+1}, \quad (72c)$$

$$\ell_{t+1} = (\Theta_t F_{t+1}^{-1} \Theta_t^\top)^{-1} (\Theta_t F_{t+1}^{-1} f_{t+1} + \theta_t), \quad (72d)$$

$$k_{t+1} = \Theta_t^\top \ell_{t+1} - f_{t+1}. \quad (72e)$$

It should be noted that the Bellman functions V_t , $t \in \mathbb{T}_d$, depend *implicitly* on the multiplier ν_d . The optimal control is given by

$$u_t^* = F_{t+1}^{-1} K_{t+1} x_t + F_{t+1}^{-1} k_{t+1}, \quad t \in \mathbb{T}_d. \quad (73)$$

Under the assumption of independence of noises, we have the relation

$$\mathcal{L}_d(x_{t_d^i}, \nu_d) = V_{t_d^i}(x_{t_d^i}). \quad (74)$$

5.3.2 Dependence on ν_d of the intraday function $\mathcal{L}_d$

As seen in §5.1, the recursion allowing the calculation of the block Bellman functions $\mathcal{V}_d$ is performed on the multipliers ν_d . We are therefore interested in the nature of the dependence on ν_d of the intraday function $\mathcal{L}_d$ and the Bellman functions V_t obtained during the calculation of this intraday function.

Dependence on ν_d of the matrices P_t . We show by recursion that, for $t \in \mathbb{T}_d$, the matrices P_t do not depend on ν_d . Indeed, the matrix $P_{t_d^f} = 0$ does not depend on ν_d . The same applies to the matrices P_t in (70a) since the matrices E_{t+1} , F_{t+1} , G_{t+1} and K_{t+1} in (71) do not depend on ν_d .

Dependence on ν_d of the vectors p_t . We show by recursion that, for $t \in \mathbb{T}_d$, the vectors p_t are of the form:

$$p_t = \alpha_t \nu_d + \beta_t. \quad (75)$$

Indeed, the vector $p_{t_d^f} = \nu_d$ is of the form (75), with $\alpha_{t_d^f} = \text{I}$ and $\beta_{t_d^f} = 0$. Assuming that p_{t+1} is also of this form, the vectors f_{t+1} , g_{t+1} , ℓ_{t+1} and k_{t+1} in (72) depend linearly on ν_d , namely

$$\begin{aligned} f_{t+1} &= \underbrace{B_t^\top \alpha_{t+1}}_{\hat{\alpha}_{t+1}} \nu_d + \underbrace{n_t + B_t^\top \beta_{t+1} + B_t^\top P_{t+1} c_t}_{\hat{\beta}_{t+1}}, \\ g_{t+1} &= \underbrace{A_t^\top \alpha_{t+1}}_{\tilde{\alpha}_{t+1}} \nu_d + \underbrace{m_t + A_t^\top \beta_{t+1} + A_t^\top P_{t+1} c_t}_{\tilde{\beta}_{t+1}}, \end{aligned}$$

and

$$\begin{aligned} \ell_{t+1} &= \underbrace{(\Theta_t F_{t+1}^{-1} \Theta_t^\top)^{-1} \Theta_t F_{t+1}^{-1} \hat{\alpha}_{t+1}}_{\check{\alpha}_{t+1}} \nu_d + \underbrace{(\Theta_t F_{t+1}^{-1} \Theta_t^\top)^{-1} (\Theta_t F_{t+1}^{-1} \hat{\beta}_{t+1} + \theta_t)}_{\check{\beta}_{t+1}}, \\ k_{t+1} &= \underbrace{(\Theta_t^\top \check{\alpha}_{t+1} - \hat{\alpha}_{t+1})}_{\dot{\alpha}_{t+1}} \nu_d + \underbrace{(\Theta_t^\top \check{\beta}_{t+1} - \hat{\beta}_{t+1})}_{\dot{\beta}_{t+1}}. \end{aligned}$$

We deduce from (70b) that the vectors p_t are of the form (75), with

$$\alpha_t = \tilde{\alpha}_{t+1} + K_{t+1}^\top F_{t+1}^{-1} \hat{\alpha}_{t+1} + K_{t+1}^\top F_{t+1}^{-1} \dot{\alpha}_{t+1} + E_{t+1}^\top F_{t+1}^{-1} \dot{\alpha}_{t+1}, \quad (76a)$$

$$\beta_t = \tilde{\beta}_{t+1} + K_{t+1}^\top F_{t+1}^{-1} \hat{\beta}_{t+1} + K_{t+1}^\top F_{t+1}^{-1} \dot{\beta}_{t+1} + E_{t+1}^\top F_{t+1}^{-1} \dot{\beta}_{t+1}. \quad (76b)$$

Dependence on ν_d of the scalars r_t . We show by recursion that, for $t \in \mathbb{T}_d$, the scalars r_t are of the form:

$$r_t = \frac{1}{2} \nu_d^\top \Delta_t \nu_d + \delta_t^\top \nu_d + \epsilon_t. \quad (77)$$

Indeed, the scalar $r_{t_d^f} = 0$ is of the form (77), with $\Delta_{t_d^f} = 0$, $\delta_{t_d^f} = 0$ and $\epsilon_{t_d^f} = 0$. Suppose that r_{t+1} is also of this form. Then, using the expression of p_{t+1} obtained previously, the scalar h_{t+1} in (72) depends on ν_d in a quadratic manner, that is

$$\begin{aligned} h_{t+1} &= \frac{1}{2} \nu_d^\top \Delta_{t+1} \nu_d + \underbrace{(\delta_{t+1} + \alpha_{t+1}^\top c_t)}_{\hat{\delta}_{t+1}}^\top \nu_d \\ &\quad + \underbrace{(\epsilon_{t+1} + d_t + c_t^\top \beta_{t+1} + \frac{1}{2} c_t^\top P_{t+1} c_t + \frac{1}{2} \text{trace}(\Sigma_{t+1} C_t^\top P_{t+1} C_t))}_{\hat{\epsilon}_{t+1}}. \end{aligned}$$

Using the relations $f_{i,t+1} = \hat{\alpha}_{i,t+1} \nu_{i,d} + \hat{\beta}_{i,t+1}$ and $k_{i,t+1} = \dot{\alpha}_{i,t+1} \nu_{i,d} + \dot{\beta}_{i,t+1}$, we have

$$\begin{aligned} \frac{1}{2} k_{t+1}^\top F_{t+1}^{-1} k_{t+1} &= \frac{1}{2} \nu_d^\top \dot{\alpha}_{t+1}^\top F_{t+1}^{-1} \dot{\alpha}_{t+1} \nu_d + \dot{\beta}_{t+1}^\top F_{t+1}^{-1} \dot{\alpha}_{t+1} \nu_d + \frac{1}{2} \dot{\beta}_{t+1}^\top F_{t+1}^{-1} \dot{\beta}_{t+1}, \\ f_{t+1}^\top F_{t+1}^{-1} k_{t+1} &= \nu_d^\top \hat{\alpha}_{t+1}^\top F_{t+1}^{-1} \dot{\alpha}_{t+1} \nu_d + \hat{\beta}_{t+1}^\top F_{t+1}^{-1} \dot{\alpha}_{t+1} \nu_d + \dot{\beta}_{t+1}^\top F_{t+1}^{-1} \hat{\alpha}_{t+1} \nu_d + \hat{\beta}_{t+1}^\top F_{t+1}^{-1} \dot{\beta}_{t+1}. \end{aligned}$$

We deduce from (70c) that r_t is of the form (77), with

$$\Delta_t = \Delta_{t+1} + \dot{\alpha}_{t+1}^\top F_{t+1}^{-1} \dot{\alpha}_{t+1} + 2 \hat{\alpha}_{t+1}^\top F_{t+1}^{-1} \dot{\alpha}_{t+1}, \quad (78a)$$

$$\delta_t = \hat{\delta}_{t+1} + \dot{\alpha}_{t+1}^\top F_{t+1}^{-1} \dot{\beta}_{t+1} + \dot{\alpha}_{t+1}^\top F_{t+1}^{-1} \hat{\beta}_{t+1} + \hat{\alpha}_{t+1}^\top F_{t+1}^{-1} \dot{\beta}_{t+1}, \quad (78b)$$

$$\epsilon_t = \hat{\epsilon}_{t+1} + \frac{1}{2} \dot{\beta}_{t+1}^\top F_{t+1}^{-1} \dot{\beta}_{t+1} + \hat{\beta}_{t+1}^\top F_{t+1}^{-1} \dot{\beta}_{t+1}. \quad (78c)$$

Expressions of the intraday function. We saw in §5.3.1 that the function intraday $\mathcal{L}_d$ is a quadratic function in its first variable $x_{t_d}^i$:

$$\mathcal{L}_d(x_{t_d}^i, \nu_d) = \frac{1}{2} x_{t_d}^{\top} P_{t_d} x_{t_d}^i + p_{t_d}^{\top} x_{t_d}^i + r_{t_d}^i. \quad (79)$$

We have just shown in §5.3.2 that it is also quadratic in its second variable ν_d :

$$\mathcal{L}_d(x_{t_d}^i, \nu_d) = \frac{1}{2} \nu_d^{\top} \Delta_{t_d}^i \nu_d + (\delta_{t_d}^i + \alpha_{t_d}^{\top} x_{t_d}^i)^{\top} \nu_d + (\epsilon_{t_d}^i + \beta_{t_d}^{\top} x_{t_d}^i + \frac{1}{2} x_{t_d}^{\top} P_{t_d} x_{t_d}^i), \quad (80)$$

and we deduce that it is quadratic in the pair $(x_{t_d}^i, \nu_d)$, with a cross term $x_{t_d}^{\top} \alpha_{t_d}^i \nu_d$.

5.4 Bellman recursion by block

Using the intraday functions (68), the dual problem (67) is rewritten in the form

$$\max_{\nu} \min_{x_{t_1}^i, \dots, x_{t_D}^i} \sum_{d \in \mathbb{D}} \left(\mathcal{L}_d(x_{t_d}^i, \nu_d) - \nu_d^{\top} x_{t_{d+1}}^i \right) + \frac{1}{2} x_{\mathcal{T}}^{\top} M_{\mathcal{T}} x_{\mathcal{T}} + m_{\mathcal{T}}^{\top} x_{\mathcal{T}} + d_{\mathcal{T}},$$

with $x_{t_0}^i = x_0$ given, and where we recall that $x_{\mathcal{T}} = x_{t_D}^i$. From §5.1, we deduce the Bellman recursion between the temporal blocks which, initialized using the final cost of the problem, namely

$$\mathcal{V}_D(x_{\mathcal{T}}) = \frac{1}{2} x_{\mathcal{T}}^{\top} M_{\mathcal{T}} x_{\mathcal{T}} + m_{\mathcal{T}}^{\top} x_{\mathcal{T}} + d_{\mathcal{T}}, \quad (81a)$$

is given by, pour $d = D-1 \dots, 0$,

$$\mathcal{V}_d(x_{t_d}^i) = \max_{\nu_d} \left(\mathcal{L}_d(x_{t_d}^i, \nu_d) - (\mathcal{V}_{d+1})^*(\nu_d) \right), \quad (81b)$$

with

$$(\mathcal{V}_{d+1})^*(\nu_d) = \max_{x_{t_{d+1}}^i} (\nu_d^{\top} x_{t_{d+1}}^i - \mathcal{V}_{d+1}(x_{t_{d+1}}^i)). \quad (81c)$$

Proposition 8. *The Bellman functions by block $\mathcal{V}_d$, $d = 0, \dots, D$, are of the form*

$$\mathcal{V}_d(x_{t_d}^i) = \frac{1}{2} x_{t_d}^{\top} \mathcal{P}_d x_{t_d}^i + \pi_d^{\top} x_{t_d}^i + \rho_d, \quad (82)$$

where matrices $\mathcal{P}_d$, vectors π_d and scalars ρ_d are given by Equations (87) with the notations (85). The optimal feedback in ν_d is given by Equations (86).

Proof. The proof is by induction. The Bellman function $\mathcal{V}_D$ is of the form (82), with $\mathcal{P}_D = M_{\mathcal{T}}$, $\pi_D = m_{\mathcal{T}}$ and $\rho_D = d_{\mathcal{T}}$. Suppose that the Bellman function at $d+1$ is also of this form. First we compute the Fenchel conjugate of $\mathcal{V}_{d+1}$:

$$(\mathcal{V}_{d+1})^*(\nu_d) = \sup_{x_{t_{d+1}}^i} \left(\nu_d^{\top} x_{t_{d+1}}^i - \frac{1}{2} x_{t_{d+1}}^{\top} \mathcal{P}_{d+1} x_{t_{d+1}}^i - \pi_{d+1}^{\top} x_{t_{d+1}}^i - \rho_{d+1} \right).$$

The optimality condition in $x_{t_{d+1}^i}$ of this problem is $\nu_d - \mathcal{P}_{d+1} x_{t_{d+1}^i}^* - \pi_{d+1} = 0$, hence its solution

$$x_{t_{d+1}^i}^* = \mathcal{P}_{d+1}^{-1}(\nu_d - \pi_{d+1}) , \quad (83)$$

which leads to

$$(\mathcal{V}_{d+1})^*(\nu) = \frac{1}{2} \nu_d^\top \mathcal{P}_{d+1}^{-1} \nu_d - \pi_{d+1}^\top \mathcal{P}_{d+1}^{-1} \nu_d + \frac{1}{2} \pi_{d+1}^\top \mathcal{P}_{d+1}^{-1} \pi_{d+1} - \rho_{d+1} . \quad (84)$$

Using the expression (80) of the intraday function $\mathcal{L}_d$ and the expression (84) of the Fenchel conjugate $(\mathcal{V}_{d+1})^*$, the optimality condition in ν_d in the calculation of the Bellman function $\mathcal{V}_d$ given by (81b) is

$$\Delta_{t_d^i} \nu_d^* + \delta_{t_d^i} + \alpha_{t_d^i}^\top x_{t_d^i} - \mathcal{P}_{d+1}^{-1} \nu_d^* + \mathcal{P}_{d+1}^{-1} \pi_{d+1} = 0 .$$

Introducing the notations

$$\Xi_{d+1} = \Delta_{t_d^i} - \mathcal{P}_{d+1}^{-1} , \quad (85a)$$

$$\xi_{d+1} = \delta_{t_d^i} + \mathcal{P}_{d+1}^{-1} \pi_{d+1} , \quad (85b)$$

$$\vartheta_{d+1} = \epsilon_{t_d^i} - \frac{1}{2} \pi_{d+1}^\top \mathcal{P}_{d+1}^{-1} \pi_{d+1} + \rho_{d+1} , \quad (85c)$$

the optimal control is given by

$$\nu_d^* = -\Xi_{d+1}^{-1} (\alpha_{t_d^i}^\top x_{t_d^i} + \xi_{d+1}) . \quad (86)$$

As seen in §5.3.2, the intraday function $\mathcal{L}_d$ is quadratic in $(x_{t_d^i}, \nu_d)$. Since the Fenchel conjugate $(\mathcal{V}_{d+1})^*$ is quadratic in ν_d and the optimal feedback ν_d^* is linear in $x_{t_d^i}$, it follows that the Bellman function $\mathcal{V}_d(x_{t_d^i}) = \mathcal{L}_d(x_{t_d^i}, \nu_d^*) - (\mathcal{V}_{d+1})^*(\nu_d^*)$ is quadratic in $x_{t_d^i}$. The expression of $\mathcal{V}_d(x_{t_d^i})$ is

$$\mathcal{V}_d(x_{t_d^i}) = \frac{1}{2} \nu_d^{*\top} \Xi_{d+1} \nu_d^* + \xi_{d+1}^\top \nu_d^* + x_{t_d^i}^\top \alpha_{t_d^i} \nu_d^* + \beta_{t_d^i}^\top x_{t_d^i} + \frac{1}{2} x_{t_d^i}^\top P_{t_d^i} x_{t_d^i} + \vartheta_{d+1} .$$

Replacing ν_d^* by its expression (86), we have

$$\begin{aligned} \frac{1}{2} \nu_d^{*\top} \Xi_{d+1} \nu_d^* &= \frac{1}{2} x_{t_d^i}^\top \alpha_{t_d^i} \Xi_{d+1}^{-1} \alpha_{t_d^i}^\top x_{t_d^i} + \xi_{d+1}^\top \Xi_{d+1}^{-1} \alpha_{t_d^i}^\top x_{t_d^i} + \frac{1}{2} \xi_{d+1}^\top \Xi_{d+1}^{-1} \xi_{d+1} , \\ \xi_{d+1}^\top \nu_d^* &= -\xi_{d+1}^\top \Xi_{d+1}^{-1} \alpha_{t_d^i}^\top x_{t_d^i} - \xi_{d+1}^\top \Xi_{d+1}^{-1} \xi_{d+1} , \\ x_{t_d^i}^\top \alpha_{t_d^i} \nu_d^* &= -x_{t_d^i}^\top \alpha_{t_d^i} \Xi_{d+1}^{-1} \alpha_{t_d^i}^\top x_{t_d^i} - \xi_{d+1}^\top \Xi_{d+1}^{-1} \alpha_{t_d^i}^\top x_{t_d^i} , \end{aligned}$$

from which we deduce that $\mathcal{V}_d(x)$ is of the form (82), with

$$\mathcal{P}_d = P_{t_d^i} - \alpha_{t_d^i} \Xi_{d+1}^{-1} \alpha_{t_d^i}^\top , \quad (87a)$$

$$\pi_d = \beta_{t_d^i} - \alpha_{t_d^i} \Xi_{d+1}^{-1} \xi_{d+1} , \quad (87b)$$

$$\rho_d = \vartheta_{d+1} - \frac{1}{2} \xi_{d+1}^\top \Xi_{d+1}^{-1} \xi_{d+1} . \quad (87c)$$

This completes the proof. $\square$

Thus, we have the Bellman functions per block $\mathcal{V}_d$, and the value of the function $\mathcal{V}_0$ at the initial state x_0 at time step 0 provides, by weak duality, a lower bound on the optimal cost of Problem (1).

5.5 Simulation of an admissible control

At the end of the recursion in §5.4, we have the Bellman function $\mathcal{V}_d$ for each of the temporal blocks d . These functions are used as substitutes for the “true” Bellman functions of Problem (1) at the end of the temporal blocks. Starting from the initial state x_0 , the simulation is then performed by chaining the temporal blocks from 0 to $D - 1$ along noise scenarios. For each temporal block d , we calculate the controls minimizing the sum of the stage costs of block d and the Bellman function $\mathcal{V}_{d+1}$:

$$\min_{x,u} \mathbb{E} \left[\sum_{t \in \mathbb{T}_d} \left(\frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} + m_t^\top x_t + n_t^\top u_t + d_t \right) \right. \\ \left. + \frac{1}{2} x_{t_d^f}^\top \mathcal{P}_{d+1} x_{t_d^f} + \pi_{d+1}^\top x_{t_d^f} + \rho_{d+1} \right],$$

subject to the dynamics constraints

$$x_{t_d^i} \text{ given, } x_{t+1} = A_t x_t + B_t u_t + C_t w_{t+1} + c_t, \quad t \in \mathbb{T}_d,$$

and the linear constraints

$$\theta_t - \Theta_t u_t = 0, \quad t \in \mathbb{T}_d,$$

The solution to this problem is obtained by solving a general constrained Riccati equation over the span $\mathbb{T}_d$ as described in §1.4, with the final cost being the Bellman function $\mathcal{V}_{d+1}$. The optimal controls are then applied along the considered noise trajectory, the final state $x_{t_d^f}$ obtained at the end of the processing of temporal block d serving as the initial state $x_{t_{d+1}^i}$ for the processing of the next block $d+1$.

This simulation, performed along a set of noise trajectories obtained by Monte Carlo sampling, provides a statistical upper bound on the optimal cost of Problem (1).

6 Mixing spatial and temporal block decompositions

To conclude on the decomposition of Problem (1), we implement *simultaneously* the spatial decomposition (as in Section 3) and the temporal block decomposition (as in Section 5). This implementation is done by first applying the spatial decomposition to obtain subproblems with a low-dimensional state,⁷ then by performing the temporal block decomposition during the resolution of the subproblems within the Uzawa algorithm. We recall that $\mathbb{D} = \{0, \dots, D-1\}$ is the set of blocks of the optimization span $\mathbb{T}$, that

⁷even if this is of little importance when dealing with LQG problems...

$\mathbb{T}_d = \{dT, \dots, (d+1)T-1\}$ is the set of time steps of block $d \in \mathbb{D}$, and that we denote $t_d^i = dT$ the initial time step of block d and $t_d^f - 1 = (d+1)T-1$ the final time step of block d , with $t_d^f = t_{d+1}^i$.

6.1 Difficulty with periodicity classes

In this case, there is a practical difficulty. In the resolution by temporal block decomposition, we start by calculating the intraday functions of each periodicity class (see Remark 7). However, these intraday functions depend on the multipliers associated with the spatially coupling constraints, and these multipliers vary over time, so the intraday functions must either be calculated for all temporal blocks at each iteration of the Uzawa algorithm, or be obtained as expressions parameterized by the multiplier of the spatial coupling constraints. This latter possibility turned out to lead to complex calculations, so we preferred to duplicate the periodicity classes so that each temporal block corresponds to its own class.

Once this difficulty is overcome, the methodology to be applied consists of first treating the coupling constraint (1c) by duality (with deterministic multipliers λ_t) and of implementing, as described in Section 3, a Uzawa algorithm. At iteration k of the algorithm, knowing the value $\lambda^{(k)}$ of the multiplier, we have to solve the LQG subproblems resulting from the spatial decomposition, namely, for each $i \in \mathbb{I}$,

$$\min_{x_i, u_i} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} \begin{pmatrix} x_{i,t}^\top & u_{i,t}^\top \end{pmatrix} \begin{pmatrix} M_{i,t} & S_{i,t} \\ S_{i,t}^\top & N_{i,t} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ u_{i,t} \end{pmatrix} + m_{i,t}^\top x_{i,t} + n_{i,t}^\top u_{i,t} - \lambda_t^{(k)\top} \Theta_{i,t} u_{i,t} + d_{i,t} \right) \right. \\ \left. + \frac{1}{2} x_{i,\mathcal{T}}^\top M_{i,\mathcal{T}} x_{i,\mathcal{T}} + m_{i,\mathcal{T}}^\top x_{i,\mathcal{T}} + d_{i,\mathcal{T}} \right], \quad (88a)$$

subject to the dynamic constraints

$$x_{i,0} \text{ given}, \quad x_{i,t+1} = A_{i,t} x_{i,t} + B_{i,t} u_{i,t} + C_{i,t} w_{t+1} + c_{i,t}, \quad t \in \mathbb{T}. \quad (88b)$$

We then apply to each of these subproblems the temporal block decomposition method described in Section 5, with the difference that there is no longer coupling constraints to be taken into account in the calculation of the intraday function (since they have been dualized during the spatial decomposition). Of course, the associated duality terms must be taken into account.

6.2 Study of intraday functions and their gradients

For the i -th subproblem (88) resulting from the spatial decomposition, we introduce for $d \in \mathbb{D}$ the intraday functions $\mathcal{L}_{i,d}$ defined by

$$\mathcal{L}_{i,d}(x_{i,t_d^i}, \nu_{i,d}) = \min_{x_i, u_i} \mathbb{E} \left[\sum_{t \in \mathbb{T}_d} \left(\frac{1}{2} \begin{pmatrix} x_{i,t}^\top & u_{i,t}^\top \end{pmatrix} \begin{pmatrix} M_{i,t} & S_{i,t} \\ S_{i,t}^\top & N_{i,t} \end{pmatrix} \begin{pmatrix} x_{i,t} \\ u_{i,t} \end{pmatrix} + m_{i,t}^\top x_{i,t} + n_{i,t}^\top u_{i,t} - \lambda_t^{(k)\top} \Theta_{i,t} u_{i,t} + d_{i,t} \right) + \nu_{i,d}^\top x_{i,t_d^f} \right], \quad (89a)$$

subject to the dynamic constraints

$$x_{i,t_d^i} \text{ given}, \quad x_{i,t+1} = A_{i,t}x_{i,t} + B_{i,t}u_{i,t} + C_{i,t}w_{t+1} + c_{i,t}, \quad t \in \mathbb{T}_d, \quad (89b)$$

6.2.1 Calculation of the intraday function $\mathcal{L}_{i,d}$ by Riccati

For any pair $(i, d) \in \mathbb{I} \times \mathbb{D}$, the intraday function $\mathcal{L}_{i,d}(x_{i,t_d^i}, \nu_{i,d})$ is obtained by a recursion producing, for $t \in \mathbb{T}_d$, Bellman functions $V_{i,t}$, with

$$V_{i,t}(x_{i,t}) = \frac{1}{2} x_{i,t}^\top P_{i,t} x_{i,t} + p_{i,t}^\top x_{i,t} + r_{i,t}, \quad (90)$$

which are calculated by solving, for $t = t_d^f - 1, \dots, t_d^i$, the general Riccati equation (see §1.3)

$$P_{i,t_d^f} = 0, \quad P_{i,t} = G_{i,t+1} - E_{i,t+1}^\top F_{i,t+1}^{-1} E_{i,t+1}, \quad (91a)$$

$$p_{i,t_d^f} = \nu_{i,d}, \quad p_{i,t} = g_{i,t+1} - E_{i,t+1}^\top F_{i,t+1}^{-1} f_{i,t+1}, \quad (91b)$$

$$r_{i,t_d^f} = 0, \quad r_{i,t} = h_{i,t+1} - \frac{1}{2} f_{i,t+1}^\top F_{i,t+1}^{-1} f_{i,t+1}, \quad (91c)$$

with the notations

$$E_{i,t+1} = S_{i,t}^\top + B_{i,t}^\top P_{i,t+1} A_{i,t}, \quad (92a)$$

$$F_{i,t+1} = N_{i,t} + B_{i,t}^\top P_{i,t+1} B_{i,t}, \quad (92b)$$

$$G_{i,t+1} = M_{i,t} + A_{i,t}^\top P_{i,t+1} A_{i,t}, \quad (92c)$$

and

$$f_{i,t+1} = n_{i,t} - \Theta_{i,t}^\top \lambda_t^{(k)} + B_{i,t}^\top p_{i,t+1} + B_{i,t}^\top P_{i,t+1} c_{i,t}, \quad (93a)$$

$$g_{i,t+1} = m_{i,t} + A_{i,t}^\top p_{i,t+1} + A_{i,t}^\top P_{i,t+1} c_{i,t}, \quad (93b)$$

$$h_{i,t+1} = \frac{1}{2} \text{trace}(\Sigma_{t+1} C_{i,t}^\top P_{i,t+1} C_{i,t}) + \frac{1}{2} c_{i,t}^\top P_{i,t+1} c_{i,t} + p_{i,t+1}^\top c_{i,t} + d_{i,t} + r_{i,t+1}. \quad (93c)$$

It should be noted that, for $t \in \mathbb{T}_d$, the Bellman functions $V_{i,t}$ depend *implicitly* on both the multipliers $(\lambda_{t_d^i}^i, \dots, \lambda_{t_d^f-1}^i)$ associated with the spatial constraint *and* the multiplier $\nu_{i,d}$ associated with the temporal constraint. For $t \in \mathbb{T}_d$, the associated optimal control is given by

$$u_{i,t}^* = -F_{i,t+1}^{-1} E_{i,t+1} x_{i,t} - F_{i,t+1}^{-1} f_{i,t+1} . \quad (94)$$

Under the assumption of independence of the noises, we have the relation

$$\mathcal{L}_{i,d}(x_{i,t_d^i}, \nu_{i,d}) = V_{i,t_d^i}(x_{i,t_d^i}) .$$

6.2.2 Dependence on $\nu_{i,d}$ of the intraday function $\mathcal{L}_{i,d}$

As in §5.3.2, we are interested in the dependence of the intraday functions $\mathcal{L}_{i,d}$ on the multiplier $\nu_{i,d}$ of the temporal block decomposition.

Dependence on $\nu_{i,d}$ of the matrices $P_{i,t}$. The matrices $P_{i,t}$ do not depend on $\nu_{i,d}$. Indeed, the matrix $P_{i,t_d^f} = 0$ does not depend on $\nu_{i,d}$. The same applies to the matrices $P_{i,t}$ in (91a) since the matrices $E_{i,t+1}$, $F_{i,t+1}$ and $G_{i,t+1}$ in (92) do not depend on $\nu_{i,d}$.

Dependence on $\nu_{i,d}$ of the vectors $p_{i,t}$. The vectors $p_{i,t}$ are of the form:

$$p_{i,t} = \alpha_{i,t} \nu_{i,d} + \beta_{i,t} . \quad (95)$$

Indeed, the vector $p_{i,t_d^f} = \nu_{i,d}$ is of the form (95), with $\alpha_{i,t_d^f} = \text{I}$ and $\beta_{i,t_d^f} = 0$. Suppose that $p_{i,t+1}$ is also of this form. Then, the vectors $f_{i,t+1}$ and $g_{i,t+1}$ in (93) depend linearly on $\nu_{i,d}$, namely

$$f_{i,t+1} = \underbrace{B_{i,t}^\top \alpha_{i,t+1}}_{\hat{\alpha}_{i,t+1}} \nu_{i,d} + \underbrace{n_{i,t} - \Theta_{i,t}^\top \lambda_t^{(k)} + B_{i,t}^\top \beta_{i,t+1} + B_{i,t}^\top P_{i,t+1} c_{i,t}}_{\hat{\beta}_{i,t+1}} , \quad (96a)$$

$$g_{i,t+1} = \underbrace{A_{i,t}^\top \alpha_{i,t+1}}_{\tilde{\alpha}_{i,t+1}} \nu_{i,d} + \underbrace{m_{i,t} + A_{i,t}^\top \beta_{i,t+1} + A_{i,t}^\top P_{i,t+1} c_{i,t}}_{\tilde{\beta}_{i,t+1}} , \quad (96b)$$

and thus

$$p_{i,t} = (\tilde{\alpha}_{i,t+1} - E_{i,t+1}^\top F_{i,t+1}^{-1} \hat{\alpha}_{i,t+1}) \nu_{i,d} + (\tilde{\beta}_{i,t+1} - E_{i,t+1}^\top F_{i,t+1}^{-1} \hat{\beta}_{i,t+1}) .$$

from which we deduce that $p_{i,t}$ in (91b) is of the form (95), with

$$\alpha_{i,t} = \tilde{\alpha}_{i,t+1} - E_{i,t+1}^\top F_{i,t+1}^{-1} \hat{\alpha}_{i,t+1} , \quad (97a)$$

$$\beta_{i,t} = \tilde{\beta}_{i,t+1} - E_{i,t+1}^\top F_{i,t+1}^{-1} \hat{\beta}_{i,t+1} . \quad (97b)$$

Dependence on $\nu_{i,d}$ of the scalars $r_{i,t}$. The scalars $r_{i,t}$ are of the form:

$$r_{i,t} = \frac{1}{2} \nu_{i,d}^\top \Delta_{i,t} \nu_{i,d} + \delta_{i,t}^\top \nu_{i,d} + \epsilon_{i,t} . \quad (98)$$

Indeed, the scalar $r_{i,t_d^f} = 0$ is of the form (98), with $\Delta_{i,t_d^f} = 0$, $\delta_{i,t_d^f} = 0$ and $\epsilon_{i,t_d^f} = 0$. Suppose that $r_{i,t+1}$ is also of this form. Then, using the expression of $p_{i,t+1}$ obtained previously, the scalar $h_{i,t+1}$ in (93) depends on $\nu_{i,d}$ quadratically, as

$$\begin{aligned} h_{i,t+1} = & \frac{1}{2} \nu_{i,d}^\top \Delta_{i,t+1} \nu_{i,d} + \underbrace{(\delta_{i,t+1}^\top + c_{i,t}^\top \alpha_{i,t+1})}_{\hat{\delta}_{i,t+1}^\top} \nu_{i,d} \\ & + \underbrace{(\epsilon_{i,t+1} + d_{i,t} + c_{i,t}^\top \beta_{i,t+1} + \frac{1}{2} c_{i,t}^\top P_{i,t+1} c_{i,t} + \frac{1}{2} \text{trace}(\Sigma_{t+1} C_{i,t}^\top P_{i,t+1} C_{i,t}))}_{\hat{\epsilon}_{i,t+1}} . \end{aligned}$$

Using the relation $f_{i,t+1} = \hat{\alpha}_{i,t+1} \nu_{i,d} + \hat{\beta}_{i,t+1}$ in (96), we obtain that $r_{i,t}$ in (91c) is of the form (98), with

$$\Delta_{i,t} = \Delta_{i,t+1} - \hat{\alpha}_{i,t+1}^\top F_{i,t+1}^{-1} \hat{\alpha}_{i,t+1} , \quad (99a)$$

$$\delta_{i,t} = \hat{\delta}_{i,t+1} - \hat{\alpha}_{i,t+1}^\top F_{i,t+1}^{-1} \hat{\beta}_{i,t+1} , \quad (99b)$$

$$\epsilon_{i,t} = \hat{\epsilon}_{i,t+1} - \frac{1}{2} \hat{\beta}_{i,t+1}^\top F_{i,t+1}^{-1} \hat{\beta}_{i,t+1} . \quad (99c)$$

Expressions of the intraday function. As in Section 5, we conclude that the intraday function $\mathcal{L}_{i,d}$, which is quadratic in its first variable x_{i,t_d^i} ,

$$\mathcal{L}_{i,d}(x_{i,t_d^i}, \nu_i) = \frac{1}{2} x_{i,t_d^i}^\top P_{i,t_d^i} x_{i,t_d^i} + p_{i,t_d^i}^\top x_{i,t_d^i} + r_{i,t_d^i} ,$$

is also quadratic in its second variable $\nu_{i,d}$,

$$\mathcal{L}_{i,d}(x_{i,t_d^i}, \nu_{i,d}) = \frac{1}{2} \nu_{i,d}^\top \Delta_{i,t_d^i} \nu_{i,d} + (\delta_{i,t_d^i}^\top + x_{i,t_d^i}^\top \alpha_{i,t_d^i}) \nu_{i,d} + (\epsilon_{i,t_d^i} + \beta_{i,t_d^i}^\top x_{i,t_d^i} + \frac{1}{2} x_{i,t_d^i}^\top P_{i,t_d^i} x_{i,t_d^i}) . \quad (100)$$

We deduce that it is quadratic in the couple $(x_{i,t_d^i}, \nu_{i,d})$, with a cross term $x_{i,t_d^i}^\top \alpha_{i,t_d^i} \nu_{i,d}$.

6.2.3 Gradients of the intraday function $\mathcal{L}_{i,d}$

The calculation of the gradients of the intraday function $\mathcal{L}_{i,d}$, $i \in \mathbb{I}$ and $d \in \mathbb{D}$, follows the same scheme as that seen in §3.3.2, by recursion on the set $\mathbb{T}_d$ of time steps of block d and taking into account the dependence on the multiplier $\nu_{i,d}$. In the case studied here, the optimal control is given by (94) which, thanks to the expression of $f_{i,t+1}$ in (96), takes the form

$$u_{i,t}^* = \Gamma_{i,t} x_{i,t} + \Psi_{i,t} \nu_{i,d} + \gamma_{i,t} , \quad (101)$$

with $\Gamma_{i,t} = -F_{t+1}^{-1} E_{t+1}$, $\Psi_{i,t} = -F_{t+1}^{-1} \hat{\alpha}_{i,t+1}$ and $\gamma_{i,t} = -F_{t+1}^{-1} \hat{\beta}_{i,t+1}$.

Proposition 9. For every subproblem $i \in \mathbb{I}$ and for every block $d \in \mathbb{D}$, the partial gradients of the Bellman functions $V_{i,t}$ in (90) are, for $t = t_d^i, \dots, t_d^f$, of the form

$$\nabla_{\lambda_\tau} V_{i,t}(x_{i,t}) = 0, \quad \tau = t_d^i, \dots, t-1, \quad (102a)$$

$$\nabla_{\lambda_\tau} V_{i,t}(x_{i,t}) = \alpha_{i,\tau,t} x_{i,t} + \sigma_{i,\tau,t} \nu_{i,d} + \beta_{i,\tau,t}, \quad \tau = t, \dots, t_d^f - 1. \quad (102b)$$

where the matrices $\alpha_{i,\tau,t}$ and the vectors $\beta_{i,\tau,t}$ are calculated by the recursions (105).

Proof. The demonstration is based on the results of [6] and is done by recursion. At time step t_d^f , the Bellman function does not depend on the parameter λ , so that

$$\nabla_{\lambda_\tau} V_{i,t_d^f}(x_{i,t_d^f}) = 0, \quad \tau = t_d^i, \dots, t_d^f - 1, \quad (103)$$

and thus of the form (102), with $\alpha_{i,\tau,t_d^f} = 0$, $\sigma_{i,\tau,t_d^f} = 0$ and $\beta_{i,\tau,t_d^f} = 0$. Suppose that at time step $t+1$, the partial gradients $\nabla_{\lambda_\tau} V_{i,t+1}$ are of the form (102).

- Since the Bellman function $V_{i,t}$ does not depend on the multipliers λ_τ for τ strictly before t , we have

$$\nabla_{\lambda_\tau} V_{i,t}(x_{i,t}) = 0, \quad \tau = t_d^i, \dots, t-1. \quad (104a)$$

- The Bellman function $V_{i,t}$ depending on λ_t only through the instantaneous cost $L_{i,t}$ and its term $\lambda_t^\top \Theta_{i,t} u_{i,t}$, we deduce from (101) that

$$\begin{aligned} \nabla_{\lambda_t} V_{i,t}(x_{i,t}) &= -\Theta_{i,t} u_{i,t}^*, \\ &= -\Theta_{i,t} \Gamma_{i,t} x_{i,t} - \Theta_{i,t} \Psi_{i,t} \nu_{i,d} - \Theta_{i,t} \gamma_{i,t}. \end{aligned} \quad (104b)$$

- The Bellman function $V_{i,t}$ depending on λ_τ only through the function $V_{i,t+1}$ and the dynamics $f_{i,t}$ for $\tau = t+1, \dots, t_d^f - 1$, we deduce that

$$\begin{aligned} \nabla_{\lambda_\tau} V_{i,t}(x_{i,t}) &= \mathbb{E} \left[\nabla_{\lambda_\tau} V_{i,t+1} (A_{i,t} x_{i,t} + B_{i,t} u_{i,t}^* + C_{i,t} w_{t+1} + c_{i,t}) \right], \\ &= \alpha_{i,\tau,t+1} (A_{i,t} + B_{i,t} \Gamma_{i,t}) x_{i,t} + (\alpha_{i,\tau,t+1} B_{i,t} \Psi_{i,t} + \sigma_{i,\tau,t+1}) \nu_{i,d} \\ &\quad + (\alpha_{i,t_d^f,t+1} (B_{i,t} \gamma_{i,t} + c_{i,t}) + \beta_{i,t_d^f,t+1}). \end{aligned} \quad (104c)$$

From this, we deduce the recurrence relations

$$\alpha_{i,\tau,t} = 0, \quad \tau = t_d^i, \dots, t-1, \quad (105a)$$

$$\alpha_{i,t,t} = -\Theta_{i,t} \Gamma_{i,t}, \quad (105b)$$

$$\alpha_{i,\tau,t} = \alpha_{i,\tau,t+1} (A_{i,t} + B_{i,t} \Gamma_{i,t}), \quad \tau = t+1, \dots, t_d^f - 1, \quad (105c)$$

$$\sigma_{i,\tau,t} = 0, \quad \tau = t_d^i, \dots, t-1, \quad (105d)$$

$$\sigma_{i,t,t} = -\Theta_{i,t} \Psi_{i,t}, \quad (105e)$$

$$\sigma_{i,\tau,t} = \sigma_{i,\tau,t+1} + \alpha_{i,\tau,t+1} B_{i,t} \Psi_{i,t}, \quad \tau = t+1, \dots, t_d^f - 1, \quad (105f)$$

$$\beta_{i,\tau,t} = 0, \quad \tau = t_d^i, \dots, t-1, \quad (105g)$$

$$\beta_{i,t,t} = -\Theta_{i,t} \gamma_{i,t}, \quad (105h)$$

$$\beta_{i,\tau,t} = \beta_{i,\tau,t+1} + \alpha_{i,\tau,t+1} (B_{i,t} \gamma_{i,t} + c_{i,t}), \quad \tau = t+1, \dots, t_d^f-1, \quad (105i)$$

which completes the proof. $\square$

The intraday function $\mathcal{L}_{i,d}$ being equal to the Bellman function V_{i,t_d^i} , we deduce the expression of its gradients with respect to the multipliers λ_t , $t \in \mathbb{T}$:

$$\nabla_{\lambda_t} \mathcal{L}_{i,d}(x_{i,t_d^i}, \nu_{i,d}) = \alpha_{i,t,t_d^i} x_{i,t_d^i} + \sigma_{i,t,t_d^i} \nu_{i,d} + \beta_{i,t,t_d^i}, \quad \forall t \in \mathbb{T}_d. \quad (106)$$

6.3 Bellman recursion by block

Once the intraday functions $\mathcal{L}_{i,d}$ are computed, the Bellman recursion to obtain the Bellman functions of the temporal blocks by subproblem is identical to that described in §5.4.

6.3.1 Bellman functions by block

The Bellman functions by temporal block $\mathcal{V}_{i,d}$ are given by

$$\mathcal{V}_{i,D}(x_{i,\mathcal{T}}) = \frac{1}{2} x_{i,\mathcal{T}}^\top M_{i,\mathcal{T}} x_{i,\mathcal{T}} + m_{i,\mathcal{T}}^\top x_{i,\mathcal{T}} + d_{i,\mathcal{T}}, \quad (107a)$$

and, for $d = D-1 \dots, 0$,

$$\begin{aligned} \mathcal{V}_{i,d}(x_{i,t_d^i}) &= \max_{\nu_{i,d}} \left(\mathcal{L}_{i,d}(x_{i,t_d^i}, \nu_{i,d}) - (\mathcal{V}_{i,d+1})^*(\nu_{i,d}) \right), \\ &= \max_{\nu_{i,d}} \min_{x_{i,t_{d+1}^i}} \left(\mathcal{L}_{i,d}(x_{i,t_d^i}, \nu_{i,d}) - \nu_d^\top x_{i,t_{d+1}^i} + \mathcal{V}_{d+1}(x_{i,t_{d+1}^i}) \right). \end{aligned} \quad (107b)$$

The Bellman functions $\mathcal{V}_{i,d}$ are of the form

$$\mathcal{V}_{i,d}(x_{i,t_d^i}) = \frac{1}{2} x_{i,t_d^i}^\top \mathcal{P}_{i,d} x_{i,t_d^i} + \pi_{i,d}^\top x_{i,t_d^i} + \rho_{i,d}, \quad (108a)$$

and the optimal controls in (107b) are given by

$$x_{i,t_{d+1}^i}^* = \mathcal{P}_{i,d+1}^{-1} \nu_{i,d}^* - \mathcal{P}_{i,d+1}^{-1} \pi_{i,d+1}, \quad (108b)$$

$$\nu_{i,d}^* = -\Xi_{i,d+1}^{-1} \alpha_{i,t_d^i}^\top x_{i,t_d^i} - \Xi_{i,d+1}^{-1} \xi_{i,d+1}, \quad (108c)$$

(see §5.4 for these calculations and the associated notations).

6.3.2 Gradients of the Bellman functions by block

We introduce new notations to simplify the writings. With

$$\begin{aligned}\Omega_{i,d+1} &= -\Xi_{i,d+1}^{-1} \alpha_{i,t_d^i}^\top, \\ \omega_{i,d+1} &= -\Xi_{i,d+1}^{-1} \xi_{i,d+1},\end{aligned}$$

the optimal controls (108b) and (108c) take the form

$$x_{i,t_{d+1}^i}^* = \mathcal{P}_{i,d+1}^{-1} \Omega_{i,d+1} x_{i,t_d^i} + \mathcal{P}_{i,d+1}^{-1} (\omega_{i,d+1} - \pi_{i,d+1}), \quad (109a)$$

$$\nu_{i,d}^* = \Omega_{i,d+1} x_{i,t_d^i} + \omega_{i,d+1}. \quad (109b)$$

Then, the gradients of the intraday function $\mathcal{L}_{i,d}$ in (106) evaluated at $\nu_{i,d}^*$ are

$$\begin{aligned}\nabla_{\lambda_t} \mathcal{L}_{i,d}(x_{i,t_d^i}, \nu_{i,d}^*) &= \alpha_{i,t,t_d^i} x_{i,t_d^i} + \sigma_{i,t,t_d^i} \nu_{i,d}^* + \beta_{i,t,t_d^i}, \\ &= (\alpha_{i,t,t_d^i} + \sigma_{i,t,t_d^i} \Omega_{i,d+1}) x_{i,t_d^i} + (\beta_{i,t,t_d^i} + \sigma_{i,t,t_d^i} \omega_{i,d+1}).\end{aligned} \quad (110)$$

The block Bellman gradients are given by the following proposition.

Proposition 10. *The partial gradients with respect to λ of the Bellman functions $\mathcal{V}_{i,d}$ are, for $d = 0, \dots, D$, given by*

$$\nabla_{\lambda_t} \mathcal{V}_{i,d}(x_{i,t_d^i}) = \mu_{i,t,d} x_{i,t_d^i} + \eta_{i,t,d}, \quad t \in \mathbb{T}. \quad (111)$$

where the matrices $\mu_{i,t,d}$ and the vectors $\eta_{i,t,d}$ are calculated by the recursions (114).

Proof. The proof is by induction. For $d = D$, the Bellman function (107a) does not depend in the parameter λ , so that

$$\nabla_{\lambda_t} \mathcal{V}_{i,D}(x_{i,\tau}) = 0, \quad t \in \mathbb{T}, \quad (112)$$

and thus of the form (111), with $\mu_{i,t,D} = 0$ and $\eta_{i,t,D} = 0$. Assume that, at block $d+1$, the partial gradients $\nabla_{\lambda_t} \mathcal{V}_{i,d+1}$ are of the form (111).

- Since the Bellman function $\mathcal{V}_{i,d}$ does not depend on the multipliers λ_t of the time steps t of the blocks d' strictly prior to d , we have

$$\nabla_{\lambda_t} \mathcal{V}_{i,d}(x_{i,t_d^i}) = 0, \quad t \in \mathbb{T}_{d'}, \quad d' = 0, \dots, d-1. \quad (113a)$$

- Following (107b), the Bellman function $\mathcal{V}_{i,d}$ depends on the multipliers λ_t , $t \in \mathbb{T}_d$, only by the intraday function $\mathcal{L}_{i,d}$, and we deduce from (110) that

$$\begin{aligned}\nabla_{\lambda_t} \mathcal{V}_{i,d}(x_{i,t_d^i}) &= \nabla_{\lambda_t} \mathcal{L}_{i,d}(x_{i,t_d^i}, \nu_{i,d}^*), \\ &= (\alpha_{i,t,t_d^i} + \sigma_{i,t,t_d^i} \Omega_{i,d+1}) x_{i,t_d^i} + (\beta_{i,t,t_d^i} + \sigma_{i,t,t_d^i} \omega_{i,d+1}).\end{aligned} \quad (113b)$$

- Following (107b), the Bellman function $\mathcal{V}_{i,d}$ depends on the multipliers λ_t of the time steps t of the blocks d' strictly posterior to d only through the function $\mathcal{V}_{i,d+1}$, hence

$$\begin{aligned}\nabla_{\lambda_t} \mathcal{V}_{i,d}(x_{i,t_d^i}) &= \nabla_{\lambda_t} \mathcal{V}_{i,d+1}(x_{i,t_{d+1}^i}^*) , \\ &= \mu_{i,t,d+1} \mathcal{P}_{i,d+1}^{-1} \Omega_{i,d+1} x_{i,t_d^i} + \mu_{i,t,d+1} \mathcal{P}_{i,d+1}^{-1} (\omega_{i,d+1} - \pi_{i,d+1}) + \eta_{i,t,d+1} ,\end{aligned}\tag{113c}$$

for $t \in \mathbb{T}_{d'}$, $d' = d+1, \dots, D-1$.

We deduce the recurrence relations

$$\mu_{i,t,d} = 0 , \quad t \in \mathbb{T}_{d'} , \quad d' = 0, \dots, d-1 , \quad (114a)$$

$$\mu_{i,t,d} = \alpha_{i,t,t_d^i} + \sigma_{i,t,t_d^i} \Omega_{i,d+1} , \quad t \in \mathbb{T}_d , \quad (114b)$$

$$\mu_{i,t,d} = \mu_{i,t,d+1} \mathcal{P}_{i,d+1}^{-1} \Omega_{i,d+1} , \quad t \in \mathbb{T}_{d'} , \quad d' = d+1, \dots, D-1 , \quad (114c)$$

$$\eta_{i,t,d} = 0 , \quad t \in \mathbb{T}_{d'} , \quad d' = 0, \dots, d-1 , \quad (114d)$$

$$\eta_{i,t,d} = \beta_{i,t,t_d^i} + \sigma_{i,t,t_d^i} \omega_{i,d+1} , \quad t \in \mathbb{T}_d , \quad (114e)$$

$$\eta_{i,\tau,t} = \eta_{i,t,d+1} + \mu_{i,t,d+1} \mathcal{P}_{i,d+1}^{-1} (\omega_{i,d+1} - \pi_{i,d+1}) , \quad t \in \mathbb{T}_{d'} , \quad d' = d+1, \dots, D-1 , \quad (114f)$$

which completes the proof. $\square$

6.4 Uzawa algorithm

As in §3.2, we implement the Uzawa algorithm. At iteration k of the algorithm, knowing the values $\lambda_t^{(k)}$ of the multipliers, it is necessary to

1. compute the intraday functions $\mathcal{L}_{i,d}$ and the Bellman functions $\mathcal{V}_{i,0}$ for all $i \in \mathbb{I}$, the optimal controls of subproblem i being denoted $u_{i,t}^{(k)}$,
2. update the multipliers by a gradient-type formula:

$$\lambda_t^{(k+1)} = \lambda_t^{(k)} + \rho \mathbb{E} \left[\theta_t - \sum_{i \in \mathbb{I}} \Theta_{i,t} u_{i,t}^{(k)} \right] , \quad t \in \mathbb{T} .$$

As in §3.3, updating the multipliers λ_t by the gradient formula requires each subproblem i to compute the terms $\mathbb{E}[\Theta_{i,t} u_{i,t}^{(k)}]$. This calculation is performed using the method presented in §6.2 and §6.3, which involves computing the gradients with respect to λ_t using an adapted Bellman recursion from [6]. This method yields

$$\mathbb{E}[\Theta_{i,t} u_{i,t}^{(k)}] = -\nabla_{\lambda_t} \mathcal{V}_{i,0}(x_{i,0}) , \tag{115a}$$

$$= -\mu_{i,t,0} x_{i,0} - \eta_{i,t,0} . \tag{115b}$$

6.5 Simulation of an admissible control

Finally, the simulation of an admissible control is done, first of all, by constituting, in the same spirit as that of §3.4, surrogate Bellman functions $\tilde{\mathcal{V}}_d$ by summing the Bellman functions obtained for each temporal block in the subproblems, namely

$$\tilde{\mathcal{V}}_d(x_{1,t_d^i}, \dots, x_{I,t_d^i}) = \sum_{i=1}^I \mathcal{V}_{i,d}(x_{i,t_d^i}) ,$$

then proceeding as described in §5.5, with the surrogate functions $\tilde{\mathcal{V}}_d$ replacing the functions $\mathcal{V}_d$.

7 Numerical results

We implement the different decomposition methods exposed in the preceding sections on a toy problem.

7.1 Problem formulation

We consider a linear quadratic Gaussian problem with a state of dimension 4 and a control of dimension 4. The optimization span contains 3 temporal blocks, each containing 24 time steps (i.e., 3 days), for a total of $\mathcal{T} = 72$ time steps. The dynamics correspond to the evolution of 4 dams which are given by the equations

$$\begin{aligned} x_{1,0} &= 2 , \quad x_{1,t+1} = x_{1,t} - u_{1,t} + c_{1,t} + w_{1,t+1} , & t \in \mathbb{T} , \\ x_{2,0} &= 3 , \quad x_{2,t+1} = x_{2,t} - u_{2,t} + c_{2,t} + w_{2,t+1} , & t \in \mathbb{T} , \\ x_{3,0} &= 4 , \quad x_{3,t+1} = x_{3,t} + u_{2,t} - u_{3,t} + c_{3,t} + w_{3,t+1} , & t \in \mathbb{T} , \\ x_{4,0} &= 5 , \quad x_{4,t+1} = x_{4,t} - u_{4,t} + c_{4,t} + w_{4,t+1} , & t \in \mathbb{T} , \end{aligned}$$

where all coefficients $c_{j,t}$ are equal to 1 and where the noises $w_{j,t+1}$ are Gaussian with zero mean and standard deviation 0.01. The coupling constraints are

$$5 - u_{1,t} - u_{2,t} - u_{3,t} - u_{4,t} = 0 , \quad t \in \mathbb{T} .$$

The final cost is

$$\sum_{j=1}^4 \left(\frac{1}{2} x_{j,72}^2 - x_{j,0} x_{j,72} \right) ,$$

and penalizes the deviations between the final states and the initial states. The instantaneous cost are

$$\sum_{j=1}^4 \left(\frac{1}{10} x_{j,t}^2 + \frac{1}{10} u_{j,t}^2 + n_{j,t} u_{j,t} + d_{j,t} \right) , \quad t \in \mathbb{T} ,$$

where the coefficients $n_{j,t}$ fluctuate around the value -1 for $j = 1, \dots, 3$ and around the value -1.5 for $j = 4$. The coefficients $d_{j,t}$ are chosen to be constant over time. The coefficients $n_{j,t}$ vary in the same way each day, such that the 3 temporal blocks of the optimization span belong to a single periodicity class.

This problem consists of 3 independent subproblems of respective dimensions 1, 2 and 1 linked by the coupling constraint. The first subproblem is associated with dam 1, the second subproblem with dams 2 and 3 which are in series, and the third subproblem is the one corresponding to dam 4. The global problem is therefore of the same nature as Problem (1).

7.2 Remarks on the problem data

It should be noted that the constant coefficients $c_{j,t}$ in the dynamics, all equal to the value 1, serve as the mean for the noises in the dynamics of the subproblems and can be interpreted as the expected inputs into the dams, i.e., a global input of 72×4 in expectation. As in Subproblem 2, the water from the first dam is turbinated twice (first by the first dam, then by the second dam) and since we have included in each subproblem a final cost penalizing the deviation between the final state and the initial state of the dams, we deduce that the global quantity turbinated in the problem must be on average $72 \times 5 = 360$. The coupling constraint between the dams requires that the sum of the turbinated quantities of the dams be equal to 5 at each time step, and is therefore compatible with the inputs and the constraints on the final state. Finally, the coefficients $d_{j,t}$ in the instantaneous cost are all equal, the value being adjusted so that the optimal cost of the original problem is equal to 100.00.

7.3 Global resolution

We compute the solution of the global problem by solving a constrained Riccati equation (see Section 2). The solution of the Riccati equation is done very quickly, and the simulation of the optimal control along 10^6 noise scenarios takes a few seconds. The agreement between the Bellman cost and the simulation cost is almost perfect. The optimal solution of the problem is described by Figure 1 and the optimal cost of this solution is 100.00.

7.4 Spatial decomposition with deterministic multipliers

We implement the methodology presented in Section 3.

7.4.1 Convergence of the Uzawa algorithm and lower bound

The Uzawa algorithm converges in just over 5000 fixed-step gradient iterations, the stopping criterion being that the norm of the gradient is less than 10^{-3} . Figure 2 illustrates the decrease in the norm of the gradient of the dual function during the iterations, as well as the satisfaction *in expectation* of the coupling constraint at convergence, the almost-sure

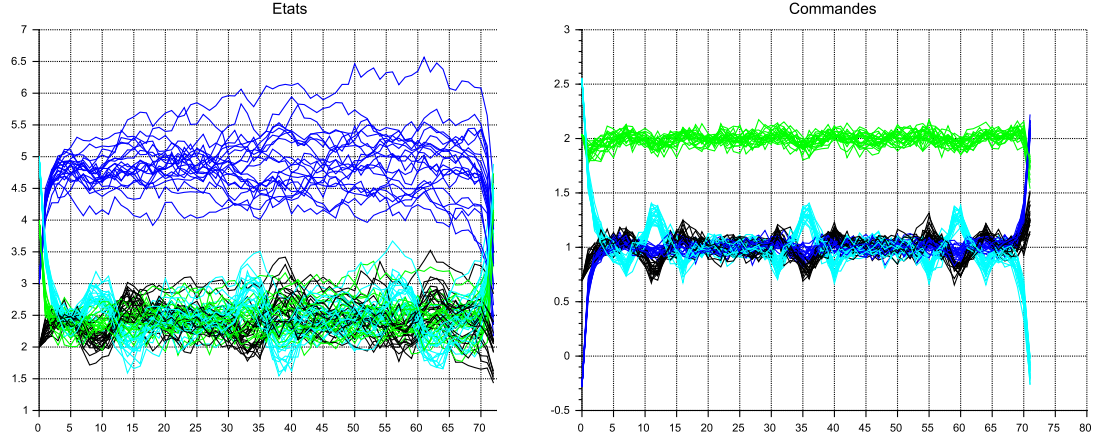


Figure 1: Solution of the global problem

constraint being of course not satisfied: this is what we expect given that we limited ourselves to deterministic multipliers. The fact that the gradient is calculated exactly and not by Monte Carlo explains the linear decrease of the logarithm of the norm of the gradient. The maximization of the dual cost by Uzawa leads to a lower bound of around 97.24.

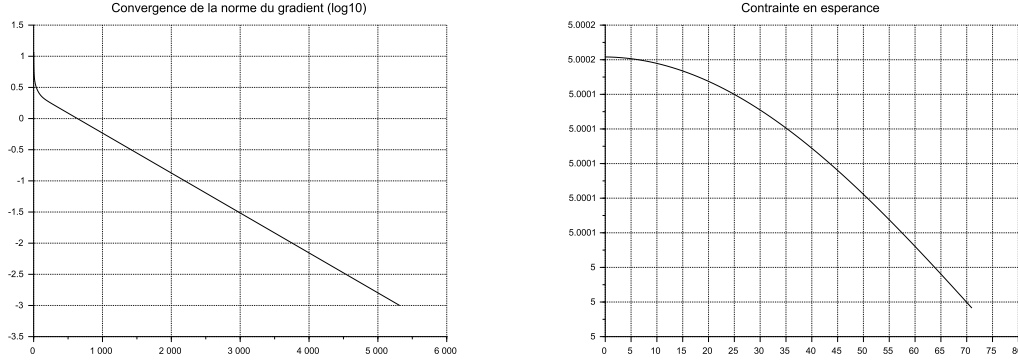


Figure 2: Convergence of the norm of the gradient (left) and expected value of the coupling constraint (right) in space decomposition

Remark 8. *It is instructive to compare the convergence of the Uzawa algorithm in the case where the gradients are computed by a Bellman recursion described in Figure 2 with that in the case where the gradients are computed by Monte Carlo simulation, described in Figure 3.*

We see that, for a nearly identical number of iterations, around 5000, the convergence in the Monte Carlo method (with 10^4 noise scenarios) is less precise than in the Bellman

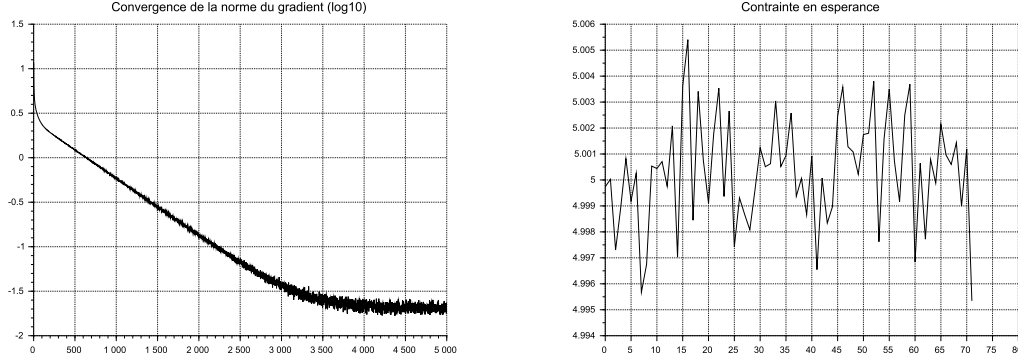


Figure 3: Convergence of the norm of the gradient (left) and expected value of the coupling constraint (right) in space decomposition where gradients are computed by Monte Carlo

method. Up to iteration 3000, the methods are similar in terms of gradient decrease, but the Monte Carlo method then exhibits oscillations linked to the fact that the gradient is computed statistically and therefore approximately. However, the lower and upper bounds of the optimal cost are very close for the two methods, as are the trajectories in simulation. Of course, the computation of gradients by Bellman recursion is much faster than that by Monte Carlo simulation

7.4.2 Obtaining an admissible solution and upper bound

Once the convergence of the Uzawa algorithm is obtained, we can construct surrogate Bellman functions for the global problem by adding the Bellman functions of the subproblems obtained by the algorithm (see §3.4). We can then simulate the global problem by minimizing at each time step t the instantaneous cost at t of the problem and the future cost at $t+1$ given by the substitute Bellman function. This simulation is done with the same 10^6 noise scenarios as those used for the simulation of the global resolution of §7.3. We thus obtain an admissible solution of the global problem described by Figure 4. The cost of this admissible solution is an upper bound on the optimal cost of the global problem, whose value is 100.06, that is, is almost identical to the optimal cost of the problem.

Remark 9. *If we limit ourselves to deterministic spatial multipliers that are constant per temporal block, convergence occurs in less than 150 iterations, and the simulation cost is 105.55, which is a limited degradation compared to the optimal cost. However, the lower bound returned by the algorithm is -164.32 , which provides a very poor lower bound on the optimal cost of the problem.*

To better understand such degradation of the lower bound when using multipliers constant per temporal block, we study the influence on the convergence of the spatial decomposition algorithm of the size $\#T$ of the temporal intervals over which we impose the multiplier to be constant:

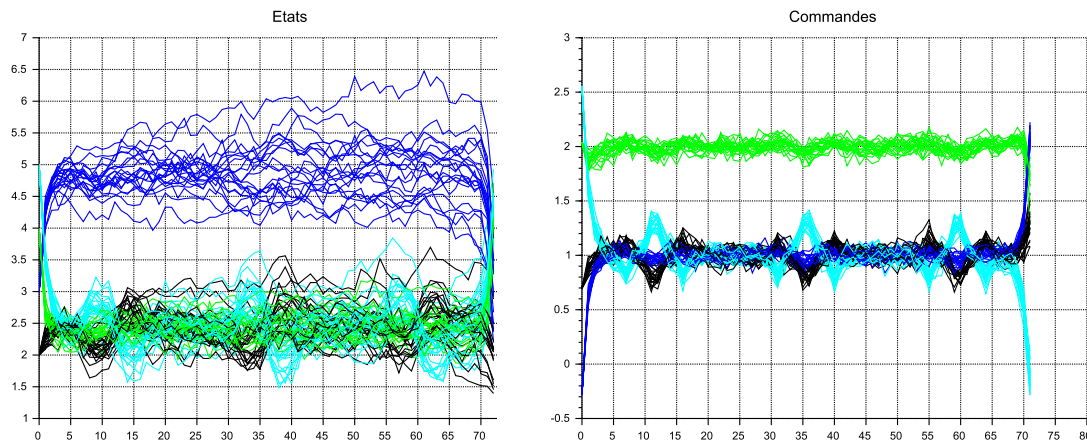


Figure 4: Feasible solution obtained by space decomposition with deterministic multipliers

- $\#T = 1$ corresponds to the multiplier per time step (solution in Figure 4),
- $\#T = 24$ corresponds to the multiplier per temporal block,
- $\#T = 72$ corresponds to the multiplier constant over the entire time span.

We provide in Table 1 the dual cost, the feasible cost as well as the number of iterations necessary for convergence for different values of $\#T$ (length of the time block).⁸ It can be

$\#T$	Dual Cost	Feasible Cost	#iter.	ρ
1	97.24	100.06	5317	0.10
2	64.93	100.23	4002	0.12
3	26.86	100.65	2797	0.15
6	-60.11	102.41	1125	0.25
12	-128.00	104.40	347	0.50
24	-164.32	105.55	116	1.00
72	-188.63	106.32	26	3.00

Table 1: Spatial decomposition with constant multiplier per interval

observed that, except for a judicious choice of intervals on which the multiplier is forced to be constant, such an approach may severely degrade the lower bound (dual cost) returned by the decomposition method. On the other hand, the upper bound (feasible cost) seems to be less affected.

⁸The last column of the table gives for reference the value of the step used in the gradient method.

7.5 Spatial decomposition with AR(1) multipliers

We implement the methodology presented in Section 4.

7.5.1 Uzawa algorithm with a general AR(1) process

We first consider the most general case where the parameters Ξ_t , ξ_t and Ω_t the AR(1) model depend on the time t . After 7500 fixed-step gradient iterations, the Uzawa algorithm stabilizes, as illustrated by Figure 5. Note however that the magnitude of the gradient norm remains relatively large, on the order 10^{-2} , whereas the stopping criterion is that this norm is less than 10^{-3} . If we perform more (and even many more) iterations, the gradient norm continues to decrease and the dual cost to increase, but extremely slowly. This seems to be due to the parameter Ξ , for which very small gradient steps must be used to avoid divergence.

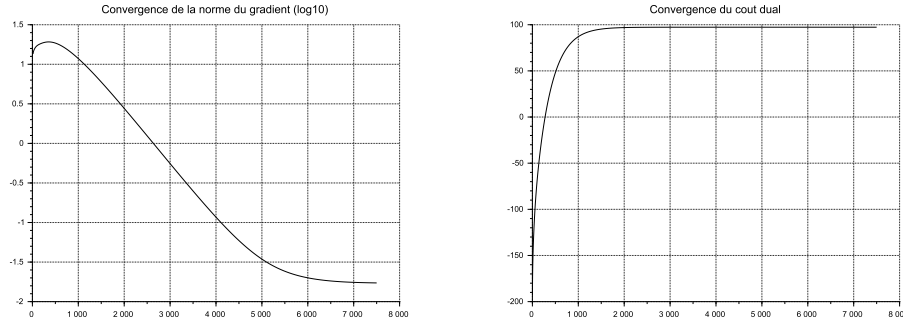


Figure 5: Convergence of the norm of the gradient (left) and of the dual cost (right) in space decomposition when using standard AR(1) multipliers

The value of the best dual cost obtained by using such AR(1) multipliers is 97.32, that is, very slightly higher than the value 97.24 obtained by using deterministic multipliers.

7.5.2 Uzawa algorithm with stationary matrix Ξ

We then consider the case where the parameters Ξ_t are stationary, that is, $\Xi_t = \Xi$ for all $t \in \mathbb{T}$. After 1500 fixed-step gradient iterations, the Uzawa algorithm stabilizes, as illustrated by Figure 6. However the magnitude of the gradient norm remains large, on the order 10^{-1} (the stopping criterion is that this norm is less than 10^{-3}). If we perform more iterations, the dual cost continues to increase very slowly, and the gradient norm slowly increases! Note that the contribution of each subproblem to the partial gradient with respect to Ξ is high (on the order of 10^4), but the sum of the contributions is low (on the order of 10^{-1}), which may lead to some numerical instability in the calculation of this partial gradient. The precise behavior of the algorithm and the interplay between the gradients of the different parameters would require further analysis.

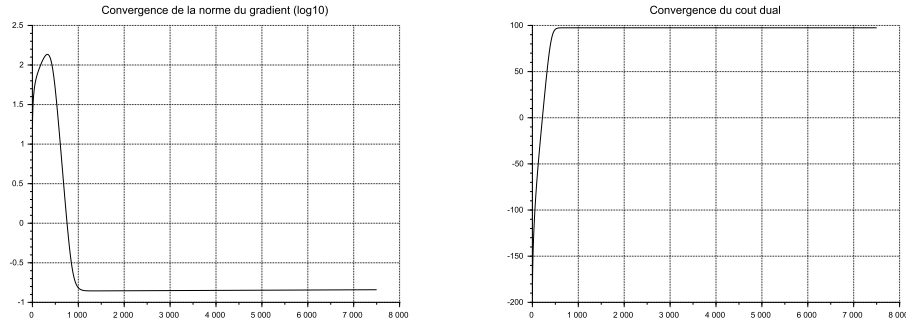


Figure 6: Convergence of the norm of the gradient (left) and of the dual cost (right) in space decomposition when using AR(1) multipliers with stationary state matrix Ξ

The value of the best dual cost obtained by using AR(1) multipliers with stationary Ξ is 97.46, that is, slightly higher than the value 97.24 obtained by using deterministic multipliers.

7.5.3 Uzawa algorithm with stationary matrix Ω

We also consider the case where the parameters Ω_t are stationary, that is, $\Omega_t = \Omega$ for all $t \in \mathbb{T}$. After 7500 fixed-step gradient iterations, the Uzawa algorithm stabilizes, as illustrated by Figure 7. The convergence behavior is very similar to the one observed at §7.5.1 for general AR(1) processes, which seems to indicate that the role of Ω is of minor importance in the algorithm convergence.

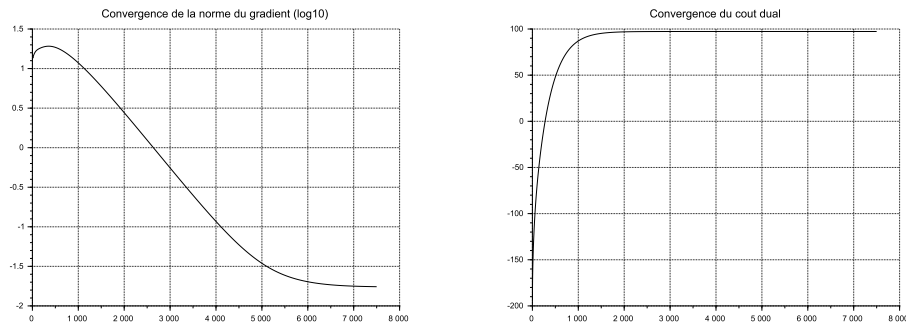


Figure 7: Convergence of the norm of the gradient (left) and of the dual cost (right) in space decomposition when using AR(1) multipliers with stationary noise matrix Ω

The value of the best dual cost obtained by using AR(1) multipliers with stationary Ω is 97.31.

7.5.4 Uzawa algorithm with stationary matrices Ξ and Ω

We finally consider the case where the parameters Ξ_t and Ω_t are stationary, that is, $\Xi_t = \Xi$ and $\Omega_t = \Omega$ for all $t \in \mathbb{T}$. After 1500 fixed-step gradient iterations, the Uzawa algorithm stabilizes, as illustrated by Figure 8. The convergence behavior is very similar to the one observed at §7.5.2 for stationary Ξ , which reinforces the impression that the role of Ω is of minor importance in the algorithm convergence.

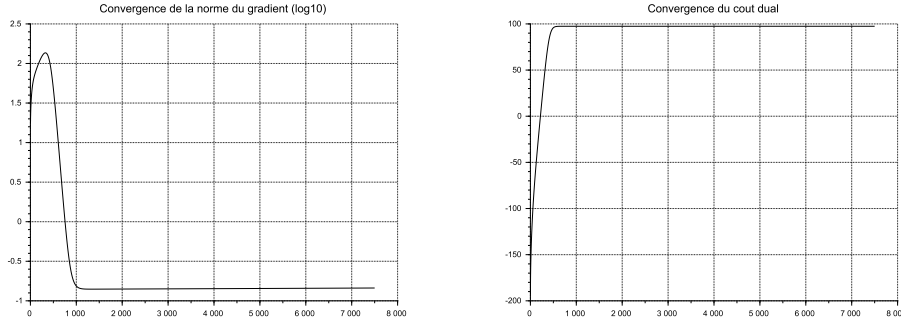


Figure 8: Convergence of the norm of the gradient (left) and of the dual cost (right) in space decomposition when using AR(1) multipliers with stationary state matrix Ξ and stationary noise matrix Ω

The value of the best dual cost obtained by using AR(1) multipliers with stationary Ω is 97.46.

7.5.5 Admissible policy and simulation

At the end of the Uzawa algorithm, we can construct surrogate Bellman functions for the global problem by adding the Bellman functions of the subproblems obtained by the algorithm. We can then simulate the global problem by minimizing at each time step t the instantaneous cost at t of the problem and the future cost at $t + 1$ given by the surrogate Bellman function (see §4.4 for further details).

The cost of a feasible solution obtained in simulation, which is therefore an upper bound of the optimal cost of the problem, has a value of 100.06, and the feasible solution is represented in Figure 9 with the same 10^6 noise scenarios as those used for the simulation of the global resolution at §7.3 and for the spatial decomposition at §7.4. Note that the four different variants of the algorithm (with or without stationary matrices Ξ and Ω) lead to the same upper bound and feasible solution. Again the solution obtained in simulation is virtually identical, in terms of cost and optimal controls, to that obtained by the global resolution of the problem.

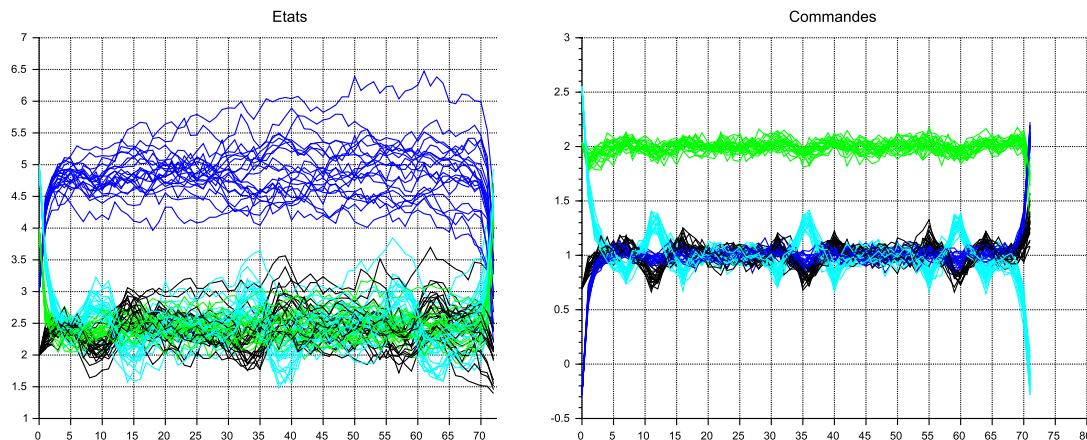


Figure 9: Feasible solution obtained by space decomposition with AR(1) multipliers

7.5.6 Interest of using AR(1) multipliers

The advantage of using AR(1) multipliers rather than deterministic multipliers for spatial decomposition of the problem is therefore very limited in the example considered here: the lower bound obtained with AR(1) multipliers is barely better than that provided by deterministic multipliers, and the feasible solutions are virtually identical. This behavior had already been observed, albeit less pronounced, in [2]. In that last study the gain from using AR(1) multipliers remained on the order of one percent, while the computational effort required to use such multipliers was much greater, since the size of the state for dynamic programming in each subproblem was increased by the size of the multiplier in the AR(1) case.

7.6 Temporal block decomposition

We implement the methodology presented in Section 5. The dual cost obtained by the calculation of the intraday functions and the Bellman recursion by temporal block, which corresponds to a lower bound of the optimal cost of the problem, has a value of 97.89. The cost of a feasible solution obtained in simulation, which is therefore an upper bound of the optimal cost of the problem, has a value of 100.00. This feasible solution is represented in Figure 10, with the same 10^6 noise scenarios as those used for the simulation of the global resolution of §7.3.

It is quite surprising to note that, while the Bellman functions are calculated by Lagrangian relaxation with deterministic multipliers on the temporal coupling constraints, the solution obtained in simulation is virtually identical, in terms of cost and optimal controls, to that obtained by the global resolution of the problem. Moreover, it seems that this property does not depend on the number of temporal blocks in the problem. Indeed, starting from the initial situation with 3 temporal blocks, if these initial blocks are arti-

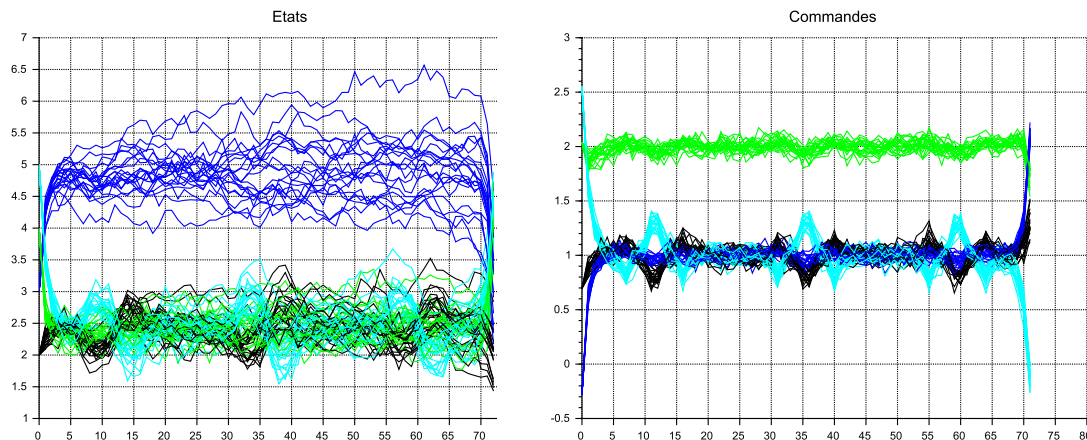


Figure 10: Feasible solution obtained by time block decomposition

ficially split into 2, 4 or 8 to move to 6, 12 or 24 temporal blocks, it can be observed in Table 2 that, although the lower bound given by the dual cost decreases slightly, the upper bound given by the admissible cost in simulation remains unchanged.

# blocks	Dual cost	Admissible cost
3	97.89	100.00
6	97.44	100.00
12	97.18	100.00
24	96.99	100.00

Table 2: Decomposition by temporal block according to the number of blocks

This surprising result probably comes from a particularity of the LQG problem considered, and it would be appropriate to explain it.

7.7 Mixed spatial and temporal block decompositions

The methodology presented in Section 6 is implemented.

7.7.1 Convergence of the Uzawa algorithm and lower bound

The Uzawa algorithm converges satisfactorily in a few thousand fixed-step gradient iterations. Figure 11 illustrates the decrease in the norm of the gradient of the dual function during the iterations, as well as the satisfaction *in expectation* of the coupling constraint at convergence (whereas the almost-sure constraint is of course not satisfied): this is what is expected given that we have limited ourselves to deterministic multipliers. The maximization of the dual cost by Uzawa leads to a lower bound of the order of 97.19.

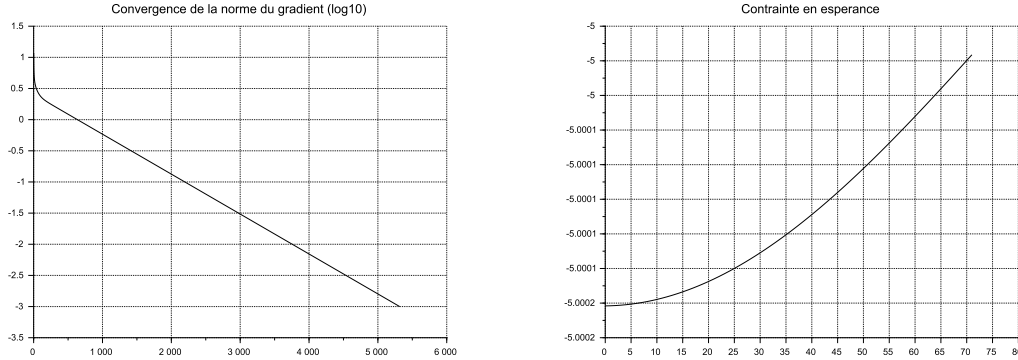


Figure 11: Convergence of the norm of the gradient (left) and expected value of the coupling constraint (right) in space/block decomposition

7.7.2 Obtaining an admissible solution and upper bound

Once the convergence of the Uzawa algorithm is obtained, substitute Bellman functions for the global problem can be constructed by adding the Bellman functions of the subproblems obtained by the algorithm. Then, the global problem can be simulated by minimizing, at each block d , the sum of the instantaneous costs of block d and the future cost at $d + 1$ given by the substitute Bellman function. This simulation is performed using the same 10^6 noise scenarios as those used for the simulation of the global solution in §7.3. Thus, an admissible solution of the global problem described by Figure 12 is obtained.

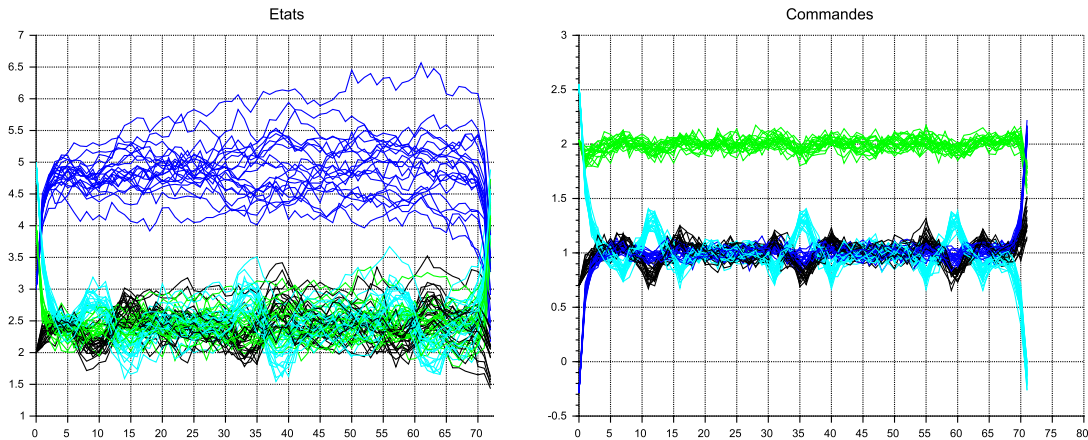


Figure 12: Admissible solution obtained following the space/block decomposition

The cost of this admissible solution is an upper bound on the optimal cost of the global problem. Its value is 100.01, very close to the optimal cost. Artificially varying

the number and size of the blocks (as presented for the temporal block decomposition in Table 2) changes this value in a very marginal manner.

Remark 10. *If we limit ourselves to deterministic spatial multipliers that are constant per temporal block, convergence occurs in less than 150 iterations, and the simulation cost is 100.24, i.e., an extremely small degradation compared to the optimal cost. In this case, the lower bound returned by the algorithm is -164.38 , which provides a very poor bound on the optimal cost of the problem. If we limit ourselves to a deterministic spatial multiplier that is constant over the entire optimization time span, the convergence occurs in less than 50 iterations, and the simulation cost is 100.24, which is therefore excellent. The lower bound given by the algorithm is -188.68 , thus extremely poor. Finally, with a spatial multiplier per time step, if we artificially split the initial temporal blocks, we observe that the lower and upper bounds vary only very marginally (less than 1%).*

7.8 Summary of obtained costs

The costs obtained by the different decompositions provide, as expected, lower and upper bounds of the optimal solution cost, as summarized in Table 3.

Dual cost (space/time decomposition)	97.19
Dual cost (space decomposition, deterministic multipliers)	97.24
Dual cost (space decomposition, AR(1) multipliers)	97.46
Dual cost (time decomposition)	97.89
Optimal cost of the global solution	100.00
Feasible cost (time decomposition)	100.00
Feasible cost (space decomposition, AR(1) multipliers)	100.06
Feasible cost (space decomposition, deterministic multipliers)	100.06
Feasible cost (space/time decomposition)	100.01

Table 3: Comparison of decomposition costs and optimal cost

We observe that if the lower bounds deviate by slightly less than 3% from the optimal cost, the upper bounds corresponding to feasible costs for the initial global problem are almost identical to the optimal cost. Therefore, the bound of the optimal cost is very good for the three decomposition methods, with a particularity that remains to be explained for all decomposition methods which provide the optimal solution of the problem in simulation.

8 Conclusion

In conclusion, for the particular instances of LQG problems studied in this paper, it appears that the spatial decomposition by prices and the temporal block decomposition by prices provide very good results, both in terms of the lower bounds obtained by the decompositions (within 3% of the optimal cost) and in terms of the costs associated with

the admissible orders induced by the decompositions (almost identical to the optimal cost for all decomposition methods). In the example considered in this paper, taking AR(1) multipliers rather than deterministic multipliers in the spatial decomposition does not significantly change the results. The temporal block decomposition by prices is particularly interesting because it does not require any specific algorithmic adjustment, whereas the spatial decomposition by prices requires the adjustment of a gradient-type algorithm, which can be slow if the problem involves many time steps.

There is a practical difficulty in implementing the simultaneous decomposition in space and temporal blocks, as there is then no advantage in considering periodicity classes. The presence of multipliers in the subproblems induced by the spatial decomposition by prices makes all temporal blocks of the optimization time span distinct. Therefore, there is no gain in considering blocks rather than considering all time steps of the time span.

Surprisingly, the best admissible solution obtained by the temporal block decomposition is identical to the optimal solution of the global problem. This result remains true even when smaller temporal blocks are used by artificially dividing the initial 3 temporal blocks of the problem's time span. The reason for this behavior needs to be understood. Equally surprising, the best admissible solution of the global problem obtained by the simultaneous decomposition in space and temporal blocks is also very close to the optimal solution of the global problem, may be of better quality than that obtained by the spatial decomposition alone. It seems that the temporal block decomposition mitigates the deviations induced by the spatial decomposition.

Finally, while considering constant multipliers per temporal block or over the entire time span significantly degrades the quality of the admissible solution in the case of spatial decomposition, this is not the case for the simultaneous decomposition in space and temporal blocks. For the latter, the admissible solution remains very close to the optimal solution. These behaviors need to be explained in the LQG case.

An extension of this work to the case of Decision-Hazard-Decision information structure [4] is currently being studied.

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A Computation of an exact upper bound

We are placing ourselves within the framework of Section 3. Once obtained the surrogate Bellman functions, the final simulation process consists in computing the expectation

$$\mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} \begin{pmatrix} x_t^\top & u_t^\top \end{pmatrix} \begin{pmatrix} M_t & S_t \\ S_t^\top & N_t \end{pmatrix} \begin{pmatrix} x_t \\ u_t \end{pmatrix} + m_t^\top x_t + n_t^\top u_t + d_t \right) + \frac{1}{2} x_{\mathcal{T}}^\top M_{\mathcal{T}} x_{\mathcal{T}} + m_{\mathcal{T}}^\top x_{\mathcal{T}} + d_{\mathcal{T}} \right],$$

subject to the dynamic constraints

$$x_0 \text{ given}, \quad x_{t+1} = A_t x_t + B_t u_t + C_t w_{t+1} + c_t, \quad t \in \mathbb{T},$$

where the control is the one given by §3.4, that is,

$$u_t = F_{t+1}^{-1} K_{t+1} x_t + F_{t+1}^{-1} k_{t+1},$$

that we express for convenience as $u_t = \Gamma_t x_t + \gamma_t$.

Replacing the last expression of the control in the cost function, we have to compute

$$\mathbb{E} \left[\sum_{t \in \mathbb{T}} \left(\frac{1}{2} x_t^\top \widetilde{M}_t x_t + \widetilde{m}_t^\top x_t + \widetilde{d}_t \right) + \frac{1}{2} x_{\mathcal{T}}^\top M_{\mathcal{T}} x_{\mathcal{T}} + m_{\mathcal{T}}^\top x_{\mathcal{T}} + d_{\mathcal{T}} \right], \quad (116a)$$

subject to

$$x_0 \text{ given}, \quad x_{t+1} = \widetilde{A}_t x_t + C_t w_{t+1} + \widetilde{c}_t, \quad t \in \mathbb{T}, \quad (116b)$$

with

$$\begin{aligned} \widetilde{M}_t &= M_t + (S_t \Gamma_t + \Gamma_t^\top S_t^\top) + \Gamma_t^\top N_t \Gamma_t, \\ \widetilde{m}_t &= S_t \gamma_t + \Gamma_t^\top N_t \gamma_t + m_t + \Gamma_t^\top n_t, \\ \widetilde{d}_t &= \frac{1}{2} \gamma_t^\top N_t \gamma_t + n_t^\top \gamma_t + d_t, \\ \widetilde{A}_t &= A_t + B_t \Gamma_t, \\ \widetilde{c}_t &= c_t + B_t \gamma_t. \end{aligned}$$

The expectation (116) can be computed by Dynamic Programming. Starting with

$$\widehat{V}_{\mathcal{T}}(x_{\mathcal{T}}) = \frac{1}{2} x_{\mathcal{T}}^\top M_{\mathcal{T}} x_{\mathcal{T}} + m_{\mathcal{T}}^\top x_{\mathcal{T}} + d_{\mathcal{T}},$$

and assuming that the Bellman function at time $t+1$ is of the form

$$\widehat{V}_{t+1}(x_{t+1}) = \frac{1}{2} x_{t+1}^\top \widehat{P}_{t+1} x_{t+1} + \widehat{p}_{t+1}^\top x_{t+1} + \widehat{r}_{t+1},$$

we easily prove that the Bellman function at time t is of the same form, with

$$\begin{aligned} \widehat{P}_t &= \widetilde{M}_t + \widetilde{A}_t^\top \widehat{P}_{t+1} \widetilde{A}_t, \\ \widehat{p}_t &= \widetilde{m}_t + \widetilde{A}_t^\top (\widehat{p}_{t+1} + \widehat{P}_{t+1} \widetilde{c}_t), \\ \widehat{r}_t &= \widetilde{d}_t + \frac{1}{2} \widetilde{c}_t^\top \widehat{P}_{t+1} \widetilde{c}_t + \frac{1}{2} \text{trace}(\Sigma_{t+1} C_t^\top \widehat{P}_{t+1} C_t) + \widehat{p}_{t+1}^\top \widetilde{c}_t + \widehat{r}_{t+1}. \end{aligned}$$

The value $\widehat{V}_0(x_0)$ is an *exact* upper bound of the optimal value of Problem (1).

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